2023 Annual Dairy Report Reissued WDR for Existing Milk Cow Dairies

No. R5-2013-0122

John A. Machado Dairy Farms, 24916 S. Manteca, CA 95337

June 26, 2024

→ **The Power of Commitment**

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: John A Machado Dairy Farms

Physical address of dairy:

Number and Street	City	County	Zip Code
24916 South Manteca RD	Manteca	San Joaquin	95337

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1938

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

2571-1004-XXXX-XXXX 2572-2001-XXXX-XXXX

B. OPERATORS

Machado, John

Operator name: Machado, John

Telephone no.: (209) 239-4144 Landline, (209) 985-5816 Cellular

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 809	Ripon	CA	95366

This operator is responsible for paying permit fees.

C. OWNERS

Machado, John

Legal owner name: Machado, John

Telephone no.: (209) 239-4144 Landline, (209) 985-5816 Cellular

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 809	Ripon	CA	95366

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	197	137	0	0
Number under roof	1,714	237	0	0	89	83
Maximum number	1,714	237	197	137	89	83
Average number	1,714	237	197	137	89	83
Avg live weight (lbs)	1,400	1,500	1,100	775		

Predominant milk cow breed: Holstein

Average milk production: 64 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 50,313.99 *tons per reporting period*

Total nitrogen from manure: 642,799.40 *lbs per reporting period*

After ammonia losses (30% loss applied): 449,959.58 *lbs per reporting period*

Total phosphorus from manure: 106,110.52 *lbs per reporting period*

Total potassium from manure: 321,490.23 *lbs per reporting period*

Total salt from manure: 861,535.05 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 16,574,171 *gallons*

Total nitrogen generated: 92,311.37 *lbs*

Total phosphorus generated: 8,474.28 *lbs*

Total potassium generated: 103,492.86 *lbs*

Total salt generated: 649,005.15 *lbs*

	16,574,171	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	16,574,171	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water
River	Surface water
Well #1	Ground water
Well #10	Ground water
Well #11	Ground water

Source Description	Type
Well #12	Ground water
Well #13	Ground water
Well #14	Ground water
Well #15	Ground water
Well #16	Ground water
Well #17	Ground water
Well #2	Ground water
Well #3	Ground water
Well #6	Ground water
Well #8	Ground water
Well #9	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Airport 1	79	79	2	manure	2413-2018-XXXX-XXXX
Airport 2	37	37	2	manure	2413-2018-XXXX-XXXX
Airport 3	39	39	2	manure	2413-2018-XXXX-XXXX
Anderson	55	55	2	both	2571-1004-XXXX-XXXX
Angyls 1-A	35	35	2	both	2571-0016-XXXX-XXXX
Angyls 2	38	38	2	both	2570-5020-XXXX-XXXX
Angyls 3	48	48	2	both	2571-0016-XXXX-XXXX
Angyls 4	52	52	2	both	2571-0016-XXXX-XXXX
Angyls 5	20	20	2	both	2571-0016-XXXX-XXXX
Angyls 7	27	27	2	both	2571-0008-XXXX-XXXX 2571-0016-XXXX-XXXX
Angyls 8	78	78	2	both	2571-0016-XXXX-XXXX
Home A	9	9	2	both	2572-2001-XXXX-XXXX
Home B	38	38	2	both	2572-2001-XXXX-XXXX 2572-2002-XXXX-XXXX
Home C	37	37	2	both	2571-4009-XXXX-XXXX
Home D	43	43	2	both	2571-4010-XXXX-XXXX 2571-4011-XXXX-XXXX
Home E/F	37	37	2	both	2571-4011-XXXX-XXXX
Home G	19	19	2	both	2571-4003-XXXX-XXXX
Hunts 1	63	63	2	manure	2412-3004-XXXX-XXXX
Hunts 2	17	17	2	manure	2412-3004-XXXX-XXXX
Manteca	80	80	2	both	2571-7005-XXXX-XXXX 2571-7006-XXXX-XXXX
Maries 1	30	30	2	both	2571-2004-XXXX-XXXX
Maries 2	80	80	2	both	2571-2004-XXXX-XXXX
Maries 3	35	35	2	both	2571-2004-XXXX-XXXX
Totals for areas that were used for application	996	996	46		
Totals for areas that were not used for application					
Land application area totals	996	996	46		

B. CROPS AND HARVESTS

Airport 1

Airport 1

Field name: Airport 1

11/05/2022: Winter Forage

Crop: Winter Forage Acres planted: 79 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/25/2023	653.30 *ton*	Dry-weight		66.7	10,700.00	2,640.00	15,800.00		8.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	10.00	110.00	17.00	83.00	0.00
Total actual harvest content	8.27	58.93	14.54	87.02	440.61

06/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 79 Plant date: 06/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/05/2023	940.80 *ton*	Dry-weight		59.2	12,800.00	1,950.00	12,000.00		4.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	120.00	22.50	99.00	0.00
Total actual harvest content	11.91	124.39	18.95	116.61	447.01

Airport 2

Field name: Airport 2

11/05/2022: Winter Forage

Crop: Winter Forage Acres planted: 37 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	285.60 *ton*	Dry-weight		56.0	10,200.00	2,140.00	13,900.00		12.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	10.00	110.00	17.00	83.00	0.00
Total actual harvest content	7.72	69.29	14.54	94.42	835.50

Airport 2

06/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 37 Plant date: 06/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/05/2023	322.20 *ton*	Dry-weight		63.2	8,160.00	2,410.00	17,000.00		5.91

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	10.00	80.00	15.00	66.00	0.00
Total actual harvest content	8.71	52.30	15.45	108.96	378.78

Airport 3

Field name: Airport 3

11/05/2022: Winter Forage

Crop: Winter Forage Acres planted: 39 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/25/2023	428.40 *ton*	Dry-weight		61.5	10,100.00	2,710.00	14,800.00		8.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	165.00	25.50	124.50	0.00
Total actual harvest content	10.98	85.43	22.92	125.18	676.65

06/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 39 Plant date: 06/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/05/2023	531.30 *ton*	Dry-weight		69.4	8,480.00	2,520.00	17,800.00		5.47

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	120.00	22.50	99.00	0.00
Total actual harvest content	13.62	70.70	21.01	148.40	456.05

Anderson

Field name: Anderson

11/06/2022: Winter Forage

Crop: Winter Forage Acres planted: 55 Plant date: 11/06/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	1,140.70 *ton*	Dry-weight		61.7	14,500.00	3,800.00	35,400.00		10.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	220.00	34.00	166.00	0.00
Total actual harvest content	20.74	230.36	60.37	562.39	1,699.89

06/07/2023: Corn, silage

Crop: Corn, silage Acres planted: 55 Plant date: 06/07/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/05/2023	1,326.80 *ton*	Dry-weight		61.0	9,390.00	2,190.00	20,400.00		5.27

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	24.12	176.69	41.21	383.86	991.63

Angyls 1-A

Field name: Angyls 1-A

10/28/2022: Winter Forage

Crop: Winter Forage Acres planted: 35 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	433.40 *ton*	Dry-weight		73.2	13,200.00	2,540.00	21,500.00		11.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	154.00	23.80	116.20	0.00
Total actual harvest content	12.38	87.61	16.86	142.70	763.28

Angyls 1-A

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	698.30 *ton*	Dry-weight		71.4	11,400.00	2,330.00	11,700.00		9.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	160.00	30.00	132.00	0.00
Total actual harvest content	19.95	130.10	26.59	133.52	1,049.92

Angyls 2

Field name: Angyls 2

10/28/2022: Winter Forage

Crop: Winter Forage Acres planted: 38 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	586.50 *ton*	Dry-weight		70.8	14,200.00	1,880.00	16,200.00		8.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	165.00	25.50	124.50	0.00
Total actual harvest content	15.43	127.99	16.95	146.02	784.18

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 38 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	646.40 *ton*	Dry-weight		70.2	10,500.00	2,070.00	8,480.00		8.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	160.00	30.00	132.00	0.00
Total actual harvest content	17.01	106.45	20.99	85.97	821.20

Angyls 3

Field name: Angyls 3

10/28/2022: Winter Forage

Crop: Winter Forage Acres planted: 48 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	681.20 *ton*	Dry-weight		74.9	13,700.00	1,840.00	14,200.00		8.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	165.00	25.50	124.50	0.00
Total actual harvest content	14.19	97.60	13.11	101.16	577.06

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 48 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	941.90 *ton*	Dry-weight		72.2	14,400.00	2,040.00	9,930.00		10.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	160.00	30.00	132.00	0.00
Total actual harvest content	19.62	157.11	22.26	108.34	1,134.68

Angyls 4

Field name: Angyls 4

10/28/2022: Winter Forage

Crop: Winter Forage Acres planted: 52 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	523.70 *ton*	Dry-weight		73.3	12,800.00	2,160.00	17,100.00		8.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	10.00	110.00	17.00	83.00	0.00
Total actual harvest content	10.07	68.84	11.62	91.96	441.00

Angyls 4

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 52 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/05/2023	1,111.10 *ton*	Dry-weight		70.8	12,500.00	1,560.00	6,980.00		9.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	21.37	155.98	19.47	87.10	1,123.07

Angyls 5

Field name: Angyls 5

10/28/2022: Winter Forage

Crop: Winter Forage Acres planted: 20 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	293.20 *ton*	Dry-weight		72.8	11,800.00	1,370.00	6,490.00		8.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	154.00	23.80	116.20	0.00
Total actual harvest content	14.66	94.11	10.93	51.76	669.90

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	479.70 *ton*	Dry-weight		72.8	11,800.00	1,370.00	6,490.00		8.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	160.00	30.00	132.00	0.00
Total actual harvest content	23.99	153.96	17.88	84.68	1,096.02

Angyls 7

Field name: Angyls 7

10/28/2022: Winter Forage

Crop: Winter Forage Acres planted: 27 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	374.50 *ton*	Dry-weight		72.9	12,400.00	1,950.00	4,340.00		8.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	154.00	23.80	116.20	0.00
Total actual harvest content	13.87	93.22	14.66	32.63	639.01

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 27 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	438.90 *ton*	Dry-weight		73.8	10,200.00	1,750.00	6,150.00		8.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	160.00	30.00	132.00	0.00
Total actual harvest content	16.26	86.88	14.91	52.39	741.06

Angyls 8

Field name: Angyls 8

10/28/2022: Winter Forage

Crop: Winter Forage Acres planted: 78 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	923.50 *ton*	Dry-weight		70.3	12,800.00	1,990.00	16,200.00		9.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	132.00	20.40	99.60	0.00
Total actual harvest content	11.84	90.02	14.00	113.93	639.99

Angyls 8

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 78 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	1,454.80 *ton*	Dry-weight		73.0	13,400.00	1,680.00	7,110.00		8.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	160.00	30.00	132.00	0.00
Total actual harvest content	18.65	134.96	16.92	71.61	815.81

Home A

Field name: Home A

11/04/2022: Winter Forage

Crop: Winter Forage Acres planted: 9 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	206.70 *ton*	Dry-weight		66.2	16,000.00	4,010.00	29,300.00		11.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	275.00	42.50	207.50	0.00
Total actual harvest content	22.97	248.41	62.26	454.90	1,816.48

06/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 9 Plant date: 06/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	183.80 *ton*	Dry-weight		70.2	10,800.00	4,220.00	28,300.00		6.12

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	160.00	30.00	132.00	0.00
Total actual harvest content	20.42	131.45	51.36	344.46	744.90

Home B

Field name: Home B

11/04/2022: Winter Forage

Crop: Winter Forage Acres planted: 38 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	914.60 *ton*	Dry-weight		70.1	15,100.00	3,970.00	35,000.00		11.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	275.00	42.50	207.50	0.00
Total actual harvest content	24.07	217.33	57.14	503.75	1,655.19

06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 38 Plant date: 06/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	518.90 *ton*	Dry-weight		69.0	12,200.00	2,690.00	17,800.00		9.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	120.00	22.50	99.00	0.00
Total actual harvest content	13.66	103.29	22.77	150.70	770.43

Home C

Field name: Home C

11/04/2022: Winter Forage

Crop: Winter Forage Acres planted: 37 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	698.30 *ton*	Dry-weight		54.3	15,700.00	3,150.00	30,300.00		8.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	220.00	34.00	166.00	0.00
Total actual harvest content	18.87	270.82	54.34	522.67	1,431.74

Home C

06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 37 Plant date: 06/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	709.90 *ton*	Dry-weight		69.2	11,400.00	2,590.00	13,300.00		9.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	160.00	30.00	132.00	0.00
Total actual harvest content	19.19	134.74	30.61	157.19	1,087.34

Home D

Field name: Home D

11/04/2022: Winter Forage

Crop: Winter Forage Acres planted: 43 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/25/2023	812.70 *ton*	Dry-weight		68.5	15,200.00	3,770.00	33,500.00		9.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	220.00	34.00	166.00	0.00
Total actual harvest content	18.90	180.99	44.89	398.88	1,166.89

06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 43 Plant date: 06/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	853.80 *ton*	Dry-weight		68.8	15,000.00	2,940.00	15,300.00		9.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	160.00	30.00	132.00	0.00
Total actual harvest content	19.86	185.85	36.43	189.57	1,189.44

Home E/F

Field name: Home E/F

11/04/2022: Winter Forage

Crop: Winter Forage Acres planted: 37 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	849.80 *ton*	Dry-weight		65.3	14,800.00	3,900.00	28,600.00		10.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	275.00	42.50	207.50	0.00
Total actual harvest content	22.97	235.90	62.16	455.87	1,673.65

06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 37 Plant date: 06/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	827.50 *ton*	Dry-weight		71.2	11,300.00	3,150.00	19,700.00		6.67

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	160.00	30.00	132.00	0.00
Total actual harvest content	22.36	145.57	40.58	253.78	859.24

Home G

Field name: Home G

11/04/2022: Winter Forage

Crop: Winter Forage Acres planted: 19 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	376.30 *ton*	Dry-weight		69.3	14,700.00	3,890.00	37,100.00		11.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	220.00	34.00	166.00	0.00
Total actual harvest content	19.81	178.76	47.30	451.15	1,447.09

Home G

06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 19 Plant date: 06/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/05/2023	508.90 *ton*	Dry-weight		67.8	11,900.00	1,990.00	11,700.00		8.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.78	205.26	34.33	201.81	1,397.17

Hunts 1

Field name: Hunts 1

11/04/2022: Winter Forage

Crop: Winter Forage Acres planted: 63 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	908.60 *ton*	Dry-weight		71.8	13,600.00	2,130.00	16,700.00		11.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	165.00	25.50	124.50	0.00
Total actual harvest content	14.42	110.62	17.33	135.84	935.43

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 63 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/05/2023	960.90 *ton*	Dry-weight		66.0	7,060.00	1,930.00	10,300.00		5.37

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	160.00	30.00	132.00	0.00
Total actual harvest content	15.25	73.22	20.02	106.83	556.96

Hunts 2

Field name: Hunts 2

11/04/2022: Winter Forage

Crop: Winter Forage Acres planted: 17 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	245.20 *ton*	Dry-weight		74.6	12,300.00	2,280.00	17,400.00		10.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	165.00	25.50	124.50	0.00
Total actual harvest content	14.42	90.12	16.71	127.49	747.37

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 17 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/05/2023	308.60 *ton*	Dry-weight		69.0	6,390.00	2,110.00	12,700.00		5.66

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	160.00	30.00	132.00	0.00
Total actual harvest content	18.15	71.92	23.75	142.94	637.02

Manteca

Field name: Manteca

11/01/2022: Winter Forage

Crop: Winter Forage Acres planted: 80 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	1,569.50 *ton*	Dry-weight		68.2	18,200.00	3,070.00	25,100.00		11.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	220.00	34.00	166.00	0.00
Total actual harvest content	19.62	227.09	38.31	313.19	1,372.53

Manteca

06/06/2023: Corn, silage

Crop: Corn, silage Acres planted: 80 Plant date: 06/06/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/05/2023	1,817.90 *ton*	Dry-weight		65.0	7,780.00	2,540.00	22,900.00		6.18

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	160.00	30.00	132.00	0.00
Total actual harvest content	22.72	123.75	40.40	364.26	983.03

Maries 1

Field name: Maries 1

11/06/2022: Winter Forage

Crop: Winter Forage Acres planted: 30 Plant date: 11/06/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/05/2023	573.00 *ton*	Dry-weight		75.2	20,800.00	3,580.00	29,900.00		11.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	220.00	34.00	166.00	0.00
Total actual harvest content	19.10	197.05	33.92	283.26	1,089.46

06/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 06/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	770.30 *ton*	Dry-weight		74.0	16,300.00	3,120.00	22,500.00		11.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.68	217.64	41.66	300.42	1,508.76

Maries 2

Field name: Maries 2

11/06/2022: Winter Forage

Crop: Winter Forage Acres planted: 80 Plant date: 11/06/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/05/2023	1,370.90 *ton*	Dry-weight		62.6	21,000.00	3,030.00	24,000.00		8.99

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	220.00	34.00	166.00	0.00
Total actual harvest content	17.14	269.18	38.84	307.63	1,152.33

06/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 80 Plant date: 06/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	1,561.80 *ton*	Dry-weight		71.8	16,800.00	3,440.00	19,000.00		11.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	160.00	30.00	132.00	0.00
Total actual harvest content	19.52	184.98	37.88	209.20	1,255.22

Maries 3

Field name: Maries 3

11/06/2022: Winter Forage

Crop: Winter Forage Acres planted: 35 Plant date: 11/06/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/05/2023	674.00 *ton*	Dry-weight		76.0	19,800.00	3,930.00	31,600.00		12.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	220.00	34.00	166.00	0.00
Total actual harvest content	19.26	183.02	36.33	292.09	1,164.67

Maries 3

06/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 06/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	890.40 *ton*	Dry-weight		71.8	13,000.00	3,240.00	17,000.00		9.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	25.44	186.53	46.49	243.92	1,334.38

NUTRIENT BUDGET

A. LAND APPLICATIONS

Airport 1 - 11/05/2022: Winter Forage

Field name: Airport 1

Crop: Winter Forage — Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Separator solids	73.85	132.73	23.10	55.93	128.50 *ton*
Application event totals		73.85	132.73	23.10	55.93	

Airport 1 - 06/13/2023: Corn, silage

Field name: Airport 1

Crop: Corn, silage — Plant date: 06/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Separator solids	158.81	23.67	54.39	0.00	410.00 *ton*
Application event totals		158.81	23.67	54.39	0.00	

05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	29.77	8,807,226.00 *gal*
Application event totals		0.00	0.00	0.00	29.77	

06/08/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3-8-11	Solid commercial fertilizer	8.12	21.66	29.79	0.00	
Application event totals		8.12	21.66	29.79	0.00	

Airport 1 - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	22.33	6,605,419.00 *gal*
Application event totals		0.00	0.00	0.00	22.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	27.29	8,073,290.00 *gal*
Application event totals		0.00	0.00	0.00	27.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	27.29	8,073,290.00 *gal*
Application event totals		0.00	0.00	0.00	27.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	27.29	8,073,290.00 *gal*
Application event totals		0.00	0.00	0.00	27.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	34.73	10,275,097.00 *gal*
Application event totals		0.00	0.00	0.00	34.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	34.73	10,275,097.00 *gal*
Application event totals		0.00	0.00	0.00	34.73	

Airport 1 - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	34.73	10,275,097.00 *gal*
Application event totals		0.00	0.00	0.00	34.73	

Airport 2 - 11/05/2022: Winter Forage

Field name: Airport 2

Crop: Winter Forage

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Separator solids	85.89	154.38	26.86	65.05	70.00 *ton*
Application event totals		85.89	154.38	26.86	65.05	

Airport 2 - 06/13/2023: Corn, silage

Field name: Airport 2

Crop: Corn, silage

Plant date: 06/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Separator solids	57.89	8.63	19.83	0.00	70.00 *ton*
Application event totals		57.89	8.63	19.83	0.00	

Airport 2 - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	29.77	4,124,903.00 *gal*
Application event totals		0.00	0.00	0.00	29.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3-8-11	Solid commercial fertilizer	8.12	21.66	29.79	0.00	
Application event totals		8.12	21.66	29.79	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	22.33	3,093,677.00 *gal*
Application event totals		0.00	0.00	0.00	22.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	27.29	3,781,161.00 *gal*
Application event totals		0.00	0.00	0.00	27.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	27.29	3,781,161.00 *gal*
Application event totals		0.00	0.00	0.00	27.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	27.29	3,781,161.00 *gal*
Application event totals		0.00	0.00	0.00	27.29	

Airport 2 - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	34.73	4,812,387.00 *gal*
Application event totals		0.00	0.00	0.00	34.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	34.73	4,812,387.00 *gal*
Application event totals		0.00	0.00	0.00	34.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	34.73	4,812,387.00 *gal*
Application event totals		0.00	0.00	0.00	34.73	

Airport 3 - 11/05/2022: Winter Forage

Field name: Airport 3

Crop: Winter Forage

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Separator solids	104.77	188.31	32.77	79.35	90.00 *ton*
Application event totals		104.77	188.31	32.77	79.35	

Airport 3 - 06/13/2023: Corn, silage

Field name: Airport 3

Crop: Corn, silage

Plant date: 06/13/2023

Airport 3 - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Separator solids	83.17	12.39	28.48	0.00	106.00 *ton*
Application event totals		83.17	12.39	28.48	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	29.77	4,347,871.00 *gal*
Application event totals		0.00	0.00	0.00	29.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3-8-11	Solid commercial fertilizer	8.12	21.66	29.79	0.00	
Application event totals		8.12	21.66	29.79	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	22.33	3,260,903.00 *gal*
Application event totals		0.00	0.00	0.00	22.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	27.29	3,985,548.00 *gal*
Application event totals		0.00	0.00	0.00	27.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	27.29	3,985,548.00 *gal*
Application event totals		0.00	0.00	0.00	27.29	

Airport 3 - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	27.29	3,985,548.00 *gal*
Application event totals		0.00	0.00	0.00	27.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	34.73	5,072,516.00 *gal*
Application event totals		0.00	0.00	0.00	34.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	34.73	5,072,516.00 *gal*
Application event totals		0.00	0.00	0.00	34.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	34.73	5,072,516.00 *gal*
Application event totals		0.00	0.00	0.00	34.73	

Anderson - 11/06/2022: Winter Forage

Field name: Anderson

Crop: Winter Forage

Plant date: 11/06/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	27.34	3.60	64.18	300.35	422,979.00 *gal*
Application event totals		27.34	3.60	64.18	300.35	

Anderson - 11/06/2022: Winter Forage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Separator solids	244.79	104.87	216.45	712.41	354.30 *ton*
Application event totals		244.79	104.87	216.45	712.41	

Anderson - 06/07/2023: Corn, silage

Field name: Anderson

Crop: Corn, silage

Plant date: 06/07/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	49.14	3.74	61.38	352.20	426,702.00 *gal*
Application event totals		49.14	3.74	61.38	352.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Corral solids	24.38	6.89	35.02	0.00	45.00 *ton*
Application event totals		24.38	6.89	35.02	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.22	0.00	0.00	25.34	5,760,000.00 *gal*
Application event totals		1.22	0.00	0.00	25.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	47.38	3.61	59.18	339.59	411,429.00 *gal*
Application event totals		47.38	3.61	59.18	339.59	

Anderson - 06/07/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.92	0.00	0.00	19.01	4,320,000.00 *gal*
Application event totals		0.92	0.00	0.00	19.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0 WECO-BOOST	Liquid commercial fertilizer	0.06	0.00	0.00	0.00	
Wastewater	Process wastewater	53.30	4.06	66.58	382.04	462,857.00 *gal*
Application event totals		53.36	4.06	66.58	382.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.92	0.00	0.00	19.01	4,320,000.00 *gal*
Application event totals		0.92	0.00	0.00	19.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	53.30	4.06	66.58	382.04	462,857.00 *gal*
River	Surface water	0.92	0.00	0.00	19.01	4,320,000.00 *gal*
Application event totals		54.22	4.06	66.58	401.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.92	0.00	0.00	19.01	4,320,000.00 *gal*
Application event totals		0.92	0.00	0.00	19.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	3.06	0.00	0.00	63.36	14,400,000.00 *gal*
Application event totals		3.06	0.00	0.00	63.36	

Anderson - 06/07/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.84	0.00	0.00	38.02	8,640,000.00 *gal*
Application event totals		1.84	0.00	0.00	38.02	

09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.84	0.00	0.00	38.02	8,640,000.00 *gal*
Application event totals		1.84	0.00	0.00	38.02	

Angyls 1-A - 10/28/2022: Winter Forage

Field name: Angyls 1-A

Crop: Winter Forage Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	27.34	3.60	64.18	300.35	269,169.00 *gal*
Application event totals		27.34	3.60	64.18	300.35	

10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Separator solids	81.43	34.89	72.00	236.98	75.00 *ton*
Application event totals		81.43	34.89	72.00	236.98	

Angyls 1-A - 06/20/2023: Corn, silage

Field name: Angyls 1-A

Crop: Corn, silage Plant date: 06/20/2023

Angyls 1-A - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	49.14	3.74	61.38	352.20	271,537.00 *gal*
Application event totals		49.14	3.74	61.38	352.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	2.03	0.00	0.00	42.10	6,088,591.00 *gal*
Application event totals		2.03	0.00	0.00	42.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Corral solids	106.43	30.07	152.86	0.00	125.00 *ton*
Application event totals		106.43	30.07	152.86	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3-8-11	Solid commercial fertilizer	8.12	21.66	29.79	0.00	
Application event totals		8.12	21.66	29.79	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.58	0.00	0.00	32.74	4,735,570.00 *gal*
Application event totals		1.58	0.00	0.00	32.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.35	0.00	0.00	28.07	4,059,060.00 *gal*
Application event totals		1.35	0.00	0.00	28.07	

Angyls 1-A - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.35	0.00	0.00	28.07	4,059,060.00 *gal*
Application event totals		1.35	0.00	0.00	28.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.35	0.00	0.00	28.07	4,059,060.00 *gal*
Application event totals		1.35	0.00	0.00	28.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.35	0.00	0.00	28.07	4,059,060.00 *gal*
Application event totals		1.35	0.00	0.00	28.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #12	Ground water	0.80	0.00	0.00	154.61	1,403,617.00 *gal*
Application event totals		0.80	0.00	0.00	154.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #13	Ground water	0.00	0.00	0.00	106.42	1,403,617.00 *gal*
Application event totals		0.00	0.00	0.00	106.42	

Angyls 2 - 10/28/2022: Winter Forage

Field name: Angyls 2

Crop: Winter Forage

Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Angyls 2 - 10/28/2022: Winter Forage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	27.34	3.60	64.18	300.35	292,240.00 *gal*
Application event totals		27.34	3.60	64.18	300.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Separator solids	137.30	58.82	121.40	399.59	137.30 *ton*
Application event totals		137.30	58.82	121.40	399.59	

Angyls 2 - 06/20/2023: Corn, silage

Field name: Angyls 2

Crop: Corn, silage

Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	49.14	3.74	61.38	352.20	294,812.00 *gal*
Application event totals		49.14	3.74	61.38	352.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	2.03	0.00	0.00	42.10	6,610,470.00 *gal*
Application event totals		2.03	0.00	0.00	42.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Corral solids	70.58	19.94	101.37	0.00	90.00 *ton*
Application event totals		70.58	19.94	101.37	0.00	

Angyls 2 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3-8-11	Solid commercial fertilizer	8.12	21.66	29.79	0.00	
Application event totals		8.12	21.66	29.79	0.00	

06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.58	0.00	0.00	32.74	5,141,477.00 *gal*
Application event totals		1.58	0.00	0.00	32.74	

07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.35	0.00	0.00	28.07	4,406,980.00 *gal*
Application event totals		1.35	0.00	0.00	28.07	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.35	0.00	0.00	28.07	4,406,980.00 *gal*
Application event totals		1.35	0.00	0.00	28.07	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.35	0.00	0.00	28.07	4,406,980.00 *gal*
Application event totals		1.35	0.00	0.00	28.07	

08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.35	0.00	0.00	28.07	4,406,980.00 *gal*
Application event totals		1.35	0.00	0.00	28.07	

Angyls 2 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #12	Ground water	0.80	0.00	0.00	154.61	1,523,928.00 *gal*
Application event totals		0.80	0.00	0.00	154.61	

09/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #13	Ground water	0.00	0.00	0.00	106.42	1,523,928.00 *gal*
Application event totals		0.00	0.00	0.00	106.42	

Angyls 3 - 10/28/2022: Winter Forage

Field name: Angyls 3

Crop: Winter Forage Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	27.34	3.60	64.18	300.35	369,145.00 *gal*
Application event totals		27.34	3.60	64.18	300.35	

10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Separator solids	100.54	43.07	88.90	292.61	127.00 *ton*
Application event totals		100.54	43.07	88.90	292.61	

Angyls 3 - 06/20/2023: Corn, silage

Field name: Angyls 3

Crop: Corn, silage Plant date: 06/20/2023

Angyls 3 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	49.14	3.74	61.38	352.20	372,394.00 *gal*
Application event totals		49.14	3.74	61.38	352.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	2.03	0.00	0.00	42.10	8,350,067.00 *gal*
Application event totals		2.03	0.00	0.00	42.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Corral solids	144.03	40.70	206.87	0.00	232.00 *ton*
Application event totals		144.03	40.70	206.87	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3-8-11	Solid commercial fertilizer	8.12	21.66	29.79	0.00	
Application event totals		8.12	21.66	29.79	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.58	0.00	0.00	32.74	6,494,497.00 *gal*
Application event totals		1.58	0.00	0.00	32.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.35	0.00	0.00	28.07	5,566,711.00 *gal*
Application event totals		1.35	0.00	0.00	28.07	

Angyls 3 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.35	0.00	0.00	28.07	5,566,711.00 *gal*
Application event totals		1.35	0.00	0.00	28.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.35	0.00	0.00	28.07	5,566,711.00 *gal*
Application event totals		1.35	0.00	0.00	28.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.35	0.00	0.00	28.07	5,566,711.00 *gal*
Application event totals		1.35	0.00	0.00	28.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #12	Ground water	0.80	0.00	0.00	154.61	1,924,961.00 *gal*
Application event totals		0.80	0.00	0.00	154.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #13	Ground water	0.00	0.00	0.00	106.42	1,924,961.00 *gal*
Application event totals		0.00	0.00	0.00	106.42	

Angyls 4 - 10/28/2022: Winter Forage

Field name: Angyls 4

Crop: Winter Forage

Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Angyls 4 - 10/28/2022: Winter Forage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	27.34	3.60	64.18	300.35	399,908.00 *gal*
Application event totals		27.34	3.60	64.18	300.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Separator solids	58.46	25.05	51.69	170.14	80.00 *ton*
Application event totals		58.46	25.05	51.69	170.14	

Angyls 4 - 06/20/2023: Corn, silage

Field name: Angyls 4

Crop: Corn, silage

Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	49.14	3.74	61.38	352.20	403,427.00 *gal*
Application event totals		49.14	3.74	61.38	352.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	2.03	0.00	0.00	42.10	9,045,906.00 *gal*
Application event totals		2.03	0.00	0.00	42.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Corral solids	96.11	27.15	138.03	0.00	167.70 *ton*
Application event totals		96.11	27.15	138.03	0.00	

Angyls 4 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3-8-11	Solid commercial fertilizer	8.12	21.66	29.79	0.00	
Application event totals		8.12	21.66	29.79	0.00	

06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.58	0.00	0.00	32.74	7,035,705.00 *gal*
Application event totals		1.58	0.00	0.00	32.74	

07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.35	0.00	0.00	28.07	6,030,604.00 *gal*
Application event totals		1.35	0.00	0.00	28.07	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.35	0.00	0.00	28.07	6,030,604.00 *gal*
Application event totals		1.35	0.00	0.00	28.07	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.35	0.00	0.00	28.07	6,030,604.00 *gal*
Application event totals		1.35	0.00	0.00	28.07	

08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.35	0.00	0.00	28.07	6,030,604.00 *gal*
Application event totals		1.35	0.00	0.00	28.07	

Angyls 4 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #12	Ground water	0.80	0.00	0.00	154.61	2,085,374.00 *gal*
Application event totals		0.80	0.00	0.00	154.61	

09/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #13	Ground water	0.00	0.00	0.00	106.42	2,085,374.00 *gal*
Application event totals		0.00	0.00	0.00	106.42	

Angyls 5 - 10/28/2022: Winter Forage

Field name: Angyls 5

Crop: Winter Forage Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	27.34	3.60	64.18	300.35	153,811.00 *gal*
Application event totals		27.34	3.60	64.18	300.35	

10/18/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Separator solids	85.50	36.63	75.60	248.83	45.00 *ton*
Application event totals		85.50	36.63	75.60	248.83	

Angyls 5 - 06/20/2023: Corn, silage

Field name: Angyls 5

Crop: Corn, silage Plant date: 06/20/2023

Angyls 5 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	49.14	3.74	61.38	352.20	155,164.00 *gal*
Application event totals		49.14	3.74	61.38	352.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	2.03	0.00	0.00	42.10	3,479,195.00 *gal*
Application event totals		2.03	0.00	0.00	42.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/04/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Corral solids	141.55	40.00	203.30	0.00	95.00 *ton*
Application event totals		141.55	40.00	203.30	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3-8-11	Solid commercial fertilizer	8.12	21.66	29.79	0.00	
Application event totals		8.12	21.66	29.79	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.58	0.00	0.00	32.74	2,706,040.00 *gal*
Application event totals		1.58	0.00	0.00	32.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.35	0.00	0.00	28.07	2,319,463.00 *gal*
Application event totals		1.35	0.00	0.00	28.07	

Angyls 5 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.35	0.00	0.00	28.07	2,319,463.00 *gal*
Application event totals		1.35	0.00	0.00	28.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.35	0.00	0.00	28.07	2,319,463.00 *gal*
Application event totals		1.35	0.00	0.00	28.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.35	0.00	0.00	28.07	2,319,463.00 *gal*
Application event totals		1.35	0.00	0.00	28.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #12	Ground water	0.80	0.00	0.00	154.61	802,067.00 *gal*
Application event totals		0.80	0.00	0.00	154.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #13	Ground water	0.00	0.00	0.00	106.42	802,067.00 *gal*
Application event totals		0.00	0.00	0.00	106.42	

Angyls 7 - 10/28/2022: Winter Forage

Field name: Angyls 7

Crop: Winter Forage

Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Angyls 7 - 10/28/2022: Winter Forage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	27.34	3.60	64.18	300.35	207,644.00 *gal*
Application event totals		27.34	3.60	64.18	300.35	

10/18/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Separator solids	77.41	33.16	68.44	225.28	55.00 *ton*
Application event totals		77.41	33.16	68.44	225.28	

Angyls 7 - 06/20/2023: Corn, silage

Field name: Angyls 7

Crop: Corn, silage

Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	2.03	0.00	0.00	42.10	4,696,913.00 *gal*
Application event totals		2.03	0.00	0.00	42.10	

05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	49.14	3.74	61.38	352.20	209,472.00 *gal*
Application event totals		49.14	3.74	61.38	352.20	

05/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Corral solids	44.15	12.47	63.41	0.00	40.00 *ton*
Application event totals		44.15	12.47	63.41	0.00	

Angyls 7 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3-8-11	Solid commercial fertilizer	8.12	21.66	29.79	0.00	
Application event totals		8.12	21.66	29.79	0.00	

06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.58	0.00	0.00	32.74	3,653,154.00 *gal*
Application event totals		1.58	0.00	0.00	32.74	

07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.35	0.00	0.00	28.07	3,131,275.00 *gal*
Application event totals		1.35	0.00	0.00	28.07	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.35	0.00	0.00	28.07	3,131,275.00 *gal*
Application event totals		1.35	0.00	0.00	28.07	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.35	0.00	0.00	28.07	3,131,275.00 *gal*
Application event totals		1.35	0.00	0.00	28.07	

08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.35	0.00	0.00	28.07	3,131,275.00 *gal*
Application event totals		1.35	0.00	0.00	28.07	

Angyls 7 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #12	Ground water	0.80	0.00	0.00	154.61	1,082,791.00 *gal*
Application event totals		0.80	0.00	0.00	154.61	

09/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #13	Ground water	0.00	0.00	0.00	106.42	1,082,791.00 *gal*
Application event totals		0.00	0.00	0.00	106.42	

Angyls 8 - 10/28/2022: Winter Forage

Field name: Angyls 8

Crop: Winter Forage Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	27.34	3.60	64.18	300.35	599,861.00 *gal*
Application event totals		27.34	3.60	64.18	300.35	

10/18/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Separator solids	78.92	33.81	69.78	229.69	162.00 *ton*
Application event totals		78.92	33.81	69.78	229.69	

Angyls 8 - 06/20/2023: Corn, silage

Field name: Angyls 8

Crop: Corn, silage Plant date: 06/20/2023

Angyls 8 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	2.03	0.00	0.00	42.10	13,568,859.00 *gal*
Application event totals		2.03	0.00	0.00	42.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	49.14	3.74	61.38	352.20	605,141.00 *gal*
Application event totals		49.14	3.74	61.38	352.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Corral solids	96.12	27.16	138.06	0.00	251.60 *ton*
Application event totals		96.12	27.16	138.06	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3-8-11	Solid commercial fertilizer	8.12	21.66	29.79	0.00	
Application event totals		8.12	21.66	29.79	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.58	0.00	0.00	32.74	10,553,557.00 *gal*
Application event totals		1.58	0.00	0.00	32.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.35	0.00	0.00	28.07	9,045,906.00 *gal*
Application event totals		1.35	0.00	0.00	28.07	

Angyls 8 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.35	0.00	0.00	28.07	9,045,906.00 *gal*
Application event totals		1.35	0.00	0.00	28.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.35	0.00	0.00	28.07	9,045,906.00 *gal*
Application event totals		1.35	0.00	0.00	28.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.35	0.00	0.00	28.07	9,045,906.00 *gal*
Application event totals		1.35	0.00	0.00	28.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #12	Ground water	0.80	0.00	0.00	154.61	3,128,062.00 *gal*
Application event totals		0.80	0.00	0.00	154.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #13	Ground water	0.00	0.00	0.00	106.42	3,128,062.00 *gal*
Application event totals		0.00	0.00	0.00	106.42	

Home A - 11/04/2022: Winter Forage

Field name: Home A

Crop: Winter Forage

Plant date: 11/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Home A - 11/04/2022: Winter Forage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Separator solids	327.89	589.33	102.56	248.33	65.00 *ton*
Application event totals		327.89	589.33	102.56	248.33	

Home A - 06/11/2023: Corn, silage

Field name: Home A

Crop: Corn, silage

Plant date: 06/11/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	29.56	4.32	14.96	58.27	23,274.00 *gal*
Application event totals		29.56	4.32	14.96	58.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Corral solids	45.05	10.57	36.35	0.00	29.00 *ton*
Application event totals		45.05	10.57	36.35	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	72.83	10.63	36.84	143.53	57,330.00 *gal*
Canal	Surface water	0.00	0.00	0.00	35.72	1,203,934.00 *gal*
Application event totals		72.83	10.63	36.84	179.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	29.42	991,475.00 *gal*
Application event totals		0.00	0.00	0.00	29.42	

Home A - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	14.36	0.25	5.21	147.09	47,213.00 *gal*
Canal	Surface water	0.00	0.00	0.00	29.42	991,475.00 *gal*
Application event totals		14.36	0.25	5.21	176.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	14.36	0.25	5.21	147.09	47,213.00 *gal*
Canal	Surface water	0.00	0.00	0.00	29.42	991,475.00 *gal*
Application event totals		14.36	0.25	5.21	176.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	29.42	991,475.00 *gal*
Application event totals		0.00	0.00	0.00	29.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	35.72	1,203,934.00 *gal*
Application event totals		0.00	0.00	0.00	35.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	35.72	1,203,934.00 *gal*
Application event totals		0.00	0.00	0.00	35.72	

Home B - 11/04/2022: Winter Forage

Field name: Home B

Crop: Winter Forage

Plant date: 11/04/2022

Home B - 11/04/2022: Winter Forage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Separator solids	242.29	435.49	75.78	183.50	202.80 *ton*
Application event totals		242.29	435.49	75.78	183.50	

Home B - 06/14/2023: Corn, silage

Field name: Home B

Crop: Corn, silage

Plant date: 06/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	14.78	2.16	7.48	29.13	49,135.00 *gal*
Application event totals		14.78	2.16	7.48	29.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Separator solids	18.95	3.21	9.21	0.00	30.00 *ton*
Application event totals		18.95	3.21	9.21	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	72.83	10.63	36.84	143.53	242,061.00 *gal*
Canal	Surface water	0.00	0.00	0.00	35.72	5,083,279.00 *gal*
Application event totals		72.83	10.63	36.84	179.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	29.42	4,186,230.00 *gal*
Application event totals		0.00	0.00	0.00	29.42	

Home B - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	14.36	0.25	5.21	147.09	199,344.00 *gal*
Canal	Surface water	0.00	0.00	0.00	29.42	4,186,230.00 *gal*
Application event totals		14.36	0.25	5.21	176.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	14.36	0.25	5.21	147.09	199,344.00 *gal*
Canal	Surface water	0.00	0.00	0.00	29.42	4,186,230.00 *gal*
Application event totals		14.36	0.25	5.21	176.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	29.42	4,186,230.00 *gal*
Application event totals		0.00	0.00	0.00	29.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	35.72	5,083,279.00 *gal*
Application event totals		0.00	0.00	0.00	35.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	35.72	5,083,279.00 *gal*
Application event totals		0.00	0.00	0.00	35.72	

Home C - 11/04/2022: Winter Forage

Field name: Home C

Crop: Winter Forage

Plant date: 11/04/2022

Home C - 11/04/2022: Winter Forage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Corral solids	278.30	51.36	77.58	685.48	185.20 *ton*
Application event totals		278.30	51.36	77.58	685.48	

Home C - 06/14/2023: Corn, silage

Field name: Home C

Crop: Corn, silage

Plant date: 06/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Separator solids	62.03	9.24	21.24	0.00	75.00 *ton*
Application event totals		62.03	9.24	21.24	0.00	

05/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	14.78	2.16	7.48	29.13	47,842.00 *gal*
Application event totals		14.78	2.16	7.48	29.13	

05/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	72.83	10.63	36.84	143.53	235,691.00 *gal*
Canal	Surface water	0.00	0.00	0.00	35.72	4,949,508.00 *gal*
Application event totals		72.83	10.63	36.84	179.25	

07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	29.42	4,076,066.00 *gal*
Application event totals		0.00	0.00	0.00	29.42	

Home C - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	14.36	0.25	5.21	147.09	194,098.00 *gal*
Canal	Surface water	0.00	0.00	0.00	29.42	4,076,066.00 *gal*
Application event totals		14.36	0.25	5.21	176.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	14.36	0.25	5.21	147.09	194,098.00 *gal*
Canal	Surface water	0.00	0.00	0.00	29.42	4,076,066.00 *gal*
Application event totals		14.36	0.25	5.21	176.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	29.42	4,076,066.00 *gal*
Application event totals		0.00	0.00	0.00	29.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	35.72	4,949,508.00 *gal*
Application event totals		0.00	0.00	0.00	35.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	35.72	4,949,508.00 *gal*
Application event totals		0.00	0.00	0.00	35.72	

Home D - 11/04/2022: Winter Forage

Field name: Home D

Crop: Winter Forage

Plant date: 11/04/2022

Home D - 11/04/2022: Winter Forage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Corral solids	200.94	37.08	56.02	494.92	155.40 *ton*
Application event totals		200.94	37.08	56.02	494.92	

Home D - 06/14/2023: Corn, silage

Field name: Home D

Crop: Corn, silage

Plant date: 06/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Corral solids	81.28	19.07	65.58	0.00	250.00 *ton*
Application event totals		81.28	19.07	65.58	0.00	

05/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	14.78	2.16	7.48	29.13	55,600.00 *gal*
Application event totals		14.78	2.16	7.48	29.13	

05/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	72.83	10.63	36.84	143.53	273,911.00 *gal*
Canal	Surface water	0.00	0.00	0.00	35.72	5,752,131.00 *gal*
Application event totals		72.83	10.63	36.84	179.25	

07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	29.42	4,737,049.00 *gal*
Application event totals		0.00	0.00	0.00	29.42	

Home D - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	59.97	8.76	30.34	118.20	225,574.00 *gal*
Canal	Surface water	0.00	0.00	0.00	29.42	4,737,049.00 *gal*
Application event totals		59.97	8.76	30.34	147.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	14.36	0.25	5.21	147.09	225,574.00 *gal*
Canal	Surface water	0.00	0.00	0.00	29.42	4,737,049.00 *gal*
Application event totals		14.36	0.25	5.21	176.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	29.42	4,737,049.00 *gal*
Application event totals		0.00	0.00	0.00	29.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	35.72	5,752,131.00 *gal*
Application event totals		0.00	0.00	0.00	35.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	35.72	5,752,131.00 *gal*
Application event totals		0.00	0.00	0.00	35.72	

Home E/F - 11/04/2022: Winter Forage

Field name: Home E/F

Crop: Winter Forage

Plant date: 11/04/2022

Home E/F - 11/04/2022: Winter Forage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Corral solids	320.98	59.23	89.48	790.60	213.60 *ton*
Application event totals		320.98	59.23	89.48	790.60	

Home E/F - 06/14/2023: Corn, silage

Field name: Home E/F

Crop: Corn, silage

Plant date: 06/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Separator solids	70.30	10.48	24.08	0.00	85.00 *ton*
Application event totals		70.30	10.48	24.08	0.00	

05/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	14.78	2.16	7.48	29.13	47,842.00 *gal*
Application event totals		14.78	2.16	7.48	29.13	

05/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	72.83	10.63	36.84	143.53	235,691.00 *gal*
Canal	Surface water	0.00	0.00	0.00	35.72	4,949,508.00 *gal*
Application event totals		72.83	10.63	36.84	179.25	

07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	29.42	4,076,066.00 *gal*
Application event totals		0.00	0.00	0.00	29.42	

Home E/F - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	14.36	0.25	5.21	147.09	194,098.00 *gal*
Canal	Surface water	0.00	0.00	0.00	29.42	4,076,066.00 *gal*
Application event totals		14.36	0.25	5.21	176.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	14.36	0.25	5.21	147.09	194,098.00 *gal*
Canal	Surface water	0.00	0.00	0.00	29.42	4,076,066.00 *gal*
Application event totals		14.36	0.25	5.21	176.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	29.42	4,076,066.00 *gal*
Application event totals		0.00	0.00	0.00	29.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	35.72	4,949,508.00 *gal*
Application event totals		0.00	0.00	0.00	35.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	35.72	4,949,508.00 *gal*
Application event totals		0.00	0.00	0.00	35.72	

Home G - 11/04/2022: Winter Forage

Field name: Home G

Crop: Winter Forage

Plant date: 11/04/2022

Home G - 11/04/2022: Winter Forage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Corral solids	239.08	44.12	66.65	588.88	81.70 *ton*
Application event totals		239.08	44.12	66.65	588.88	

Home G - 06/14/2023: Corn, silage

Field name: Home G

Crop: Corn, silage

Plant date: 06/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Corral solids	39.73	9.32	32.06	0.00	54.00 *ton*
Application event totals		39.73	9.32	32.06	0.00	

05/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	88.70	12.95	44.87	174.80	147,406.00 *gal*
Application event totals		88.70	12.95	44.87	174.80	

05/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	109.24	15.95	55.26	215.29	181,546.00 *gal*
Canal	Surface water	0.00	0.00	0.00	35.72	2,541,639.00 *gal*
Application event totals		109.24	15.95	55.26	251.01	

07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	29.42	2,093,115.00 *gal*
Application event totals		0.00	0.00	0.00	29.42	

Home G - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	21.54	0.38	7.81	220.64	149,508.00 *gal*
Canal	Surface water	0.00	0.00	0.00	29.42	2,093,115.00 *gal*
Application event totals		21.54	0.38	7.81	250.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	21.54	0.38	7.81	220.64	149,508.00 *gal*
Canal	Surface water	0.00	0.00	0.00	29.42	2,093,115.00 *gal*
Application event totals		21.54	0.38	7.81	250.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	29.42	2,093,115.00 *gal*
Application event totals		0.00	0.00	0.00	29.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	35.72	2,541,639.00 *gal*
Application event totals		0.00	0.00	0.00	35.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	35.72	2,541,639.00 *gal*
Application event totals		0.00	0.00	0.00	35.72	

Hunts 1 - 11/04/2022: Winter Forage

Field name: Hunts 1

Crop: Winter Forage

Plant date: 11/04/2022

Hunts 1 - 11/04/2022: Winter Forage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Separator solids	137.34	58.84	121.44	399.71	227.70 *ton*
Application event totals		137.34	58.84	121.44	399.71	

Hunts 1 - 06/08/2023: Corn, silage

Field name: Hunts 1

Crop: Corn, silage

Plant date: 06/08/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/06/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Corral solids	80.41	22.72	115.49	0.00	170.00 *ton*
Application event totals		80.41	22.72	115.49	0.00	

05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #17	Ground water	0.89	0.00	0.00	227.64	8,592,885.00 *gal*
Application event totals		0.89	0.00	0.00	227.64	

06/08/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3-8-11	Solid commercial fertilizer	8.12	21.66	29.79	0.00	
Application event totals		8.12	21.66	29.79	0.00	

06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #17	Ground water	0.57	0.00	0.00	146.34	5,523,998.00 *gal*
Application event totals		0.57	0.00	0.00	146.34	

Hunts 1 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #17	Ground water	0.57	0.00	0.00	146.34	5,523,998.00 *gal*
Application event totals		0.57	0.00	0.00	146.34	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #17	Ground water	0.57	0.00	0.00	146.34	5,523,998.00 *gal*
Application event totals		0.57	0.00	0.00	146.34	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #17	Ground water	0.57	0.00	0.00	146.34	5,523,998.00 *gal*
Application event totals		0.57	0.00	0.00	146.34	

08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #17	Ground water	0.70	0.00	0.00	178.86	6,751,553.00 *gal*
Application event totals		0.70	0.00	0.00	178.86	

08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #17	Ground water	0.70	0.00	0.00	178.86	6,751,553.00 *gal*
Application event totals		0.70	0.00	0.00	178.86	

09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #17	Ground water	0.70	0.00	0.00	178.86	6,751,553.00 *gal*
Application event totals		0.70	0.00	0.00	178.86	

Hunts 2 - 11/04/2022: Winter Forage

Hunts 2 - 11/04/2022: Winter Forage

Field name: Hunts 2

Crop: Winter Forage

Plant date: 11/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Separator solids	111.76	47.88	98.82	325.27	50.00 *ton*
Application event totals		111.76	47.88	98.82	325.27	

Hunts 2 - 06/08/2023: Corn, silage

Field name: Hunts 2

Crop: Corn, silage

Plant date: 06/08/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/06/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Separator solids	49.41	8.36	24.01	0.00	35.00 *ton*
Application event totals		49.41	8.36	24.01	0.00	

05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #16	Ground water	5.69	0.00	0.00	398.38	2,318,715.00 *gal*
Application event totals		5.69	0.00	0.00	398.38	

06/08/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3-8-11	Solid commercial fertilizer	8.12	21.66	29.79	0.00	
Application event totals		8.12	21.66	29.79	0.00	

Hunts 2 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #16	Ground water	3.66	0.00	0.00	256.10	1,490,603.00 *gal*
Application event totals		3.66	0.00	0.00	256.10	

07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #16	Ground water	3.66	0.00	0.00	256.10	1,490,603.00 *gal*
Application event totals		3.66	0.00	0.00	256.10	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #16	Ground water	3.66	0.00	0.00	256.10	1,490,603.00 *gal*
Application event totals		3.66	0.00	0.00	256.10	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #16	Ground water	3.66	0.00	0.00	256.10	1,490,603.00 *gal*
Application event totals		3.66	0.00	0.00	256.10	

08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #16	Ground water	4.47	0.00	0.00	313.01	1,821,848.00 *gal*
Application event totals		4.47	0.00	0.00	313.01	

08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #16	Ground water	4.47	0.00	0.00	313.01	1,821,848.00 *gal*
Application event totals		4.47	0.00	0.00	313.01	

Hunts 2 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #16	Ground water	4.47	0.00	0.00	313.01	1,821,848.00 *gal*
Application event totals		4.47	0.00	0.00	313.01	

Manteca - 11/01/2022: Winter Forage

Field name: Manteca

Crop: Winter Forage

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	27.34	3.60	64.18	300.35	615,242.00 *gal*
Application event totals		27.34	3.60	64.18	300.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Separator solids	188.67	80.83	166.82	549.09	397.20 *ton*
Application event totals		188.67	80.83	166.82	549.09	

Manteca - 06/06/2023: Corn, silage

Field name: Manteca

Crop: Corn, silage

Plant date: 06/06/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Manteca - 06/06/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	8.19	0.62	10.23	58.70	103,442.00 *gal*
Application event totals		8.19	0.62	10.23	58.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Separator solids	58.50	9.90	28.42	0.00	195.00 *ton*
Application event totals		58.50	9.90	28.42	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	43.43	3.31	54.25	311.29	548,571.00 *gal*
River	Surface water	1.68	0.00	0.00	34.85	11,520,000.00 *gal*
Application event totals		45.11	3.31	54.25	346.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.68	0.00	0.00	34.85	11,520,000.00 *gal*
Application event totals		1.68	0.00	0.00	34.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.84	0.00	0.00	17.42	5,760,000.00 *gal*
Application event totals		0.84	0.00	0.00	17.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.26	0.00	0.00	26.14	8,640,000.00 *gal*
Application event totals		1.26	0.00	0.00	26.14	

Manteca - 06/06/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0 WECO-BOOST	Liquid commercial fertilizer	0.06	0.00	0.00	0.00	
Wastewater	Process wastewater	27.82	0.51	13.73	349.06	685,714.00 *gal*
Application event totals		27.88	0.51	13.73	349.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	16.69	0.31	8.24	209.44	411,429.00 *gal*
River	Surface water	1.26	0.00	0.00	26.14	8,640,000.00 *gal*
Application event totals		17.96	0.31	8.24	235.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.26	0.00	0.00	26.14	8,640,000.00 *gal*
Application event totals		1.26	0.00	0.00	26.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.47	0.00	0.00	30.49	10,080,000.00 *gal*
Application event totals		1.47	0.00	0.00	30.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.47	0.00	0.00	30.49	10,080,000.00 *gal*
Application event totals		1.47	0.00	0.00	30.49	

Maries 1 - 11/06/2022: Winter Forage

Field name: Maries 1

Crop: Winter Forage

Plant date: 11/06/2022

Maries 1 - 11/06/2022: Winter Forage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Separator solids	201.91	86.50	178.53	587.61	159.40 *ton*
Application event totals		201.91	86.50	178.53	587.61	

Maries 1 - 06/17/2023: Corn, silage

Field name: Maries 1

Crop: Corn, silage

Plant date: 06/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Corral solids	136.38	38.54	195.88	0.00	137.30 *ton*
Wastewater	Process wastewater	49.14	3.74	61.38	352.20	232,746.00 *gal*
Application event totals		185.52	42.28	257.26	352.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	35.94	2.74	44.89	257.62	170,246.00 *gal*
River	Surface water	0.93	0.00	0.00	19.23	2,383,448.00 *gal*
Application event totals		36.87	2.74	44.89	276.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.04	0.00	0.00	21.63	2,681,379.00 *gal*
Application event totals		1.04	0.00	0.00	21.63	

Maries 1 - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	40.44	3.08	50.51	289.82	191,527.00 *gal*
River	Surface water	1.04	0.00	0.00	21.63	2,681,379.00 *gal*
Application event totals		41.48	3.08	50.51	311.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	20.72	0.38	10.23	259.99	191,527.00 *gal*
River	Surface water	1.04	0.00	0.00	21.63	2,681,379.00 *gal*
Application event totals		21.77	0.38	10.23	281.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.04	0.00	0.00	21.63	2,681,379.00 *gal*
Application event totals		1.04	0.00	0.00	21.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.04	0.00	0.00	21.63	2,681,379.00 *gal*
Application event totals		1.04	0.00	0.00	21.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.04	0.00	0.00	21.63	2,681,379.00 *gal*
Application event totals		1.04	0.00	0.00	21.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.39	0.00	0.00	28.84	3,575,172.00 *gal*
Application event totals		1.39	0.00	0.00	28.84	

Maries 1 - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.39	0.00	0.00	28.84	3,575,172.00 *gal*
Application event totals		1.39	0.00	0.00	28.84	

Maries 2 - 11/06/2022: Winter Forage

Field name: Maries 2

Crop: Winter Forage

Plant date: 11/06/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Separator solids	190.00	81.40	168.00	552.96	400.00 *ton*
Application event totals		190.00	81.40	168.00	552.96	

Maries 2 - 06/17/2023: Corn, silage

Field name: Maries 2

Crop: Corn, silage

Plant date: 06/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Corral solids	96.14	27.17	138.08	0.00	258.10 *ton*
Wastewater	Process wastewater	49.14	3.74	61.38	352.20	620,657.00 *gal*
Application event totals		145.28	30.91	199.46	352.20	

Maries 2 - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	35.94	2.74	44.89	257.62	453,990.00 *gal*
River	Surface water	0.93	0.00	0.00	19.23	6,355,862.00 *gal*
Application event totals		36.87	2.74	44.89	276.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.04	0.00	0.00	21.63	7,150,345.00 *gal*
Application event totals		1.04	0.00	0.00	21.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	40.44	3.08	50.51	289.82	510,739.00 *gal*
River	Surface water	1.04	0.00	0.00	21.63	7,150,345.00 *gal*
Application event totals		41.48	3.08	50.51	311.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	20.72	0.38	10.23	259.99	510,739.00 *gal*
River	Surface water	1.04	0.00	0.00	21.63	7,150,345.00 *gal*
Application event totals		21.77	0.38	10.23	281.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.04	0.00	0.00	21.63	7,150,345.00 *gal*
Application event totals		1.04	0.00	0.00	21.63	

Maries 2 - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.04	0.00	0.00	21.63	7,150,345.00 *gal*
Application event totals		1.04	0.00	0.00	21.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.04	0.00	0.00	21.63	7,150,345.00 *gal*
Application event totals		1.04	0.00	0.00	21.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.39	0.00	0.00	28.84	9,533,793.00 *gal*
Application event totals		1.39	0.00	0.00	28.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.39	0.00	0.00	28.84	9,533,793.00 *gal*
Application event totals		1.39	0.00	0.00	28.84	

Maries 3 - 11/06/2022: Winter Forage

Field name: Maries 3

Crop: Winter Forage

Plant date: 11/06/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Separator solids	149.72	64.14	132.38	435.73	137.90 *ton*
Application event totals		149.72	64.14	132.38	435.73	

Maries 3 - 06/17/2023: Corn, silage

Field name: Maries 3

Crop: Corn, silage

Plant date: 06/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Corral solids	96.13	27.16	138.06	0.00	112.90 *ton*
Wastewater	Process wastewater	49.14	3.74	61.38	352.20	271,537.00 *gal*
Application event totals		145.27	30.90	199.44	352.20	

05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	35.94	2.74	44.89	257.62	198,621.00 *gal*
River	Surface water	0.93	0.00	0.00	19.23	2,780,690.00 *gal*
Application event totals		36.87	2.74	44.89	276.85	

06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.04	0.00	0.00	21.63	3,128,276.00 *gal*
Application event totals		1.04	0.00	0.00	21.63	

06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	40.44	3.08	50.51	289.82	223,448.00 *gal*
River	Surface water	1.04	0.00	0.00	21.63	3,128,276.00 *gal*
Application event totals		41.48	3.08	50.51	311.45	

07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	21.00	0.38	10.37	263.48	226,448.00 *gal*
River	Surface water	1.04	0.00	0.00	21.63	3,128,276.00 *gal*
Application event totals		22.05	0.38	10.37	285.11	

Maries 3 - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.04	0.00	0.00	21.63	3,128,276.00 *gal*
Application event totals		1.04	0.00	0.00	21.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.04	0.00	0.00	21.63	3,128,276.00 *gal*
Application event totals		1.04	0.00	0.00	21.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.04	0.00	0.00	21.63	3,128,276.00 *gal*
Application event totals		1.04	0.00	0.00	21.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.39	0.00	0.00	28.84	4,171,034.00 *gal*
Application event totals		1.39	0.00	0.00	28.84	

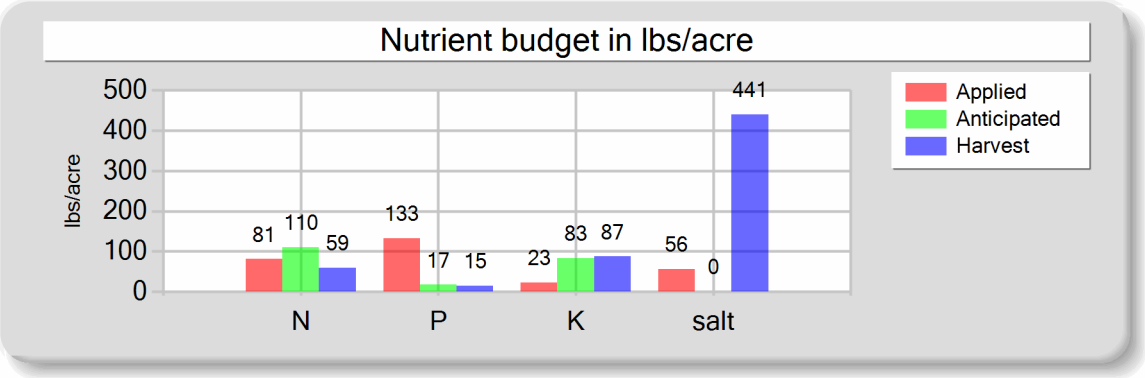
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.39	0.00	0.00	28.84	4,171,034.00 *gal*
Application event totals		1.39	0.00	0.00	28.84	

B. NUTRIENT BUDGET

Airport 1 - 11/05/2022: Winter Forage

Field name: Airport 1 Crop: Winter Forage Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	73.85	132.73	23.10	55.93
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	80.85	132.73	23.10	55.93
Anticipated crop nutrient removal	110.00	17.00	83.00	0.00
Actual crop nutrient removal	58.93	14.54	87.02	440.61
Nutrient balance	21.92	118.19	-63.92	-384.68
Applied to removed ratio	1.37	9.13	0.27	0.13

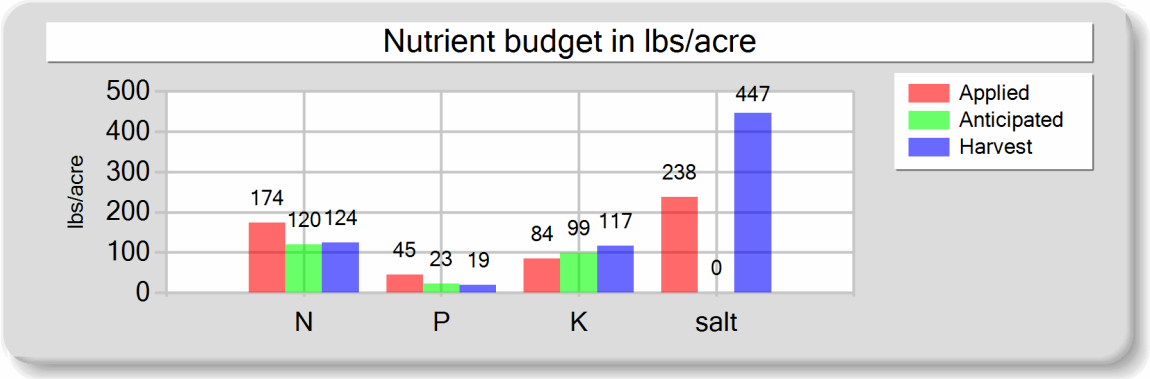
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Airport 1 - 06/13/2023: Corn, silage

Field name: Airport 1 Crop: Corn, silage Plant date: 06/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.12	21.66	29.79	0.00
Dry manure	158.81	23.67	54.39	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	238.17
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	173.93	45.33	84.18	238.17
Anticipated crop nutrient removal	120.00	22.50	99.00	0.00
Actual crop nutrient removal	124.39	18.95	116.61	447.01
Nutrient balance	49.54	26.38	-32.43	-208.85
Applied to removed ratio	1.40	2.39	0.72	0.53

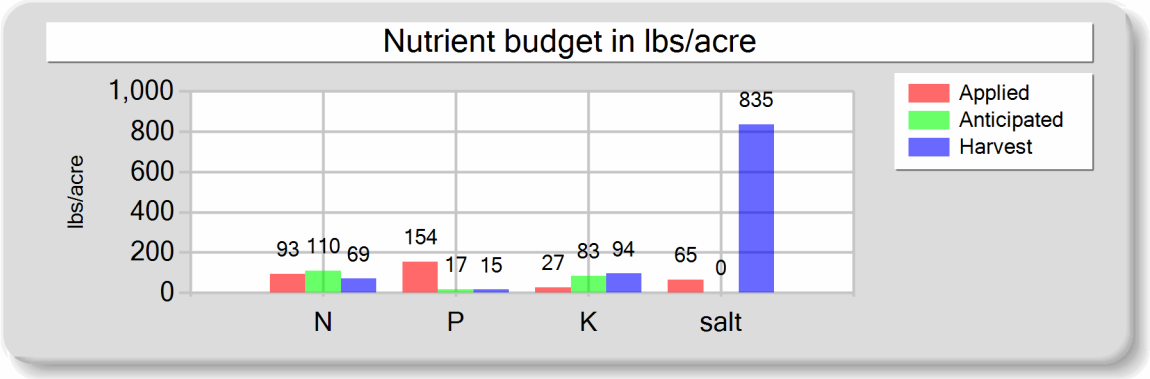
Fresh water applied	
70,457,806.00	*gallons*
2,594.72	*acre-inches*
32.84	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Airport 2 - 11/05/2022: Winter Forage

Field name: Airport 2 Crop: Winter Forage Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	85.89	154.38	26.86	65.05
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	92.89	154.38	26.86	65.05
Anticipated crop nutrient removal	110.00	17.00	83.00	0.00
Actual crop nutrient removal	69.29	14.54	94.42	835.50
Nutrient balance	23.61	139.84	-67.55	-770.44
Applied to removed ratio	1.34	10.62	0.28	0.08

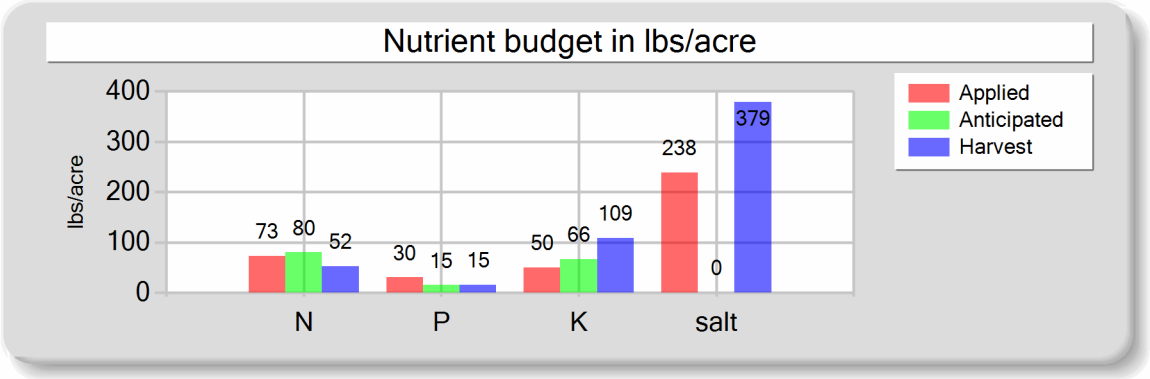
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Airport 2 - 06/13/2023: Corn, silage

Field name: Airport 2 Crop: Corn, silage Plant date: 06/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.12	21.66	29.79	0.00
Dry manure	57.89	8.63	19.83	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	238.17
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	73.01	30.29	49.62	238.17
Anticipated crop nutrient removal	80.00	15.00	66.00	0.00
Actual crop nutrient removal	52.30	15.45	108.96	378.78
Nutrient balance	20.71	14.84	-59.34	-140.62
Applied to removed ratio	1.40	1.96	0.46	0.63

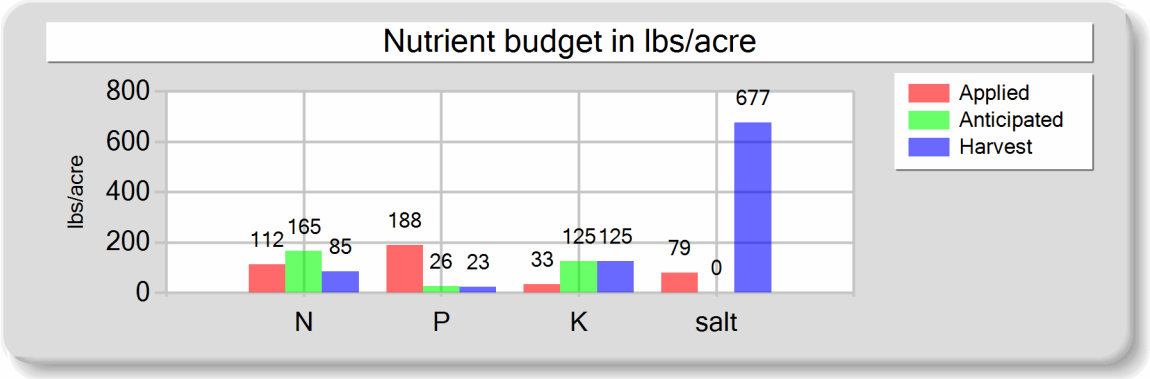
Fresh water applied	
32,999,224.00	*gallons*
1,215.25	*acre-inches*
32.84	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Airport 3 - 11/05/2022: Winter Forage

Field name: Airport 3 Crop: Winter Forage Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	104.77	188.31	32.77	79.35
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	111.77	188.31	32.77	79.35
Anticipated crop nutrient removal	165.00	25.50	124.50	0.00
Actual crop nutrient removal	85.43	22.92	125.18	676.65
Nutrient balance	26.34	165.39	-92.41	-597.30
Applied to removed ratio	1.31	8.22	0.26	0.12

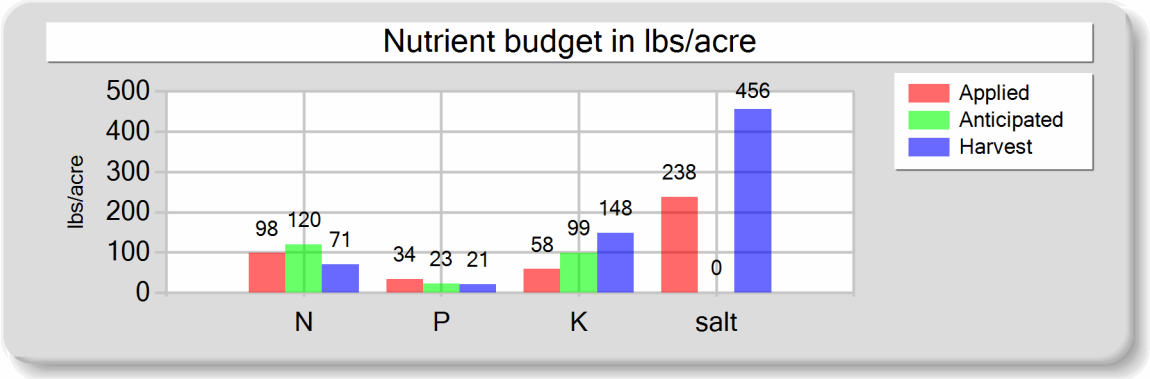
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Airport 3 - 06/13/2023: Corn, silage

Field name: Airport 3 Crop: Corn, silage Plant date: 06/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.12	21.66	29.79	0.00
Dry manure	83.17	12.39	28.48	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	238.17
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	98.29	34.05	58.27	238.17
Anticipated crop nutrient removal	120.00	22.50	99.00	0.00
Actual crop nutrient removal	70.70	21.01	148.40	456.05
Nutrient balance	27.59	13.04	-90.13	-217.89
Applied to removed ratio	1.39	1.62	0.39	0.52

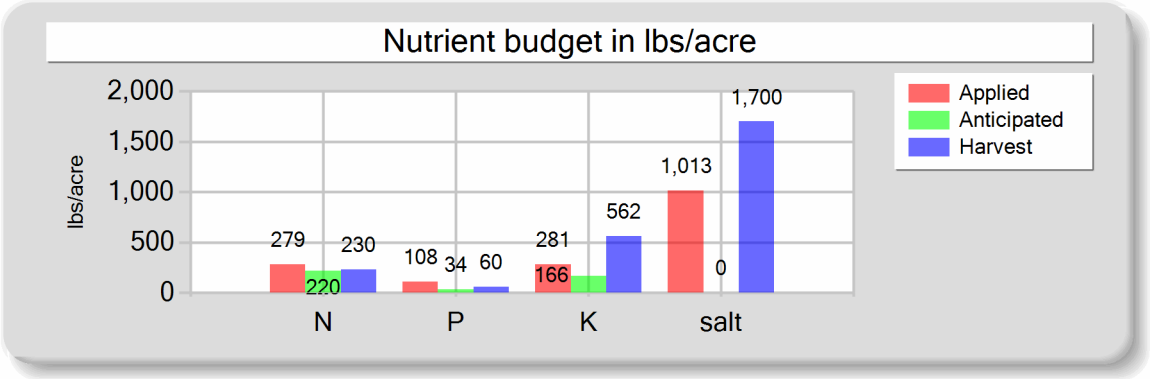
Fresh water applied	
34,782,966.00	*gallons*
1,280.94	*acre-inches*
32.84	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Anderson - 11/06/2022: Winter Forage

Field name: Anderson Crop: Winter Forage Plant date: 11/06/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	244.79	104.87	216.45	712.41
Process wastewater	27.34	3.60	64.18	300.35
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	279.13	108.47	280.62	1,012.76
Anticipated crop nutrient removal	220.00	34.00	166.00	0.00
Actual crop nutrient removal	230.36	60.37	562.39	1,699.89
Nutrient balance	48.77	48.10	-281.77	-687.13
Applied to removed ratio	1.21	1.80	0.50	0.60

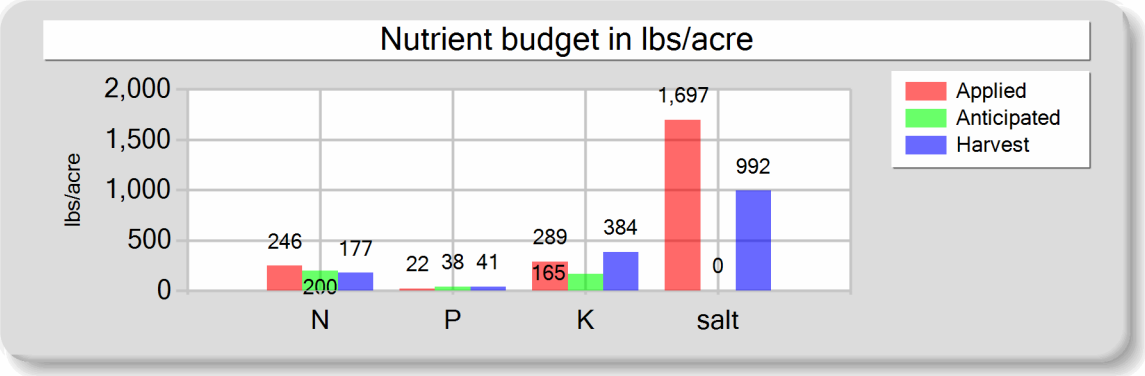
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
422,979.00	*gallons*
15.58	*acre-inches*
0.28	*inches/acre*

Total harvests for the crop	
1	*harvests*

Anderson - 06/07/2023: Corn, silage

Field name: Anderson Crop: Corn, silage Plant date: 06/07/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.06	0.00	0.00	0.00
Dry manure	24.38	6.89	35.02	0.00
Process wastewater	203.13	15.47	253.71	1,455.87
Fresh water	11.62	0.00	0.00	240.77
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	246.19	22.36	288.73	1,696.64
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	176.69	41.21	383.86	991.63
Nutrient balance	69.51	-18.85	-95.13	705.02
Applied to removed ratio	1.39	0.54	0.75	1.71

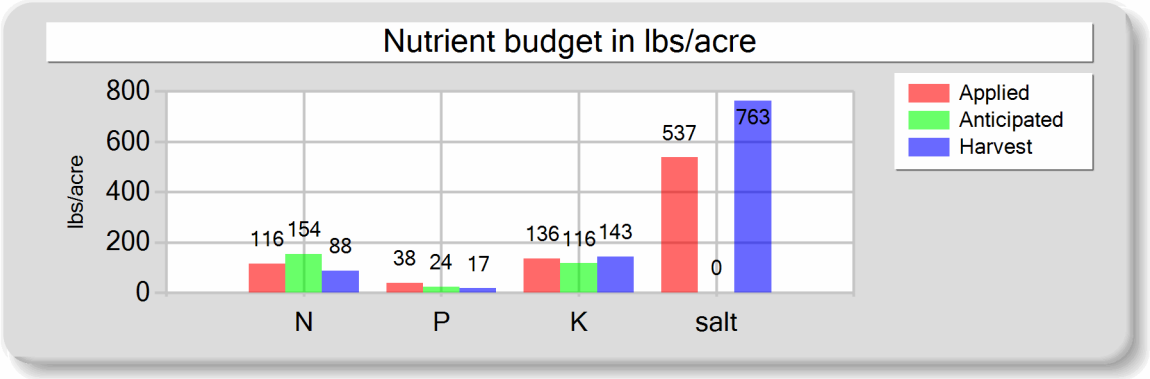
Fresh water applied	
54,720,000.00	*gallons*
2,015.15	*acre-inches*
36.64	*inches/acre*

Process wastewater applied	
1,763,845.00	*gallons*
64.96	*acre-inches*
1.18	*inches/acre*

Total harvests for the crop	
1	*harvests*

Angyls 1-A - 10/28/2022: Winter Forage

Field name: Angyls 1-A Crop: Winter Forage Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	81.43	34.89	72.00	236.98
Process wastewater	27.34	3.60	64.18	300.35
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	115.77	38.49	136.18	537.33
Anticipated crop nutrient removal	154.00	23.80	116.20	0.00
Actual crop nutrient removal	87.61	16.86	142.70	763.28
Nutrient balance	28.16	21.63	-6.52	-225.95
Applied to removed ratio	1.32	2.28	0.95	0.70

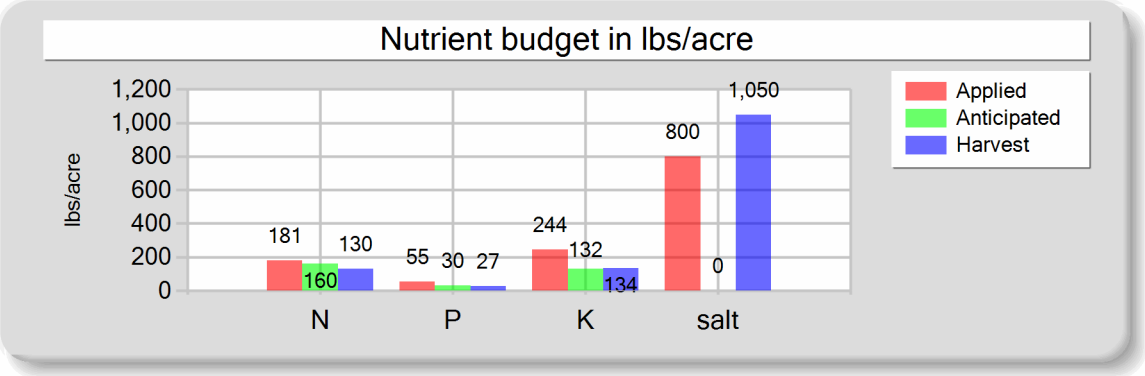
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
269,169.00 *gallons*
9.91 *acre-inches*
0.28 *inches/acre*

Total harvests for the crop
1 *harvests*

Angyls 1-A - 06/20/2023: Corn, silage

Field name: Angyls 1-A Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.12	21.66	29.79	0.00
Dry manure	106.43	30.07	152.86	0.00
Process wastewater	49.14	3.74	61.38	352.20
Fresh water	9.84	0.00	0.00	448.14
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	180.52	55.47	244.02	800.34
Anticipated crop nutrient removal	160.00	30.00	132.00	0.00
Actual crop nutrient removal	130.10	26.59	133.52	1,049.92
Nutrient balance	50.42	28.88	110.50	-249.58
Applied to removed ratio	1.39	2.09	1.83	0.76

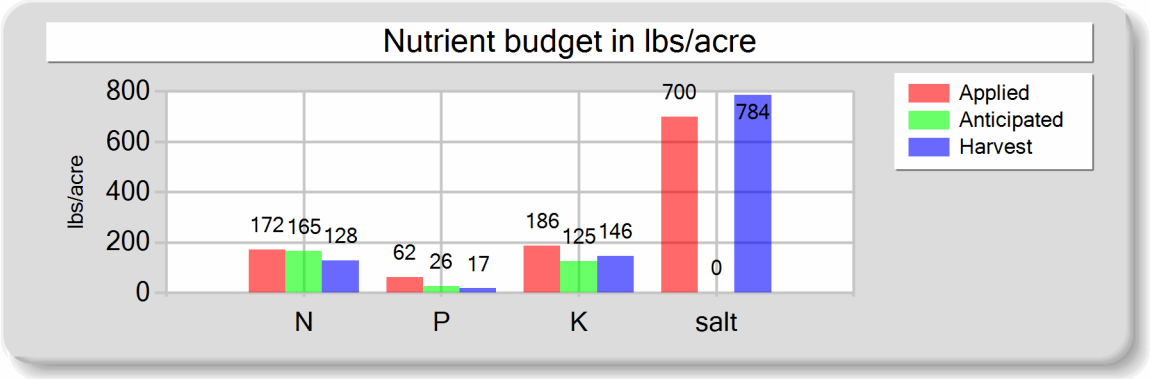
Fresh water applied	
29,867,635.00	*gallons*
1,099.92	*acre-inches*
31.43	*inches/acre*

Process wastewater applied	
271,537.00	*gallons*
10.00	*acre-inches*
0.29	*inches/acre*

Total harvests for the crop	
1	*harvests*

Angyls 2 - 10/28/2022: Winter Forage

Field name: Angyls 2 Crop: Winter Forage Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	137.30	58.82	121.40	399.59
Process wastewater	27.34	3.60	64.18	300.35
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	171.64	62.42	185.58	699.94
Anticipated crop nutrient removal	165.00	25.50	124.50	0.00
Actual crop nutrient removal	127.99	16.95	146.02	784.18
Nutrient balance	43.65	45.48	39.56	-84.24
Applied to removed ratio	1.34	3.68	1.27	0.89

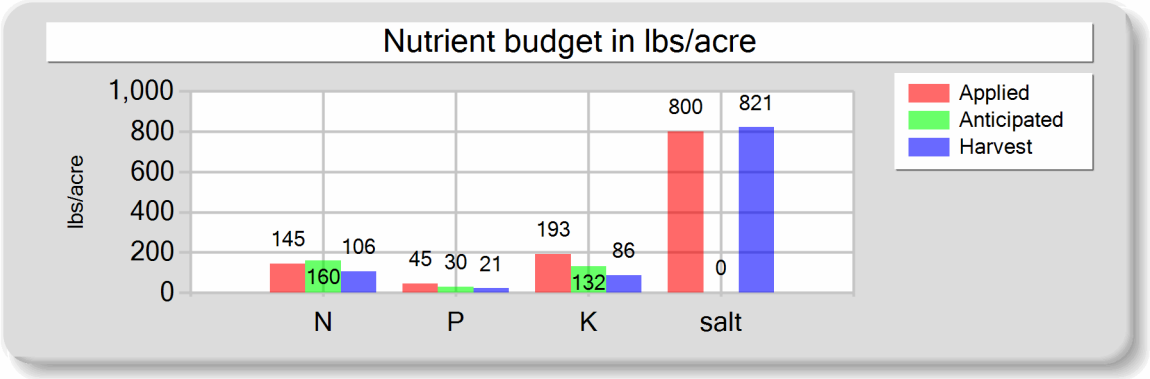
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
292,240.00 *gallons*
10.76 *acre-inches*
0.28 *inches/acre*

Total harvests for the crop
1 *harvests*

Angyls 2 - 06/20/2023: Corn, silage

Field name: Angyls 2 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.12	21.66	29.79	0.00
Dry manure	70.58	19.94	101.37	0.00
Process wastewater	49.14	3.74	61.38	352.20
Fresh water	9.84	0.00	0.00	448.14
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	144.67	45.34	192.53	800.34
Anticipated crop nutrient removal	160.00	30.00	132.00	0.00
Actual crop nutrient removal	106.45	20.99	85.97	821.20
Nutrient balance	38.22	24.36	106.56	-20.86
Applied to removed ratio	1.36	2.16	2.24	0.97

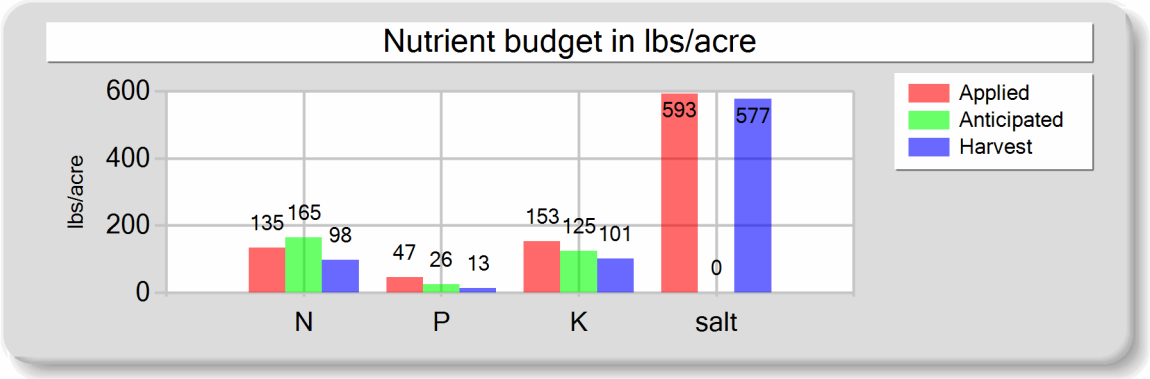
Fresh water applied	
32,427,723.00	*gallons*
1,194.20	*acre-inches*
31.43	*inches/acre*

Process wastewater applied	
294,812.00	*gallons*
10.86	*acre-inches*
0.29	*inches/acre*

Total harvests for the crop	
1	*harvests*

Angyls 3 - 10/28/2022: Winter Forage

Field name: Angyls 3 Crop: Winter Forage Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	100.54	43.07	88.90	292.61
Process wastewater	27.34	3.60	64.18	300.35
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	134.88	46.67	153.08	592.96
Anticipated crop nutrient removal	165.00	25.50	124.50	0.00
Actual crop nutrient removal	97.60	13.11	101.16	577.06
Nutrient balance	37.28	33.57	51.91	15.90
Applied to removed ratio	1.38	3.56	1.51	1.03

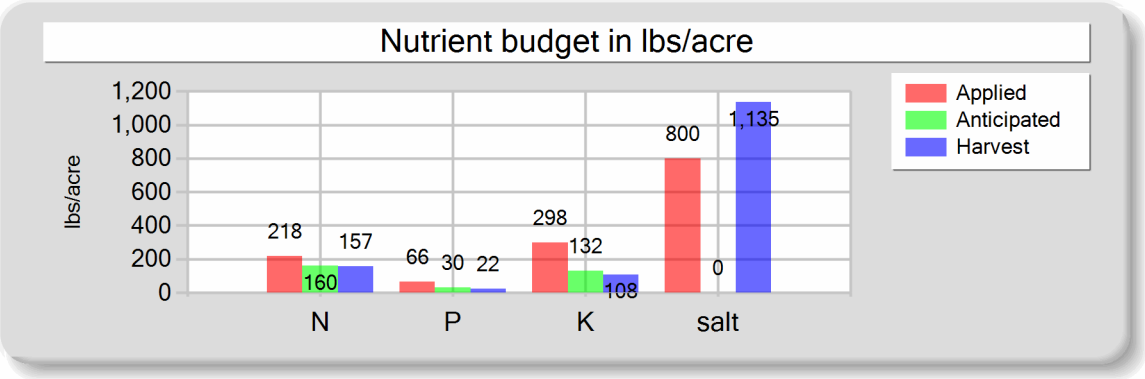
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
369,145.00 *gallons*
13.59 *acre-inches*
0.28 *inches/acre*

Total harvests for the crop
1 *harvests*

Angyls 3 - 06/20/2023: Corn, silage

Field name: Angyls 3 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.12	21.66	29.79	0.00
Dry manure	144.03	40.70	206.87	0.00
Process wastewater	49.14	3.74	61.38	352.20
Fresh water	9.84	0.00	0.00	448.14
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	218.13	66.10	298.03	800.34
Anticipated crop nutrient removal	160.00	30.00	132.00	0.00
Actual crop nutrient removal	157.11	22.26	108.34	1,134.68
Nutrient balance	61.02	43.84	189.69	-334.33
Applied to removed ratio	1.39	2.97	2.75	0.71

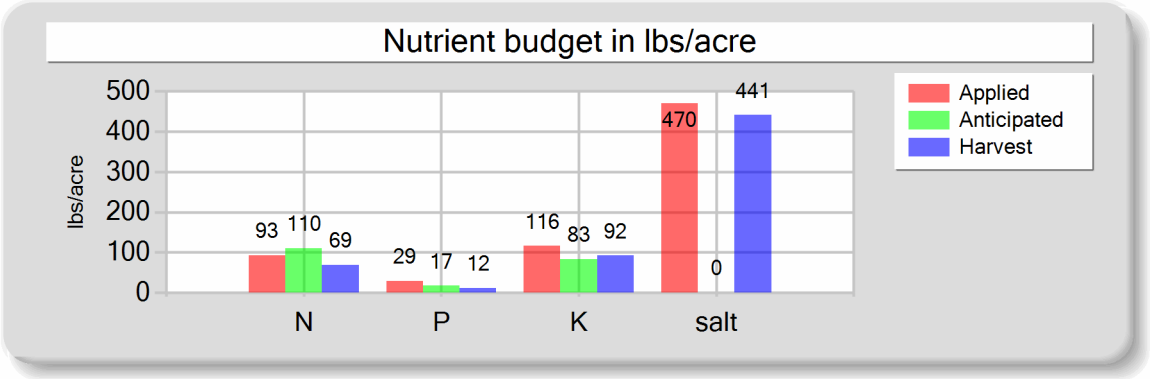
Fresh water applied	
40,961,330.00	*gallons*
1,508.47	*acre-inches*
31.43	*inches/acre*

Process wastewater applied	
372,394.00	*gallons*
13.71	*acre-inches*
0.29	*inches/acre*

Total harvests for the crop	
1	*harvests*

Angyls 4 - 10/28/2022: Winter Forage

Field name: Angyls 4 Crop: Winter Forage Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	58.46	25.05	51.69	170.14
Process wastewater	27.34	3.60	64.18	300.35
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	92.80	28.65	115.87	470.49
Anticipated crop nutrient removal	110.00	17.00	83.00	0.00
Actual crop nutrient removal	68.84	11.62	91.96	441.00
Nutrient balance	23.96	17.03	23.91	29.50
Applied to removed ratio	1.35	2.47	1.26	1.07

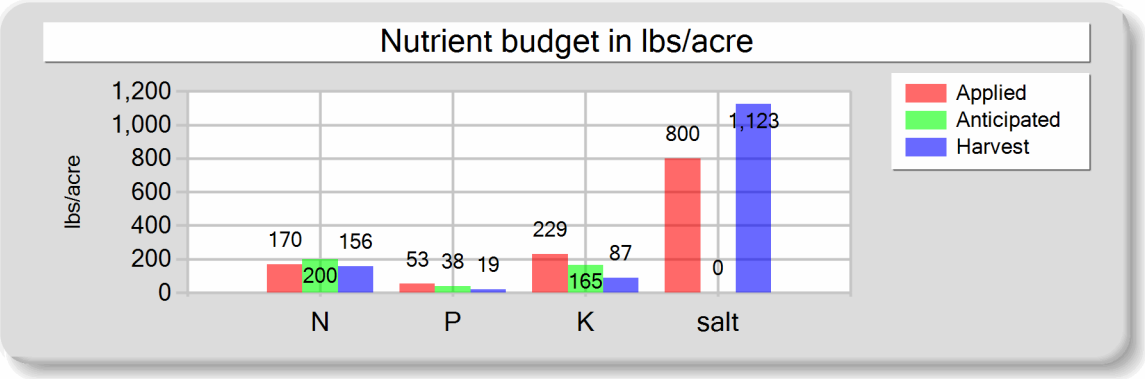
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
399,908.00 *gallons*
14.73 *acre-inches*
0.28 *inches/acre*

Total harvests for the crop
1 *harvests*

Angyls 4 - 06/20/2023: Corn, silage

Field name: Angyls 4 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.12	21.66	29.79	0.00
Dry manure	96.11	27.15	138.03	0.00
Process wastewater	49.14	3.74	61.38	352.20
Fresh water	9.84	0.00	0.00	448.14
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	170.20	52.56	229.20	800.34
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	155.98	19.47	87.10	1,123.07
Nutrient balance	14.22	33.09	142.10	-322.72
Applied to removed ratio	1.09	2.70	2.63	0.71

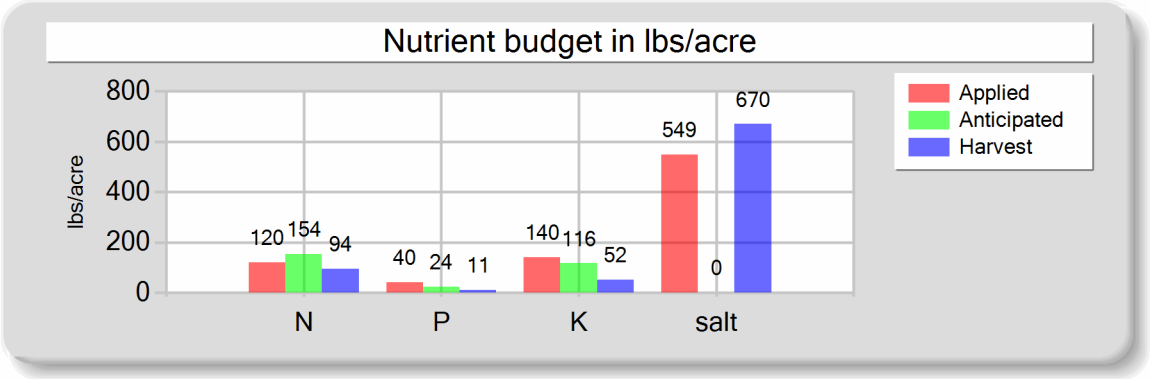
Fresh water applied	
44,374,775.00	*gallons*
1,634.17	*acre-inches*
31.43	*inches/acre*

Process wastewater applied	
403,427.00	*gallons*
14.86	*acre-inches*
0.29	*inches/acre*

Total harvests for the crop	
1	*harvests*

Angyls 5 - 10/28/2022: Winter Forage

Field name: Angyls 5 Crop: Winter Forage Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	85.50	36.63	75.60	248.83
Process wastewater	27.34	3.60	64.18	300.35
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	119.84	40.23	139.78	549.18
Anticipated crop nutrient removal	154.00	23.80	116.20	0.00
Actual crop nutrient removal	94.11	10.93	51.76	669.90
Nutrient balance	25.73	29.30	88.02	-120.72
Applied to removed ratio	1.27	3.68	2.70	0.82

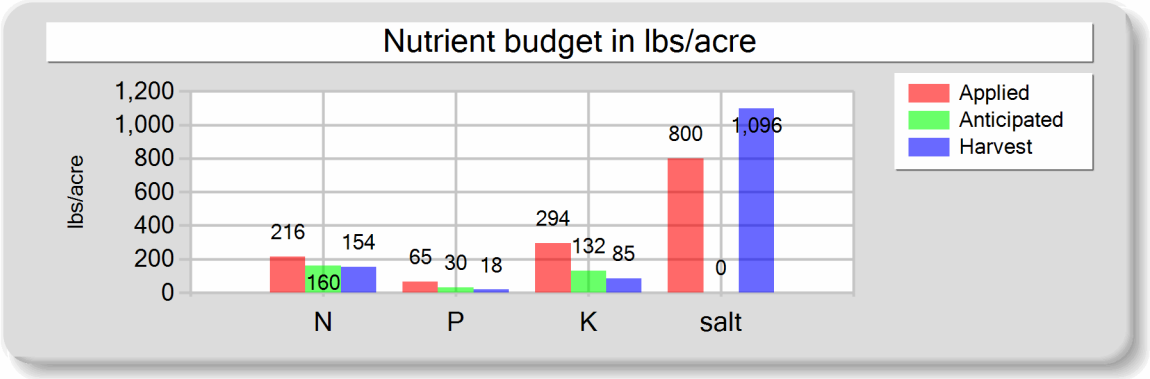
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
153,811.00 *gallons*
5.66 *acre-inches*
0.28 *inches/acre*

Total harvests for the crop
1 *harvests*

Angyls 5 - 06/20/2023: Corn, silage

Field name: Angyls 5 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.12	21.66	29.79	0.00
Dry manure	141.55	40.00	203.30	0.00
Process wastewater	49.14	3.74	61.38	352.20
Fresh water	9.84	0.00	0.00	448.14
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	215.65	65.40	294.47	800.34
Anticipated crop nutrient removal	160.00	30.00	132.00	0.00
Actual crop nutrient removal	153.96	17.88	84.68	1,096.02
Nutrient balance	61.68	47.52	209.79	-295.68
Applied to removed ratio	1.40	3.66	3.48	0.73

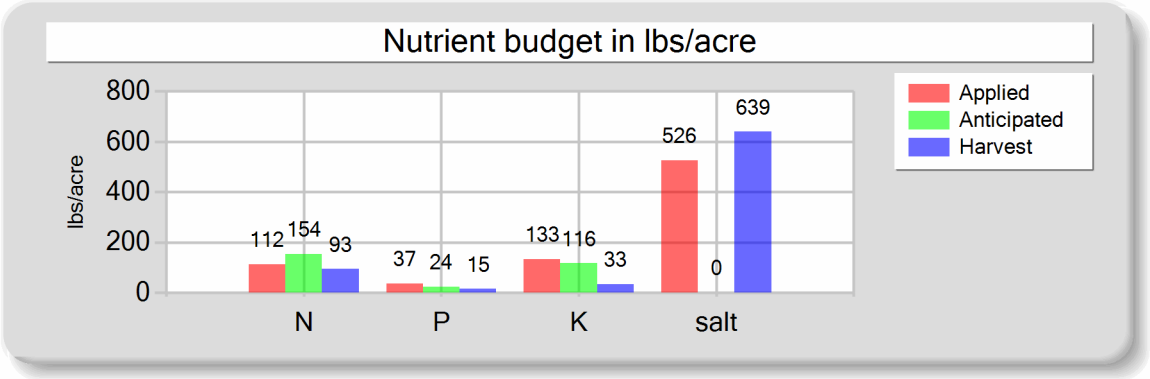
Fresh water applied	
17,067,221.00	*gallons*
628.53	*acre-inches*
31.43	*inches/acre*

Process wastewater applied	
155,164.00	*gallons*
5.71	*acre-inches*
0.29	*inches/acre*

Total harvests for the crop	
1	*harvests*

Angyls 7 - 10/28/2022: Winter Forage

Field name: Angyls 7 Crop: Winter Forage Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	77.41	33.16	68.44	225.28
Process wastewater	27.34	3.60	64.18	300.35
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	111.75	36.76	132.62	525.63
Anticipated crop nutrient removal	154.00	23.80	116.20	0.00
Actual crop nutrient removal	93.22	14.66	32.63	639.01
Nutrient balance	18.53	22.10	99.99	-113.38
Applied to removed ratio	1.20	2.51	4.06	0.82

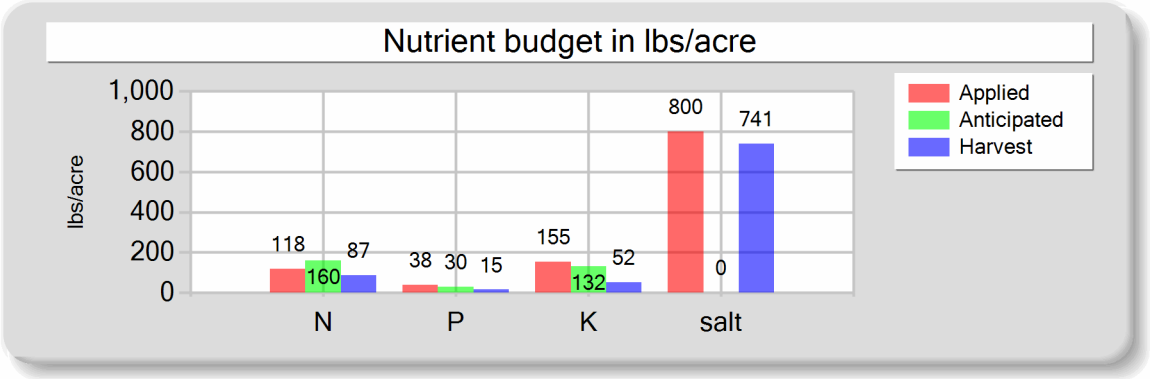
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
207,644.00 *gallons*
7.65 *acre-inches*
0.28 *inches/acre*

Total harvests for the crop
1 *harvests*

Angyls 7 - 06/20/2023: Corn, silage

Field name: Angyls 7 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.12	21.66	29.79	0.00
Dry manure	44.15	12.47	63.41	0.00
Process wastewater	49.14	3.74	61.38	352.20
Fresh water	9.84	0.00	0.00	448.14
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	118.24	37.88	154.57	800.34
Anticipated crop nutrient removal	160.00	30.00	132.00	0.00
Actual crop nutrient removal	86.88	14.91	52.39	741.06
Nutrient balance	31.36	22.97	102.19	59.28
Applied to removed ratio	1.36	2.54	2.95	1.08

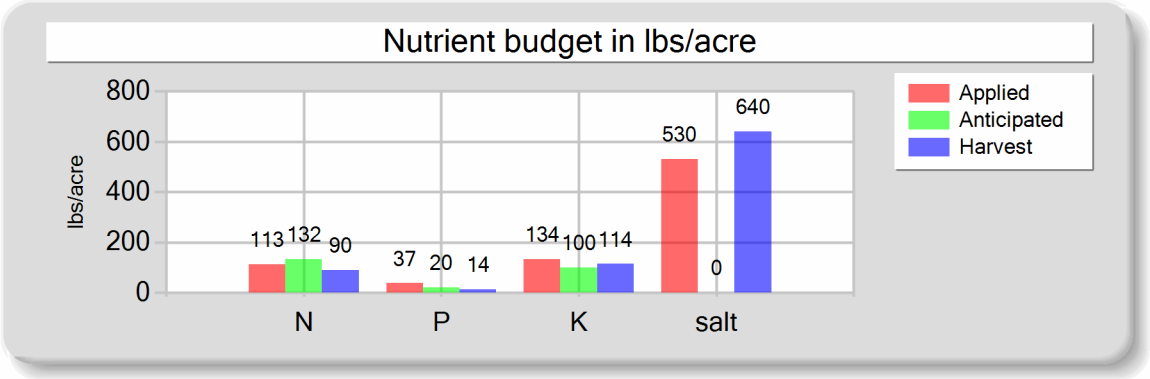
Fresh water applied	
23,040,749.00	*gallons*
848.51	*acre-inches*
31.43	*inches/acre*

Process wastewater applied	
209,472.00	*gallons*
7.71	*acre-inches*
0.29	*inches/acre*

Total harvests for the crop	
1	*harvests*

Angyls 8 - 10/28/2022: Winter Forage

Field name: Angyls 8 Crop: Winter Forage Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	78.92	33.81	69.78	229.69
Process wastewater	27.34	3.60	64.18	300.35
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	113.26	37.41	133.96	530.04
Anticipated crop nutrient removal	132.00	20.40	99.60	0.00
Actual crop nutrient removal	90.02	14.00	113.93	639.99
Nutrient balance	23.24	23.42	20.03	-109.94
Applied to removed ratio	1.26	2.67	1.18	0.83

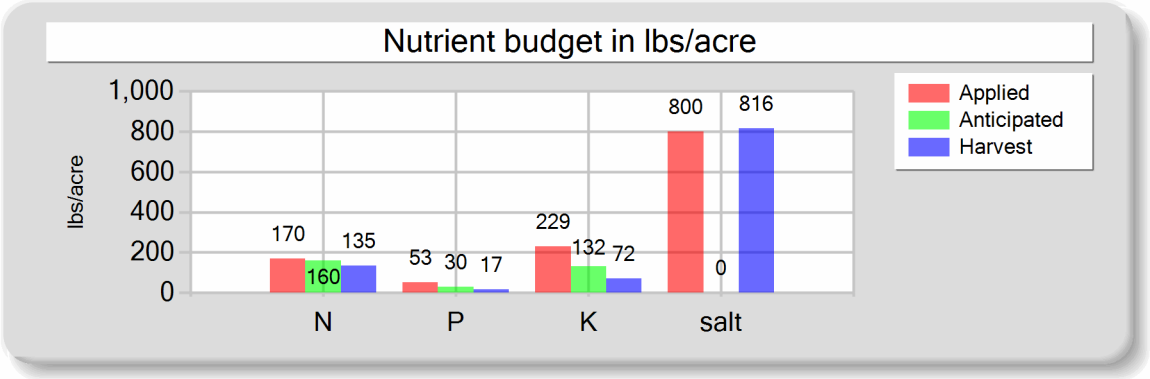
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
599,861.00	*gallons*
22.09	*acre-inches*
0.28	*inches/acre*

Total harvests for the crop	
1	*harvests*

Angyls 8 - 06/20/2023: Corn, silage

Field name: Angyls 8 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.12	21.66	29.79	0.00
Dry manure	96.12	27.16	138.06	0.00
Process wastewater	49.14	3.74	61.38	352.20
Fresh water	9.84	0.00	0.00	448.14
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	170.22	52.56	229.22	800.34
Anticipated crop nutrient removal	160.00	30.00	132.00	0.00
Actual crop nutrient removal	134.96	16.92	71.61	815.81
Nutrient balance	35.26	35.64	157.61	-15.46
Applied to removed ratio	1.26	3.11	3.20	0.98

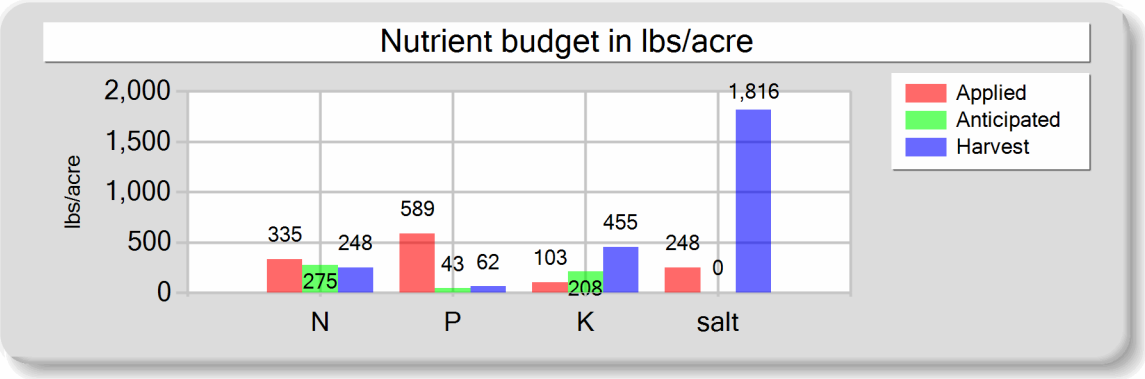
Fresh water applied	
66,562,164.00	*gallons*
2,451.26	*acre-inches*
31.43	*inches/acre*

Process wastewater applied	
605,141.00	*gallons*
22.29	*acre-inches*
0.29	*inches/acre*

Total harvests for the crop	
1	*harvests*

Home A - 11/04/2022: Winter Forage

Field name: Home A Crop: Winter Forage Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	327.89	589.33	102.56	248.33
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	334.89	589.33	102.56	248.33
Anticipated crop nutrient removal	275.00	42.50	207.50	0.00
Actual crop nutrient removal	248.41	62.26	454.90	1,816.48
Nutrient balance	86.48	527.08	-352.34	-1,568.15
Applied to removed ratio	1.35	9.47	0.23	0.14

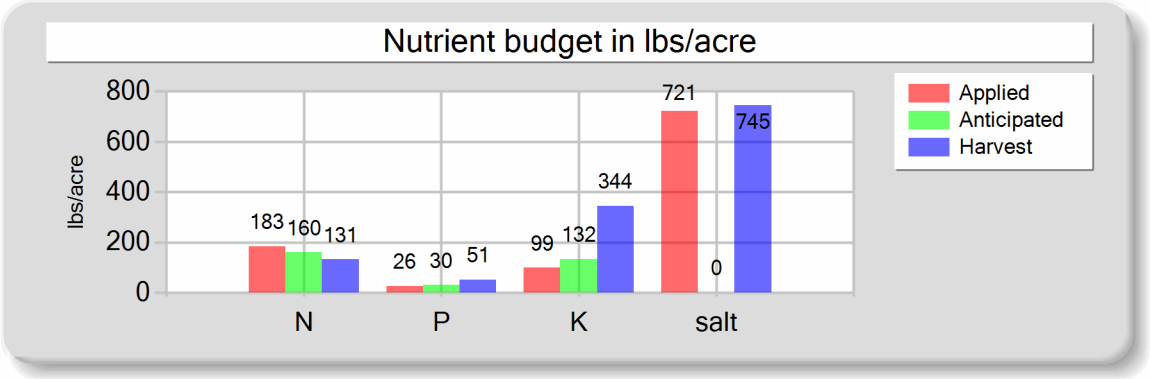
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Home A - 06/11/2023: Corn, silage

Field name: Home A | Crop: Corn, silage | Plant date: 06/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	45.05	10.57	36.35	0.00
Process wastewater	131.11	15.45	62.21	495.97
Fresh water	0.00	0.00	0.00	224.84
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	183.16	26.02	98.56	720.81
Anticipated crop nutrient removal	160.00	30.00	132.00	0.00
Actual crop nutrient removal	131.45	51.36	344.46	744.90
Nutrient balance	51.70	-25.35	-245.90	-24.09
Applied to removed ratio	1.39	0.51	0.29	0.97

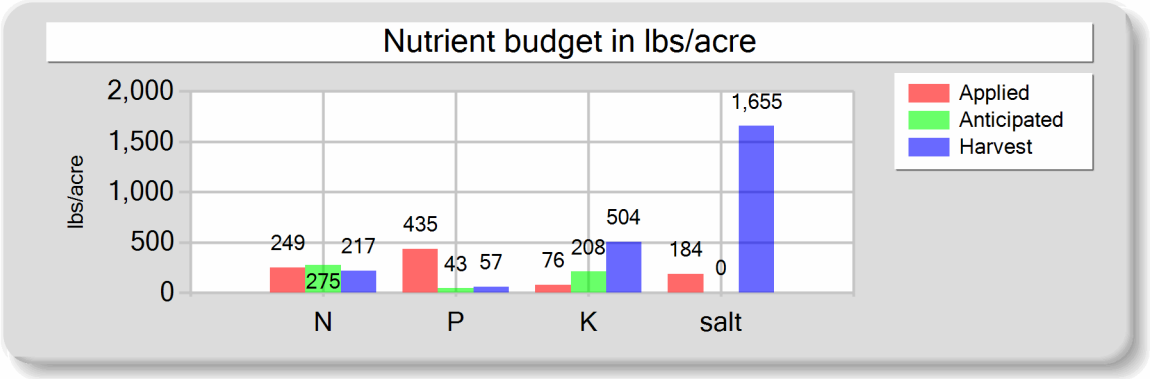
Fresh water applied	
7,577,702.00	*gallons*
279.06	*acre-inches*
31.01	*inches/acre*

Process wastewater applied	
175,030.00	*gallons*
6.45	*acre-inches*
0.72	*inches/acre*

Total harvests for the crop	
1	*harvests*

Home B - 11/04/2022: Winter Forage

Field name: Home B Crop: Winter Forage Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	242.29	435.49	75.78	183.50
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	249.29	435.49	75.78	183.50
Anticipated crop nutrient removal	275.00	42.50	207.50	0.00
Actual crop nutrient removal	217.33	57.14	503.75	1,655.19
Nutrient balance	31.96	378.35	-427.97	-1,471.68
Applied to removed ratio	1.15	7.62	0.15	0.11

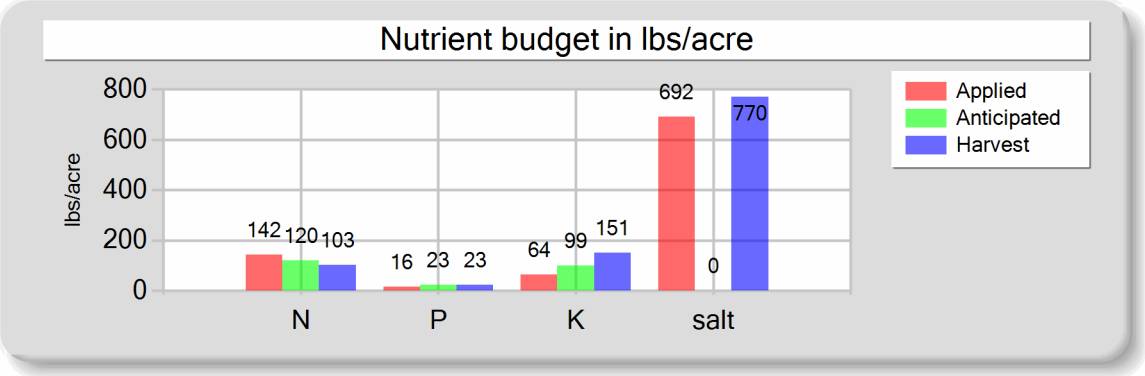
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Home B - 06/14/2023: Corn, silage

Field name: Home B Crop: Corn, silage Plant date: 06/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	18.95	3.21	9.21	0.00
Process wastewater	116.33	13.29	54.74	466.84
Fresh water	0.00	0.00	0.00	224.84
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	142.27	16.50	63.94	691.68
Anticipated crop nutrient removal	120.00	22.50	99.00	0.00
Actual crop nutrient removal	103.29	22.77	150.70	770.43
Nutrient balance	38.99	-6.28	-86.76	-78.75
Applied to removed ratio	1.38	0.72	0.42	0.90

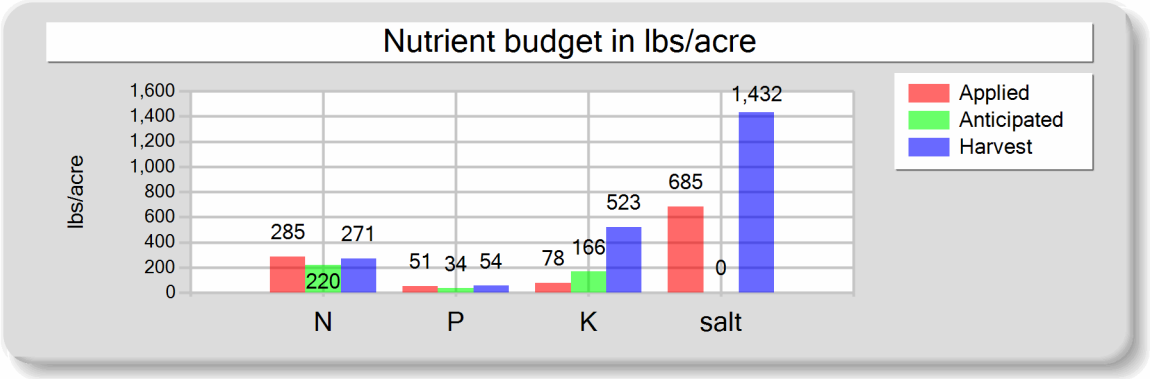
Fresh water applied	
31,994,757.00	*gallons*
1,178.26	*acre-inches*
31.01	*inches/acre*

Process wastewater applied	
689,884.00	*gallons*
25.41	*acre-inches*
0.67	*inches/acre*

Total harvests for the crop	
1	*harvests*

Home C - 11/04/2022: Winter Forage

Field name: Home C | Crop: Winter Forage | Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	278.30	51.36	77.58	685.48
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	285.30	51.36	77.58	685.48
Anticipated crop nutrient removal	220.00	34.00	166.00	0.00
Actual crop nutrient removal	270.82	54.34	522.67	1,431.74
Nutrient balance	14.48	-2.98	-445.09	-746.26
Applied to removed ratio	1.05	0.95	0.15	0.48

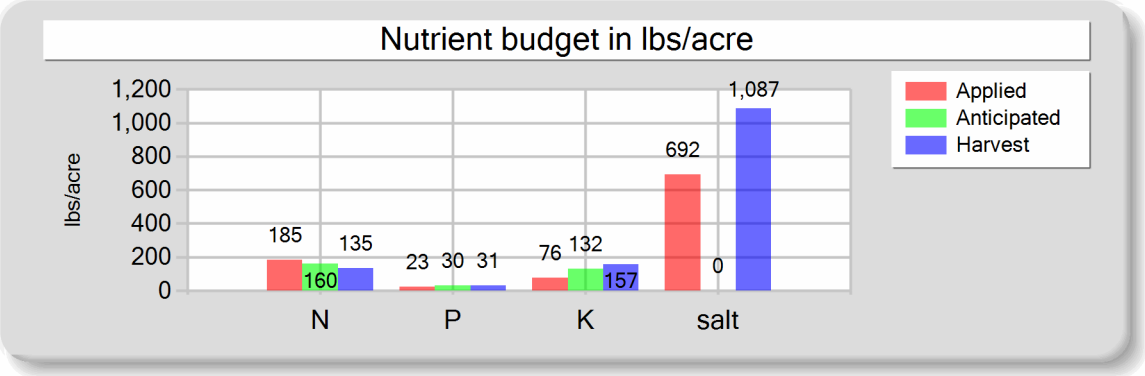
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Home C - 06/14/2023: Corn, silage

Field name: Home C Crop: Corn, silage Plant date: 06/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	62.03	9.24	21.24	0.00
Process wastewater	116.33	13.29	54.74	466.84
Fresh water	0.00	0.00	0.00	224.84
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	185.35	22.54	75.98	691.68
Anticipated crop nutrient removal	160.00	30.00	132.00	0.00
Actual crop nutrient removal	134.74	30.61	157.19	1,087.34
Nutrient balance	50.62	-8.08	-81.21	-395.66
Applied to removed ratio	1.38	0.74	0.48	0.64

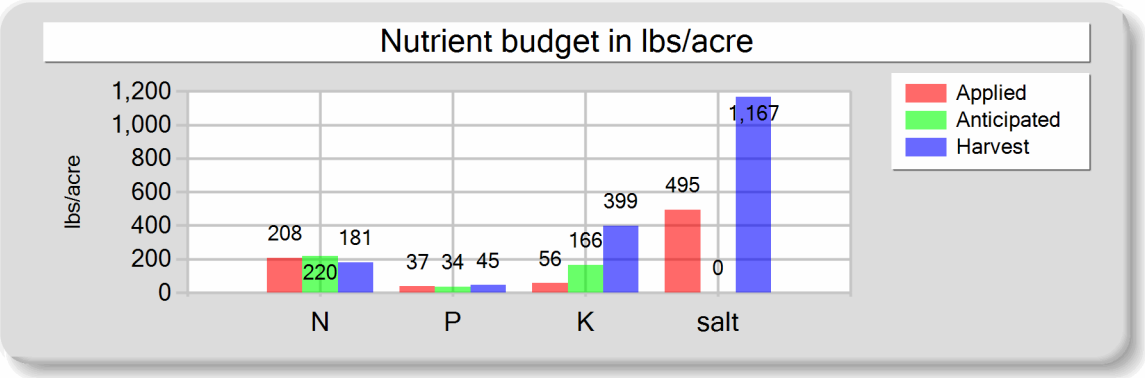
Fresh water applied	
31,152,788.00	*gallons*
1,147.25	*acre-inches*
31.01	*inches/acre*

Process wastewater applied	
671,729.00	*gallons*
24.74	*acre-inches*
0.67	*inches/acre*

Total harvests for the crop	
1	*harvests*

Home D - 11/04/2022: Winter Forage

Field name: Home D | Crop: Winter Forage | Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	200.94	37.08	56.02	494.92
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	207.94	37.08	56.02	494.92
Anticipated crop nutrient removal	220.00	34.00	166.00	0.00
Actual crop nutrient removal	180.99	44.89	398.88	1,166.89
Nutrient balance	26.95	-7.81	-342.87	-671.96
Applied to removed ratio	1.15	0.83	0.14	0.42

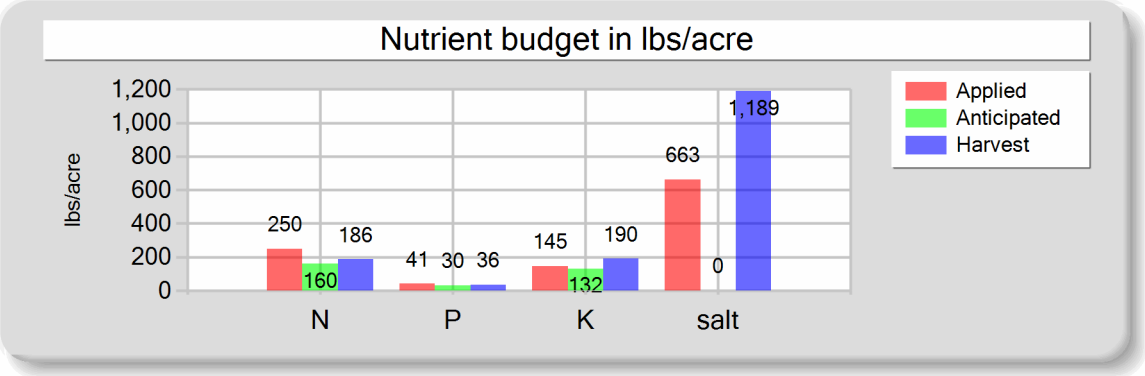
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Home D - 06/14/2023: Corn, silage

Field name: Home D Crop: Corn, silage Plant date: 06/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	81.28	19.07	65.58	0.00
Process wastewater	161.94	21.80	79.86	437.95
Fresh water	0.00	0.00	0.00	224.84
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	250.22	40.87	145.44	662.79
Anticipated crop nutrient removal	160.00	30.00	132.00	0.00
Actual crop nutrient removal	185.85	36.43	189.57	1,189.44
Nutrient balance	64.37	4.44	-44.12	-526.65
Applied to removed ratio	1.35	1.12	0.77	0.56

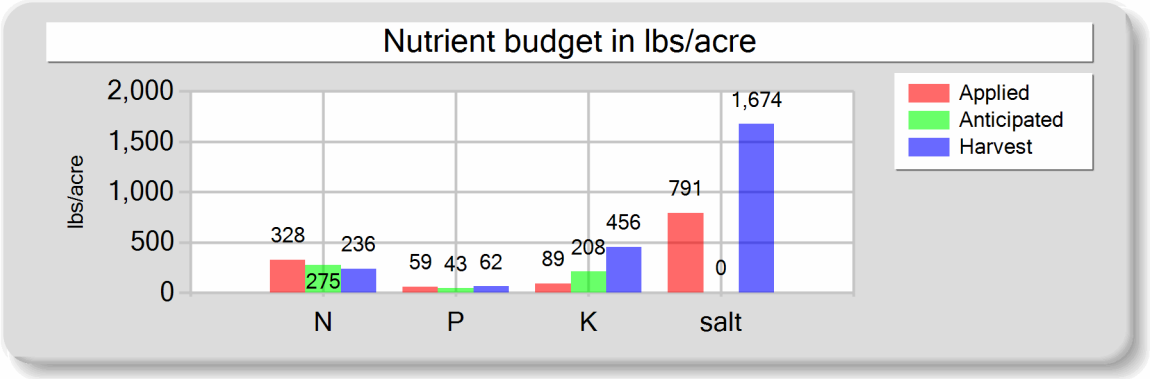
Fresh water applied	
36,204,589.00	*gallons*
1,333.29	*acre-inches*
31.01	*inches/acre*

Process wastewater applied	
780,659.00	*gallons*
28.75	*acre-inches*
0.67	*inches/acre*

Total harvests for the crop	
1	*harvests*

Home E/F - 11/04/2022: Winter Forage

Field name: Home E/F Crop: Winter Forage Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	320.98	59.23	89.48	790.60
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	327.98	59.23	89.48	790.60
Anticipated crop nutrient removal	275.00	42.50	207.50	0.00
Actual crop nutrient removal	235.90	62.16	455.87	1,673.65
Nutrient balance	92.07	-2.93	-366.39	-883.05
Applied to removed ratio	1.39	0.95	0.20	0.47

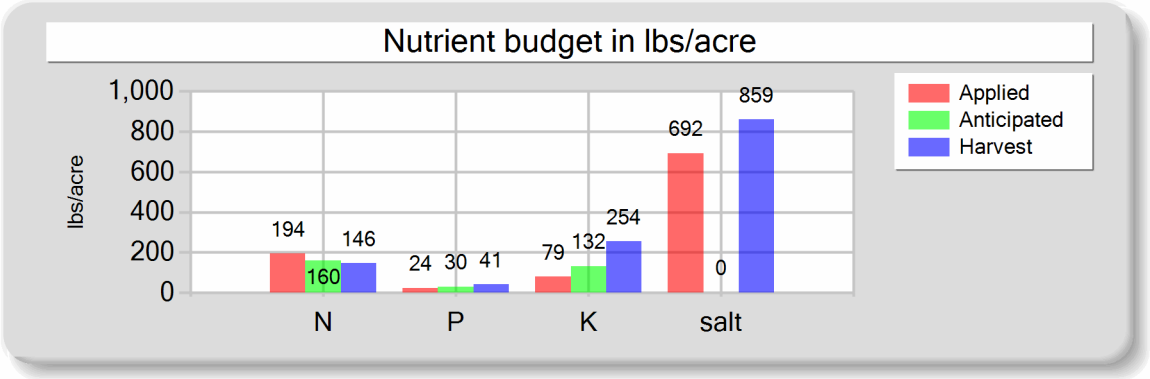
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Home E/F - 06/14/2023: Corn, silage

Field name: Home E/F Crop: Corn, silage Plant date: 06/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	70.30	10.48	24.08	0.00
Process wastewater	116.33	13.29	54.74	466.84
Fresh water	0.00	0.00	0.00	224.84
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	193.62	23.77	78.81	691.68
Anticipated crop nutrient removal	160.00	30.00	132.00	0.00
Actual crop nutrient removal	145.57	40.58	253.78	859.24
Nutrient balance	48.06	-16.81	-174.97	-167.56
Applied to removed ratio	1.33	0.59	0.31	0.80

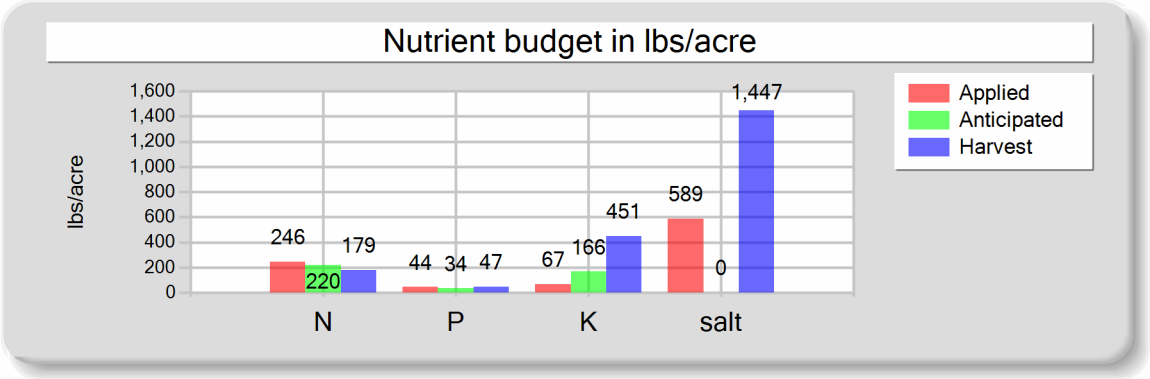
Fresh water applied	
31,152,788.00	*gallons*
1,147.25	*acre-inches*
31.01	*inches/acre*

Process wastewater applied	
671,729.00	*gallons*
24.74	*acre-inches*
0.67	*inches/acre*

Total harvests for the crop	
1	*harvests*

Home G - 11/04/2022: Winter Forage

Field name: Home G Crop: Winter Forage Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	239.08	44.12	66.65	588.88
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	246.08	44.12	66.65	588.88
Anticipated crop nutrient removal	220.00	34.00	166.00	0.00
Actual crop nutrient removal	178.76	47.30	451.15	1,447.09
Nutrient balance	67.32	-3.19	-384.50	-858.21
Applied to removed ratio	1.38	0.93	0.15	0.41

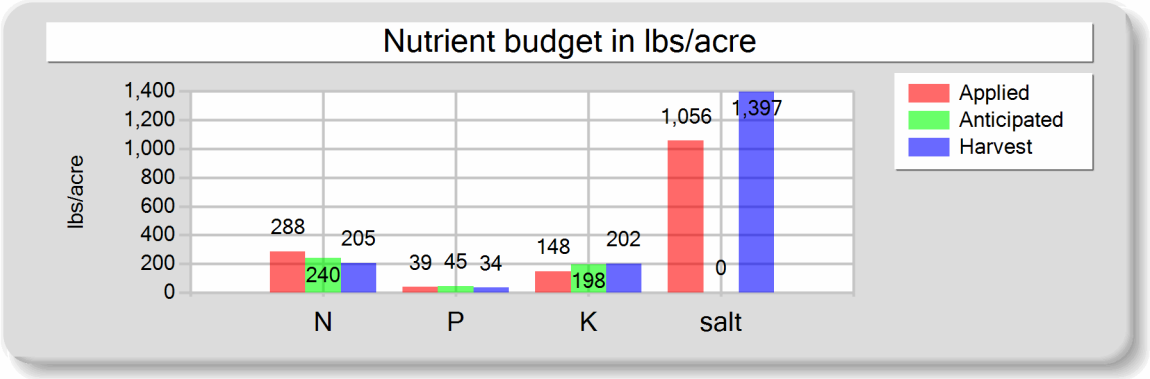
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Home G - 06/14/2023: Corn, silage

Field name: Home G | Crop: Corn, silage | Plant date: 06/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	39.73	9.32	32.06	0.00
Process wastewater	241.01	29.65	115.75	831.37
Fresh water	0.00	0.00	0.00	224.84
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	287.75	38.97	147.81	1,056.20
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	205.26	34.33	201.81	1,397.17
Nutrient balance	82.48	4.65	-54.00	-340.97
Applied to removed ratio	1.40	1.14	0.73	0.76

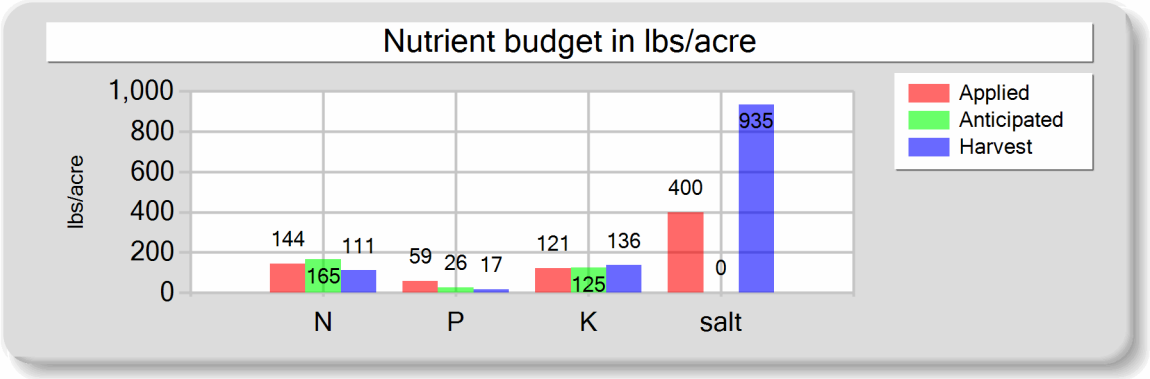
Fresh water applied	
15,997,377.00	*gallons*
589.13	*acre-inches*
31.01	*inches/acre*

Process wastewater applied	
627,968.00	*gallons*
23.13	*acre-inches*
1.22	*inches/acre*

Total harvests for the crop	
1	*harvests*

Hunts 1 - 11/04/2022: Winter Forage

Field name: Hunts 1 Crop: Winter Forage Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	137.34	58.84	121.44	399.71
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	144.34	58.84	121.44	399.71
Anticipated crop nutrient removal	165.00	25.50	124.50	0.00
Actual crop nutrient removal	110.62	17.33	135.84	935.43
Nutrient balance	33.72	41.51	-14.40	-535.71
Applied to removed ratio	1.30	3.40	0.89	0.43

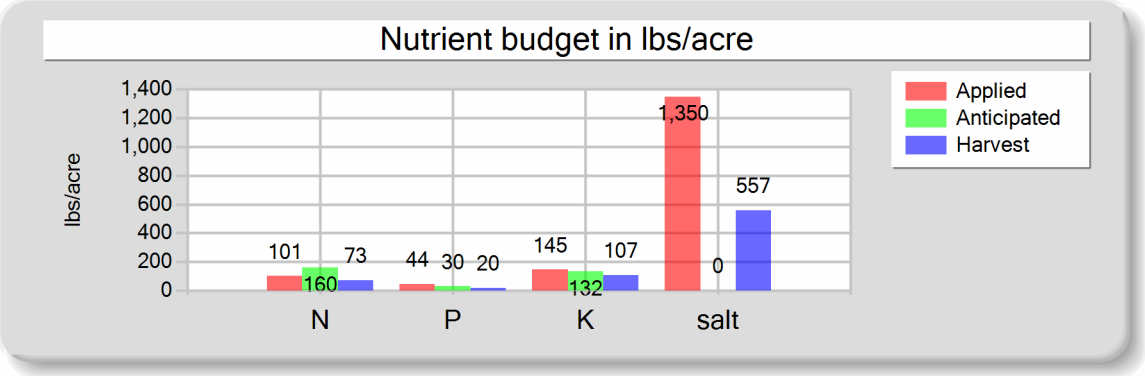
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Hunts 1 - 06/08/2023: Corn, silage

Field name: Hunts 1 Crop: Corn, silage Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.12	21.66	29.79	0.00
Dry manure	80.41	22.72	115.49	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	5.26	0.00	0.00	1,349.60
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	100.80	44.38	145.28	1,349.60
Anticipated crop nutrient removal	160.00	30.00	132.00	0.00
Actual crop nutrient removal	73.22	20.02	106.83	556.96
Nutrient balance	27.57	24.36	38.45	792.64
Applied to removed ratio	1.38	2.22	1.36	2.42

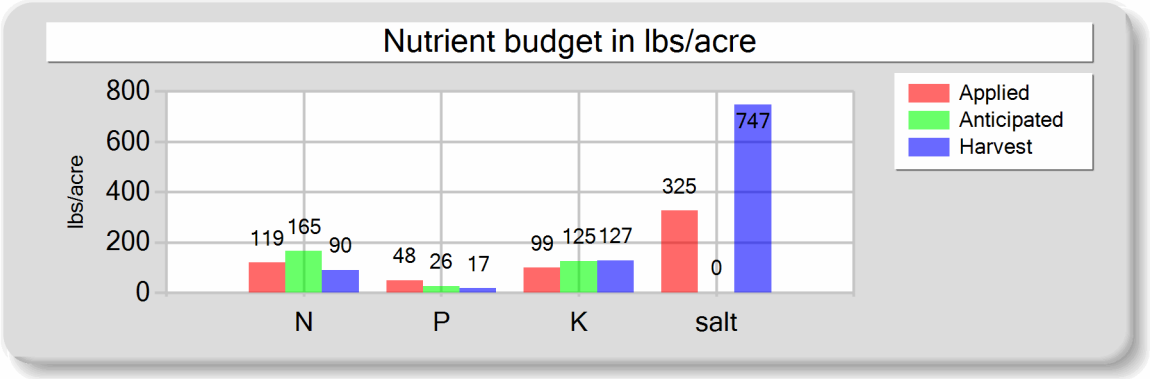
Fresh water applied	
50,943,536.00	*gallons*
1,876.08	*acre-inches*
29.78	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Hunts 2 - 11/04/2022: Winter Forage

Field name: Hunts 2 Crop: Winter Forage Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	111.76	47.88	98.82	325.27
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	118.76	47.88	98.82	325.27
Anticipated crop nutrient removal	165.00	25.50	124.50	0.00
Actual crop nutrient removal	90.12	16.71	127.49	747.37
Nutrient balance	28.64	31.18	-28.67	-422.10
Applied to removed ratio	1.32	2.87	0.78	0.44

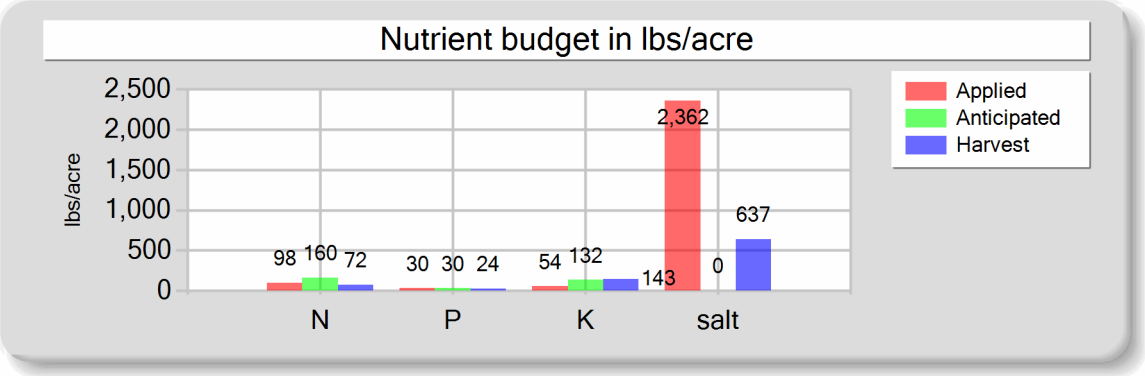
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Hunts 2 - 06/08/2023: Corn, silage

Field name: Hunts 2 Crop: Corn, silage Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.12	21.66	29.79	0.00
Dry manure	49.41	8.36	24.01	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	33.74	0.00	0.00	2,361.80
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	98.27	30.02	53.80	2,361.80
Anticipated crop nutrient removal	160.00	30.00	132.00	0.00
Actual crop nutrient removal	71.92	23.75	142.94	637.02
Nutrient balance	26.35	6.27	-89.14	1,724.78
Applied to removed ratio	1.37	1.26	0.38	3.71

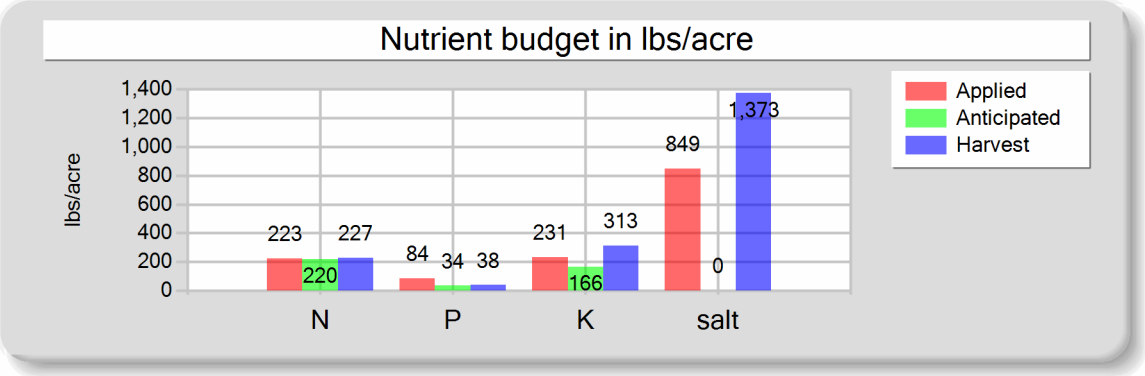
Fresh water applied	
13,746,671.00	*gallons*
506.24	*acre-inches*
29.78	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Manteca - 11/01/2022: Winter Forage

Field name: Manteca | Crop: Winter Forage | Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	188.67	80.83	166.82	549.09
Process wastewater	27.34	3.60	64.18	300.35
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	223.01	84.43	231.00	849.44
Anticipated crop nutrient removal	220.00	34.00	166.00	0.00
Actual crop nutrient removal	227.09	38.31	313.19	1,372.53
Nutrient balance	-4.08	46.12	-82.18	-523.09
Applied to removed ratio	0.98	2.20	0.74	0.62

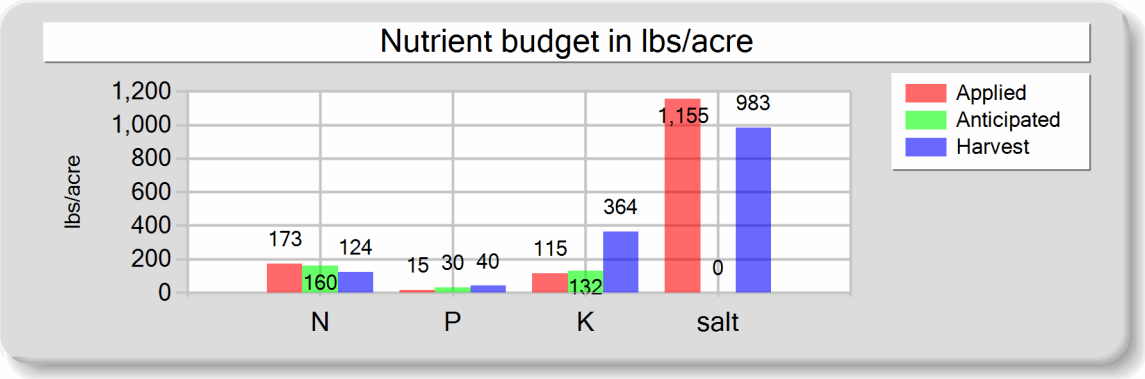
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
615,242.00	*gallons*
22.66	*acre-inches*
0.28	*inches/acre*

Total harvests for the crop	
1	*harvests*

Manteca - 06/06/2023: Corn, silage

Field name: Manteca Crop: Corn, silage Plant date: 06/06/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.06	0.00	0.00	0.00
Dry manure	58.50	9.90	28.42	0.00
Process wastewater	96.14	4.75	86.45	928.49
Fresh water	10.94	0.00	0.00	226.52
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	172.64	14.64	114.87	1,155.00
Anticipated crop nutrient removal	160.00	30.00	132.00	0.00
Actual crop nutrient removal	123.75	40.40	364.26	983.03
Nutrient balance	48.88	-25.76	-249.39	171.97
Applied to removed ratio	1.40	0.36	0.32	1.17

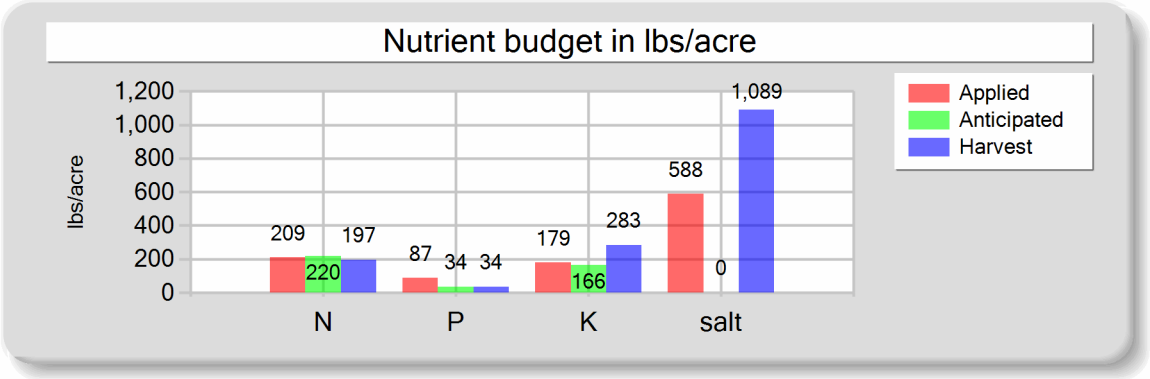
Fresh water applied	
74,880,000.00	*gallons*
2,757.58	*acre-inches*
34.47	*inches/acre*

Process wastewater applied	
1,749,156.00	*gallons*
64.42	*acre-inches*
0.81	*inches/acre*

Total harvests for the crop	
1	*harvests*

Maries 1 - 11/06/2022: Winter Forage

Field name: Maries 1 Crop: Winter Forage Plant date: 11/06/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	201.91	86.50	178.53	587.61
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	208.91	86.50	178.53	587.61
Anticipated crop nutrient removal	220.00	34.00	166.00	0.00
Actual crop nutrient removal	197.05	33.92	283.26	1,089.46
Nutrient balance	11.86	52.59	-104.73	-501.85
Applied to removed ratio	1.06	2.55	0.63	0.54

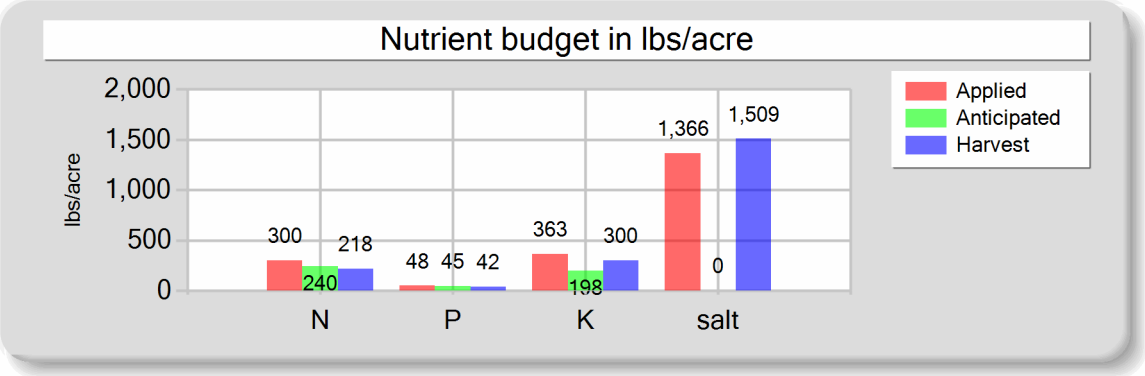
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Maries 1 - 06/17/2023: Corn, silage

Field name: Maries 1 Crop: Corn, silage Plant date: 06/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	136.38	38.54	195.88	0.00
Process wastewater	146.24	9.94	167.00	1,159.63
Fresh water	9.98	0.00	0.00	206.69
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	299.61	48.47	362.89	1,366.32
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	217.64	41.66	300.42	1,508.76
Nutrient balance	81.97	6.82	62.47	-142.44
Applied to removed ratio	1.38	1.16	1.21	0.91

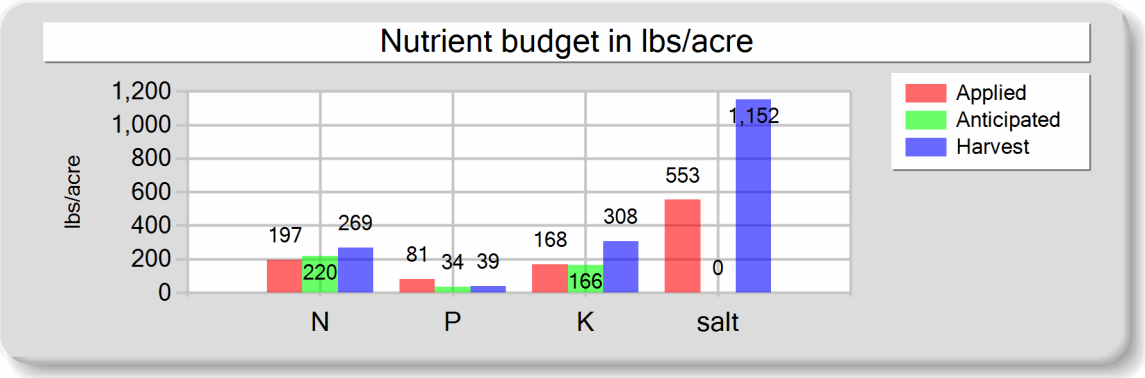
Fresh water applied	
25,622,066.00	*gallons*
943.57	*acre-inches*
31.45	*inches/acre*

Process wastewater applied	
786,046.00	*gallons*
28.95	*acre-inches*
0.96	*inches/acre*

Total harvests for the crop	
1	*harvests*

Maries 2 - 11/06/2022: Winter Forage

Field name: Maries 2 Crop: Winter Forage Plant date: 11/06/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	190.00	81.40	168.00	552.96
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	197.00	81.40	168.00	552.96
Anticipated crop nutrient removal	220.00	34.00	166.00	0.00
Actual crop nutrient removal	269.18	38.84	307.63	1,152.33
Nutrient balance	-72.18	42.56	-139.63	-599.37
Applied to removed ratio	0.73	2.10	0.55	0.48

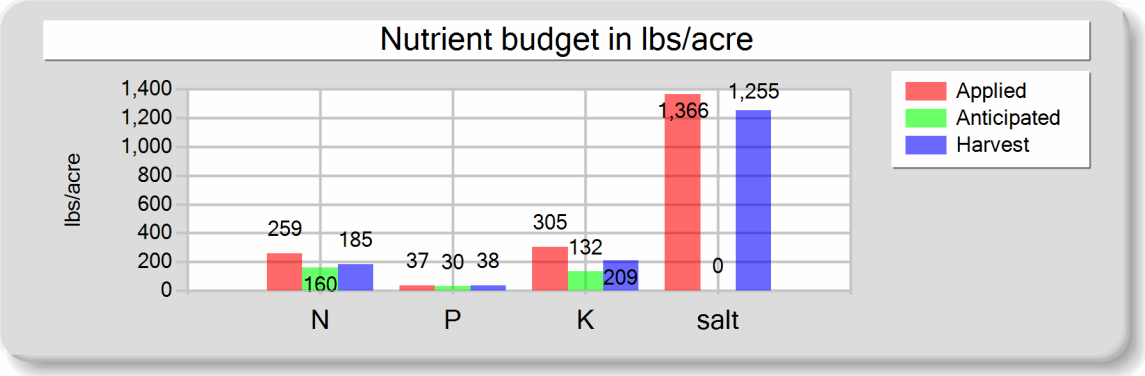
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Maries 2 - 06/17/2023: Corn, silage

Field name: Maries 2 Crop: Corn, silage Plant date: 06/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	96.14	27.17	138.08	0.00
Process wastewater	146.24	9.94	167.01	1,159.63
Fresh water	9.98	0.00	0.00	206.69
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	259.36	37.10	305.09	1,366.32
Anticipated crop nutrient removal	160.00	30.00	132.00	0.00
Actual crop nutrient removal	184.98	37.88	209.20	1,255.22
Nutrient balance	74.39	-0.77	95.89	111.10
Applied to removed ratio	1.40	0.98	1.46	1.09

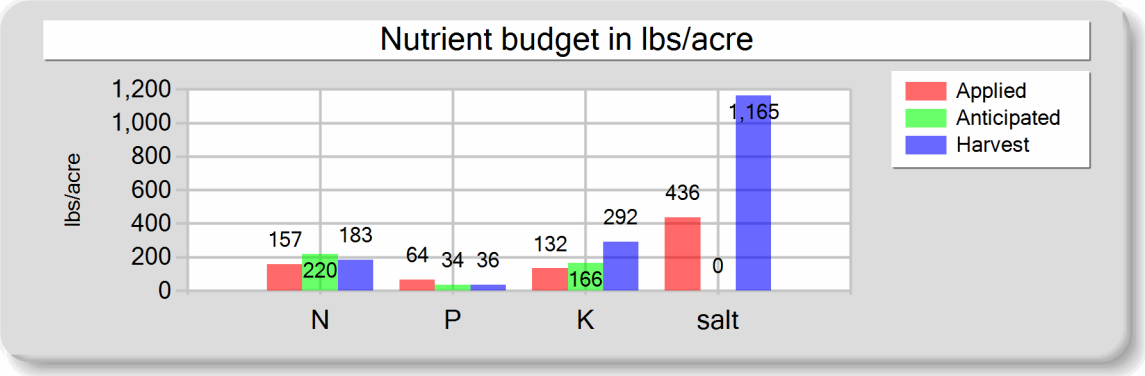
Fresh water applied	
68,325,518.00	*gallons*
2,516.20	*acre-inches*
31.45	*inches/acre*

Process wastewater applied	
2,096,125.00	*gallons*
77.19	*acre-inches*
0.96	*inches/acre*

Total harvests for the crop	
1	*harvests*

Maries 3 - 11/06/2022: Winter Forage

Field name: Maries 3 Crop: Winter Forage Plant date: 11/06/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	149.72	64.14	132.38	435.73
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	156.72	64.14	132.38	435.73
Anticipated crop nutrient removal	220.00	34.00	166.00	0.00
Actual crop nutrient removal	183.02	36.33	292.09	1,164.67
Nutrient balance	-26.30	27.82	-159.71	-728.94
Applied to removed ratio	0.86	1.77	0.45	0.37

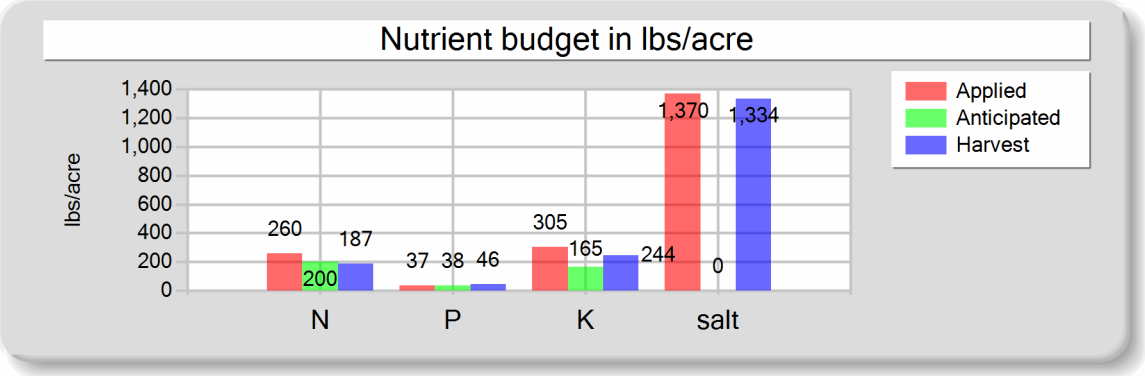
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Maries 3 - 06/17/2023: Corn, silage

Field name: Maries 3 Crop: Corn, silage Plant date: 06/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	96.13	27.16	138.06	0.00
Process wastewater	146.52	9.94	167.14	1,163.12
Fresh water	9.98	0.00	0.00	206.69
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	259.63	37.10	305.20	1,369.81
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	186.53	46.49	243.92	1,334.38
Nutrient balance	73.10	-9.38	61.28	35.43
Applied to removed ratio	1.39	0.80	1.25	1.03

Fresh water applied	
29,892,414.00	*gallons*
1,100.84	*acre-inches*
31.45	*inches/acre*

Process wastewater applied	
920,054.00	*gallons*
33.88	*acre-inches*
0.97	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

ST-11218902-081122-ZJB-001

Sample and source description: ST-11218902-081122-ZJB-001

Sample date: 08/11/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,700.00	40,800.00	7,100.00	10,800.00	3,430.00	13,600.00	3,700.00	692.00		5.60
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	10.00		0.01

ST-11218902-081122-ZJB-002

Sample and source description: ST-11218902-081122-ZJB-002

Sample date: 08/11/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 6.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,800.00	5,130.00	7,750.00	10,900.00	4,030.00	1,630.00	4,200.00	714.00		7.30
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	10.00		0.01

ST-11218902-081122-ZJB-004

Sample and source description: ST-11218902-081122-ZJB-004

Sample date: 08/11/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,000.00	8,140.00	16,800.00	23,900.00	8,060.00	3,220.00	5,500.00	1,550.00		21.60
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	10.00		0.01

ST-11218902-081122-ZJB-005

Sample and source description: ST-11218902-081122-ZJB-005

Sample date: 08/11/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 54.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,200.00	5,310.00	8,210.00	12,800.00	3,870.00	1,510.00	5,100.00	816.00		9.60
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	10.00		0.01

ST-11218902-051023-ZJB-007

Sample and source description: ST-11218902-051023-ZJB-007

Sample date: 05/10/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 82.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,300.00	2,280.00	5,240.00							
DL	100.00	100.00	100.00							

ST-11218902-051023-ZJB-008

Sample and source description: ST-11218902-051023-ZJB-008

Sample date: 05/10/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 4.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,990.00	1,640.00	5,640.00							
DL	100.00	100.00	100.00							

ST-11218902-051023-ZJB-010

Sample and source description: ST-11218902-051023-ZJB-010

Sample date: 05/10/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,000.00	2,030.00	5,830.00							
DL	100.00	100.00	100.00							

ST-11218902-051023-ZJB-011

Sample and source description: ST-11218902-051023-ZJB-011

Sample date: 05/10/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 16.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,900.00	4,210.00	21,400.00							
DL	100.00	100.00	100.00							

ST-11218902-082423-ZJB-001

Sample and source description: ST-11218902-082423-ZJB-001

Sample date: 08/24/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 78.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,100.00	2,710.00	5,010.00							
DL	100.00	100.00	100.00							

ST-11218902-082423-ZJB-002

Sample and source description: ST-11218902-082423-ZJB-002

Sample date: 08/24/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 5.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,400.00	4,140.00	15,000.00							
DL	100.00	100.00	100.00							

ST-11218902-082423-ZJB-004

Sample and source description: ST-11218902-082423-ZJB-004

Sample date: 08/24/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,300.00	2,220.00	6,680.00							
DL	100.00	100.00	100.00							

ST-11218902-082423-ZJB-005

Sample and source description: ST-11218902-082423-ZJB-005

Sample date: 08/24/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 5.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,400.00	6,160.00	43,700.00							
DL	100.00	100.00	100.00							

B. PROCESS WASTEWATER ANALYSES

WW-11218902-121422-ZJB-002

Sample and source description: WW-11218902-121422-ZJB-002

Sample date: 12/14/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	426.00	258.00	0.00	0.00	56.10	1,000.00								5,720.00	4,680
DL	10.00	10.00	0.10	0.14	10.00	10.00								100.00	100

WW-11218902-032423-ZJB-001

Sample and source description: WW-11218902-032423-ZJB-001

Sample date: 03/23/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	582.00		3.30		109.00	862.00								5,490.00	4,620
DL	10.00		0.10		10.00	10.00								100.00	10

WW-11218902-032423-ZJB-002

Sample and source description: WW-11218902-032423-ZJB-002

Sample date: 03/23/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	426.00		1.80		125.00	1,050.00								5,910.00	5,440
DL	10.00		0.10		10.00	10.00								100.00	10

WW-11218902-051023-ZJB-009

Sample and source description: WW-11218902-051023-ZJB-009

Sample date: 05/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,370.00		13.20		200.00	693.00								6,620.00	2,700
DL	10.00		0.10		10.00	10.00								100.00	10

WW-11218902-051023-ZJB-012

Sample and source description: WW-11218902-051023-ZJB-012

Sample date: 05/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	759.00		11.60		57.80	948.00								6,760.00	5,440
DL	10.00		0.10		10.00	10.00								100.00	10

WW-11218902-082423-ZJB-003

Sample and source description: WW-11218902-082423-ZJB-003

Sample date: 08/24/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	328.00		0.69		5.74	119.00								4,660.00	3,360
DL	10.00		0.10		1.00	10.00								100.00	10

WW-11218902-082423-ZJB-006

Sample and source description: WW-11218902-082423-ZJB-006

Sample date: 08/24/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	389.00		0.81		7.13	192.00								6,320.00	4,880
DL	10.00		0.10		1.00	10.00								100.00	10

WW-11218902-103023-ZJB-001

Sample and source description: WW-11218902-103023-ZJB-001

Sample date: 10/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	375.00	256.00		0.31	45.10	635.00								4,900.00	3,160
DL	10.00	10.00		0.10	10.00	10.00								100.00	10

WW-11218902-103023-ZJB-002

Sample and source description: WW-11218902-103023-ZJB-002

Sample date: 10/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.90

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	552.00	299.00		0.29	40.20	1,150.00								6,920.00	4,800
DL	10.00	10.00		0.10	10.00	10.00								100.00	10

C. FRESH WATER ANALYSES

Canal

AGG3793-01

Sample description: AGG3793-01

Sample date: 07/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00		0.00								64.00	32
DL	1.00		0.23								1.00	5

River

AGG3793-02

Sample description: AGG3793-02

Sample date: 07/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.40		0.00								62.00	29
DL	1.00		0.23								1.00	5

Well #1

Well #1

AGG3796-12

Sample description: AGG3796-12

Sample date: 07/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			18.00								810.00	
DL			0.23								1.00	

Well #10

AGG3796-06

Sample description: AGG3796-06

Sample date: 07/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			0.00								490.00	
DL			0.23								1.00	

Well #11

AGG3796-05

Sample description: AGG3796-05

Sample date: 07/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			12.00								620.00	
DL			0.23								1.00	

Well #12

Well #12

AGG3796-04

Sample description: AGG3796-04

Sample date: 07/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			2.40								770.00	
DL			0.23								1.00	

Well #13

AGG3796-03

Sample description: AGG3796-03

Sample date: 07/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			0.00								530.00	
DL			0.23								1.00	

Well #14

AGG3796-02

Sample description: AGG3796-02

Sample date: 07/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			5.80								370.00	
DL			0.23								1.00	

Well #15

Well #15

AGG3796-01

Sample description: AGG3796-01

Sample date: 07/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			17.00								610.00	
DL			0.23								1.00	

Well #16

AGG3796-13

Sample description: AGG3796-13

Sample date: 07/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			5.00	52.00	14.00	70.00	230.00	0.00	35.00	30.00	630.00	350
DL			0.23	0.10	0.10	1.00	3.00	3.00	1.00	1.00	1.00	5

Well #17

AGG3796-14

Sample description: AGG3796-14

Sample date: 07/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			0.78	31.00	7.70	30.00	120.00	0.00	21.00	15.00	330.00	200
DL			0.23	0.10	0.10	1.00	3.00	3.00	1.00	1.00	1.00	5

Well #2

Well #2

AGG3796-11

Sample description: AGG3796-11

Sample date: 07/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			40.00								1,200.00	
DL			0.23								1.00	

Well #3

AGG3796-10

Sample description: AGG3796-10

Sample date: 07/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			9.70								1,000.00	
DL			0.23								1.00	

Well #6

AGG3796-09

Sample description: AGG3796-09

Sample date: 07/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			2.00								440.00	
DL			0.23								1.00	

Well #8

Well #8

AGG3796-08

Sample description: AGG3796-08

Sample date: 07/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			28.00								1,200.00	
DL			0.23								1.00	

Well #9

AGG3796-07

Sample description: AGG3796-07

Sample date: 07/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			0.00								280.00	
DL			0.23								1.00	

D. SOIL ANALYSES

Angyls 7

So-11218902-102023-ZJB-005

Sample and source description: So-11218902-102023-ZJB-005

Sample date: 10/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			9.00				
DL			1.00				

Angyls 8

So-11218902-102023-ZJB-003

Sample and source description: So-11218902-102023-ZJB-003

Sample date: 10/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			23.00				
DL			1.00				

So-11218902-102023-ZJB-004

Sample and source description: So-11218902-102023-ZJB-004

Sample date: 10/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			9.00				
DL			1.00				

Home A

So-11218902-102023-ZJB-001

Sample and source description: So-11218902-102023-ZJB-001

Sample date: 10/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			90.00				
DL			1.00				

Home B

Home B

So-11218902-102023-ZJB-002

Sample and source description: So-11218902-102023-ZJB-002

Sample date: 10/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			119.00				
DL			1.00				

E. PLANT TISSUE ANALYSES

Airport 1 - 11/05/2022: Winter Forage

TP-11218902-050423-MCB-009

Sample and source description: TP-11218902-050423-MCB-009

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,700.00	2,640.00	15,800.00		8.00
DL	1,000.00	20.00	20.00		0.01

TP-11218902-050423-MCB-010

Sample and source description: TP-11218902-050423-MCB-010

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,150.00	2,510.00	15,400.00		8.30
DL	1,000.00	20.00	20.00		0.01

Airport 1 - 06/13/2023: Corn, silage

Airport 1 - 06/13/2023: Corn, silage

TP-11218902-100523-ZJB-001

Sample and source description: TP-11218902-100523-ZJB-001

Sample date: 10/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,800.00	1,950.00	12,000.00		4.60
DL	1,000.00	20.00	20.00		0.01

TP-11218902-100523-ZJB-002

Sample and source description: TP-11218902-100523-ZJB-002

Sample date: 10/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,980.00	1,520.00	10,200.00		3.88
DL	1,000.00	20.00	20.00		0.01

Airport 2 - 11/05/2022: Winter Forage

TP-11218902-050423-MCB-012

Sample and source description: TP-11218902-050423-MCB-012

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 56.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,200.00	2,140.00	13,900.00		12.30
DL	1,000.00	20.00	20.00		0.01

Airport 2 - 06/13/2023: Corn, silage

Airport 2 - 06/13/2023: Corn, silage

TP-11218902-100523-ZJB-003

Sample and source description: TP-11218902-100523-ZJB-003

Sample date: 10/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,160.00	2,410.00	17,000.00		5.91
DL	1,000.00	20.00	20.00		0.01

Airport 3 - 11/05/2022: Winter Forage

TP-11218902-050423-MCB-008

Sample and source description: TP-11218902-050423-MCB-008

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,100.00	2,710.00	14,800.00		8.00
DL	1,000.00	20.00	20.00		0.01

Airport 3 - 06/13/2023: Corn, silage

TP-11218902-100523-ZJB-004

Sample and source description: TP-11218902-100523-ZJB-004

Sample date: 10/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,480.00	2,520.00	17,800.00		5.47
DL	1,000.00	20.00	20.00		0.01

Anderson - 11/06/2022: Winter Forage

TP-11218902-050423-MCB-017

Sample and source description: TP-11218902-050423-MCB-017

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,500.00	3,800.00	35,400.00		10.70
DL	1,000.00	20.00	20.00		0.01

TP-11218902-050423-MCB-018

Sample and source description: TP-11218902-050423-MCB-018

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,300.00	3,030.00	26,900.00		10.10
DL	1,000.00	20.00	20.00		0.01

Anderson - 06/07/2023: Corn, silage

TP-11218902-100523-ZJB-005

Sample and source description: TP-11218902-100523-ZJB-005

Sample date: 10/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,470.00	2,220.00	20,300.00		5.11
DL	1,000.00	20.00	20.00		0.01

Anderson - 06/07/2023: Corn, silage

TP-11218902-100523-ZJB-006

Sample and source description: TP-11218902-100523-ZJB-006

Sample date: 10/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,390.00	2,190.00	20,400.00		5.27
DL	1,000.00	20.00	20.00		0.01

Angyls 1-A - 10/28/2022: Winter Forage

TP-11218902-050423-MCB-007

Sample and source description: TP-11218902-050423-MCB-007

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,200.00	2,540.00	21,500.00		11.50
DL	1,000.00	20.00	20.00		0.01

Angyls 1-A - 06/20/2023: Corn, silage

TP-11218902-102023-ZJB-015

Sample and source description: TP-11218902-102023-ZJB-015

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,400.00	2,330.00	11,700.00		9.20
DL	1,000.00	20.00	20.00		0.01

Angyls 2 - 10/28/2022: Winter Forage

TP-11218902-050423-MCB-003

Sample and source description: TP-11218902-050423-MCB-003

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,200.00	1,880.00	16,200.00		8.70
DL	1,000.00	20.00	20.00		0.01

Angyls 2 - 06/20/2023: Corn, silage

TP-11218902-102023-ZJB-016

Sample and source description: TP-11218902-102023-ZJB-016

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,500.00	2,070.00	8,480.00		8.10
DL	1,000.00	20.00	20.00		0.01

Angyls 3 - 10/28/2022: Winter Forage

TP-11218902-050423-MCB-001

Sample and source description: TP-11218902-050423-MCB-001

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,700.00	1,840.00	14,200.00		8.10
DL	1,000.00	20.00	20.00		0.01

Angyls 3 - 10/28/2022: Winter Forage

TP-11218902-050423-MCB-002

Sample and source description: TP-11218902-050423-MCB-002

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,000.00	2,150.00	17,100.00		12.00
DL	1,000.00	20.00	20.00		0.01

Angyls 3 - 06/20/2023: Corn, silage

TP-11218902-102023-ZJB-017

Sample and source description: TP-11218902-102023-ZJB-017

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,800.00	1,320.00	7,290.00		8.60
DL	1,000.00	20.00	20.00		0.01

TP-11218902-102023-ZJB-018

Sample and source description: TP-11218902-102023-ZJB-018

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,400.00	2,040.00	9,930.00		10.40
DL	1,000.00	20.00	20.00		0.01

Angyls 4 - 10/28/2022: Winter Forage

Angyls 4 - 10/28/2022: Winter Forage

TP-11218902-050423-MCB-019

Sample and source description: TP-11218902-050423-MCB-019

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,400.00	2,070.00	17,200.00		9.10
DL	1,000.00	20.00	20.00		0.01

TP-11218902-050423-MCB-020

Sample and source description: TP-11218902-050423-MCB-020

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,800.00	2,160.00	17,100.00		8.20
DL	1,000.00	20.00	20.00		0.01

Angyls 4 - 06/20/2023: Corn, silage

TP-11218902-102023-ZJB-019

Sample and source description: TP-11218902-102023-ZJB-019

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,500.00	1,560.00	6,980.00		9.00
DL	1,000.00	20.00	20.00		0.01

Angyls 4 - 06/20/2023: Corn, silage

TP-11218902-102023-ZJB-020

Sample and source description: TP-11218902-102023-ZJB-020

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,300.00	1,970.00	6,390.00		7.70
DL	1,000.00	20.00	20.00		0.01

Angyls 5 - 10/28/2022: Winter Forage

TP-11218902-050423-MCB-021

Sample and source description: TP-11218902-050423-MCB-021

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,900.00	2,290.00	19,100.00		9.80
DL	1,000.00	20.00	20.00		0.01

Angyls 5 - 06/20/2023: Corn, silage

TP-11218902-102023-ZJB-021

Sample and source description: TP-11218902-102023-ZJB-021

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,800.00	1,370.00	6,490.00		8.40
DL	1,000.00	20.00	20.00		0.01

Angyls 7 - 10/28/2022: Winter Forage

TP-11218902-050423-MCB-004

Sample and source description: TP-11218902-050423-MCB-004

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,400.00	1,950.00	4,340.00		8.50
DL	1,000.00	20.00	20.00		0.01

Angyls 7 - 06/20/2023: Corn, silage

TP-11218902-102023-ZJB-022

Sample and source description: TP-11218902-102023-ZJB-022

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,200.00	1,750.00	6,150.00		8.70
DL	1,000.00	20.00	20.00		0.01

Angyls 8 - 10/28/2022: Winter Forage

TP-11218902-050423-MCB-022

Sample and source description: TP-11218902-050423-MCB-022

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,100.00	2,330.00	16,000.00		9.00
DL	1,000.00	20.00	20.00		0.01

Angyls 8 - 10/28/2022: Winter Forage

TP-11218902-050423-MCB-023

Sample and source description: TP-11218902-050423-MCB-023

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,800.00	1,990.00	16,200.00		9.10
DL	1,000.00	20.00	20.00		0.01

Angyls 8 - 06/20/2023: Corn, silage

TP-11218902-102023-ZJB-023

Sample and source description: TP-11218902-102023-ZJB-023

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,400.00	1,680.00	7,110.00		8.10
DL	1,000.00	20.00	20.00		0.01

TP-11218902-102023-ZJB-024

Sample and source description: TP-11218902-102023-ZJB-024

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,000.00	1,700.00	7,470.00		9.40
DL	1,000.00	20.00	20.00		0.01

Home A - 11/04/2022: Winter Forage

Home A - 11/04/2022: Winter Forage

TP-11218902-050423-MCB-015

Sample and source description: TP-11218902-050423-MCB-015

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,000.00	4,010.00	29,300.00		11.70
DL	1,000.00	20.00	20.00		0.01

Home A - 06/11/2023: Corn, silage

TP-11218902-100523-ZJB-007

Sample and source description: TP-11218902-100523-ZJB-007

Sample date: 10/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,800.00	4,220.00	28,300.00		6.12
DL	1,000.00	20.00	20.00		0.01

Home B - 11/04/2022: Winter Forage

TP-11218902-050423-MCB-005

Sample and source description: TP-11218902-050423-MCB-005

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,100.00	3,970.00	35,000.00		11.50
DL	1,000.00	20.00	20.00		0.01

Home B - 06/14/2023: Corn, silage

TP-11218902-102023-ZJB-010

Sample and source description: TP-11218902-102023-ZJB-010

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,200.00	2,690.00	17,800.00		9.10
DL	1,000.00	20.00	20.00		0.01

Home C - 11/04/2022: Winter Forage

TP-11218902-050423-MCB-006

Sample and source description: TP-11218902-050423-MCB-006

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 54.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,700.00	3,150.00	30,300.00		8.30
DL	1,000.00	20.00	20.00		0.01

Home C - 06/14/2023: Corn, silage

TP-11218902-102023-ZJB-011

Sample and source description: TP-11218902-102023-ZJB-011

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,400.00	2,590.00	13,300.00		9.20
DL	1,000.00	20.00	20.00		0.01

Home D - 11/04/2022: Winter Forage

TP-11218902-050423-MCB-013

Sample and source description: TP-11218902-050423-MCB-013

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,300.00	2,930.00	18,800.00		6.40
DL	1,000.00	20.00	20.00		0.01

TP-11218902-050423-MCB-014

Sample and source description: TP-11218902-050423-MCB-014

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,200.00	3,770.00	33,500.00		9.80
DL	1,000.00	20.00	20.00		0.01

Home D - 06/14/2023: Corn, silage

TP-11218902-102023-ZJB-012

Sample and source description: TP-11218902-102023-ZJB-012

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,300.00	2,880.00	17,000.00		10.20
DL	1,000.00	20.00	20.00		0.01

Home D - 06/14/2023: Corn, silage

TP-11218902-102023-ZJB-013

Sample and source description: TP-11218902-102023-ZJB-013

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,000.00	2,940.00	15,300.00		9.60
DL	1,000.00	20.00	20.00		0.01

Home E/F - 11/04/2022: Winter Forage

TP-11218902-050423-MCB-016

Sample and source description: TP-11218902-050423-MCB-016

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,800.00	3,900.00	28,600.00		10.50
DL	1,000.00	20.00	20.00		0.01

Home E/F - 06/14/2023: Corn, silage

TP-11218902-100523-ZJB-008

Sample and source description: TP-11218902-100523-ZJB-008

Sample date: 10/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,300.00	3,150.00	19,700.00		6.67
DL	1,000.00	20.00	20.00		0.01

Home G - 11/04/2022: Winter Forage

TP-11218902-050423-MCB-011

Sample and source description: TP-11218902-050423-MCB-011

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,700.00	3,890.00	37,100.00		11.90
DL	1,000.00	20.00	20.00		0.01

Home G - 06/14/2023: Corn, silage

TP-11218902-102023-ZJB-014

Sample and source description: TP-11218902-102023-ZJB-014

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,900.00	1,990.00	11,700.00		8.10
DL	1,000.00	20.00	20.00		0.01

Hunts 1 - 11/04/2022: Winter Forage

TP-11218902-050423-MCB-024

Sample and source description: TP-11218902-050423-MCB-024

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,400.00	1,960.00	15,500.00		11.20
DL	1,000.00	20.00	20.00		0.01

Hunts 1 - 11/04/2022: Winter Forage

TP-11218902-050423-MCB-025

Sample and source description: TP-11218902-050423-MCB-025

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,600.00	2,130.00	16,700.00		11.50
DL	1,000.00	20.00	20.00		0.01

Hunts 1 - 06/08/2023: Corn, silage

TP-11218902-100523-ZJB-011

Sample and source description: TP-11218902-100523-ZJB-011

Sample date: 10/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,300.00	2,080.00	12,400.00		5.39
DL	1,000.00	20.00	20.00		0.01

TP-11218902-100523-ZJB-012

Sample and source description: TP-11218902-100523-ZJB-012

Sample date: 10/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,060.00	1,930.00	10,300.00		5.37
DL	1,000.00	20.00	20.00		0.01

Hunts 2 - 11/04/2022: Winter Forage

Hunts 2 - 11/04/2022: Winter Forage

TP-11218902-050423-MCB-026

Sample and source description: TP-11218902-050423-MCB-026

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,300.00	2,280.00	17,400.00		10.20
DL	1,000.00	20.00	20.00		0.01

Hunts 2 - 06/08/2023: Corn, silage

TP-11218902-100523-ZJB-013

Sample and source description: TP-11218902-100523-ZJB-013

Sample date: 10/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,390.00	2,110.00	12,700.00		5.66
DL	1,000.00	20.00	20.00		0.01

Manteca - 11/01/2022: Winter Forage

TP-11218902-051023-ZJB-001

Sample and source description: TP-11218902-051023-ZJB-001

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,200.00	3,070.00	25,100.00		11.00
DL	1,000.00	20.00	20.00		0.01

Manteca - 11/01/2022: Winter Forage

TP-11218902-051023-ZJB-002

Sample and source description: TP-11218902-051023-ZJB-002

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,900.00	3,620.00	26,600.00		11.70
DL	1,000.00	20.00	20.00		0.01

Manteca - 06/06/2023: Corn, silage

TP-11218902-100523-ZJB-009

Sample and source description: TP-11218902-100523-ZJB-009

Sample date: 10/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,610.00	2,780.00	23,800.00		8.11
DL	1,000.00	20.00	20.00		0.01

TP-11218902-100523-ZJB-010

Sample and source description: TP-11218902-100523-ZJB-010

Sample date: 10/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,780.00	2,540.00	22,900.00		6.18
DL	1,000.00	20.00	20.00		0.01

Maries 1 - 11/06/2022: Winter Forage

Maries 1 - 11/06/2022: Winter Forage

TP-11218902-051023-ZJB-003

Sample and source description: TP-11218902-051023-ZJB-003

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,800.00	3,580.00	29,900.00		11.50
DL	1,000.00	20.00	20.00		0.01

Maries 1 - 06/17/2023: Corn, silage

TP-11218902-102023-ZJB-008

Sample and source description: TP-11218902-102023-ZJB-008

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,300.00	3,120.00	22,500.00		11.30
DL	1,000.00	20.00	20.00		0.01

Maries 2 - 11/06/2022: Winter Forage

TP-11218902-051023-ZJB-004

Sample and source description: TP-11218902-051023-ZJB-004

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,000.00	3,030.00	24,000.00		8.99
DL	1,000.00	20.00	20.00		0.01

Maries 2 - 11/06/2022: Winter Forage

TP-11218902-051023-ZJB-005

Sample and source description: TP-11218902-051023-ZJB-005

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,300.00	3,040.00	23,900.00		11.30
DL	1,000.00	20.00	20.00		0.01

Maries 2 - 06/17/2023: Corn, silage

TP-11218902-102023-ZJB-006

Sample and source description: TP-11218902-102023-ZJB-006

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,800.00	3,440.00	19,000.00		11.40
DL	1,000.00	20.00	20.00		0.01

TP-11218902-102023-ZJB-007

Sample and source description: TP-11218902-102023-ZJB-007

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,500.00	3,600.00	20,300.00		10.30
DL	1,000.00	20.00	20.00		0.01

Maries 3 - 11/06/2022: Winter Forage

Maries 3 - 11/06/2022: Winter Forage

TP-11218902-051023-ZJB-006

Sample and source description: TP-11218902-051023-ZJB-006

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,800.00	3,930.00	31,600.00		12.60
DL	1,000.00	20.00	20.00		0.01

Maries 3 - 06/17/2023: Corn, silage

TP-11218902-102023-ZJB-009

Sample and source description: TP-11218902-102023-ZJB-009

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,000.00	3,240.00	17,000.00		9.30
DL	1,000.00	20.00	20.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

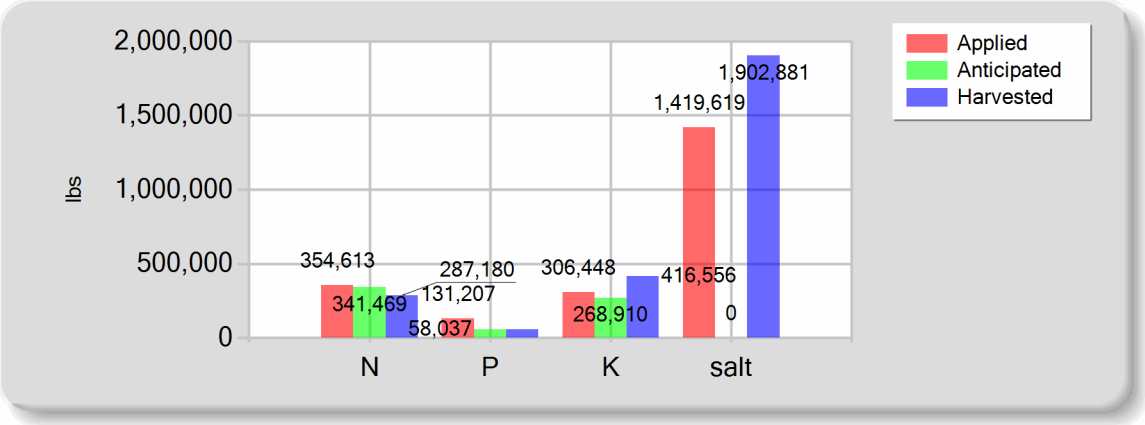
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

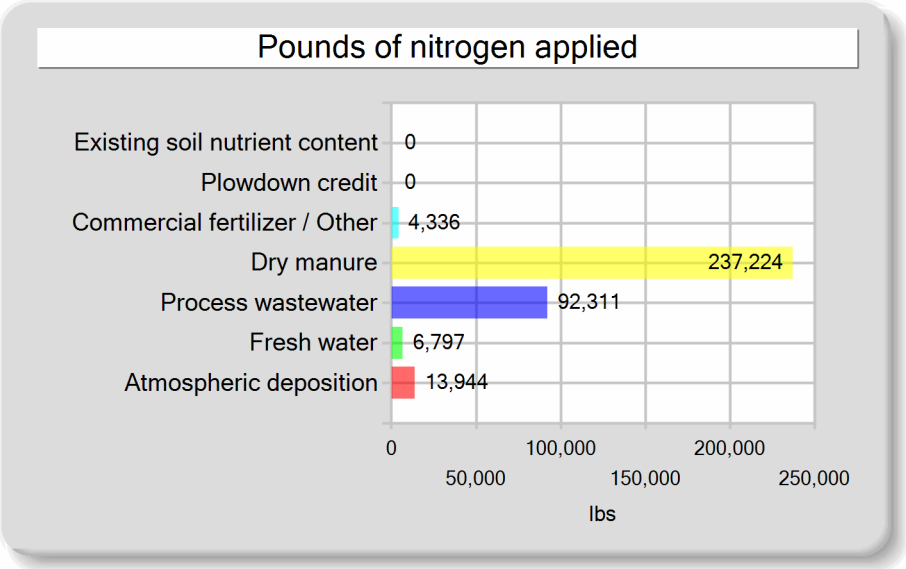
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

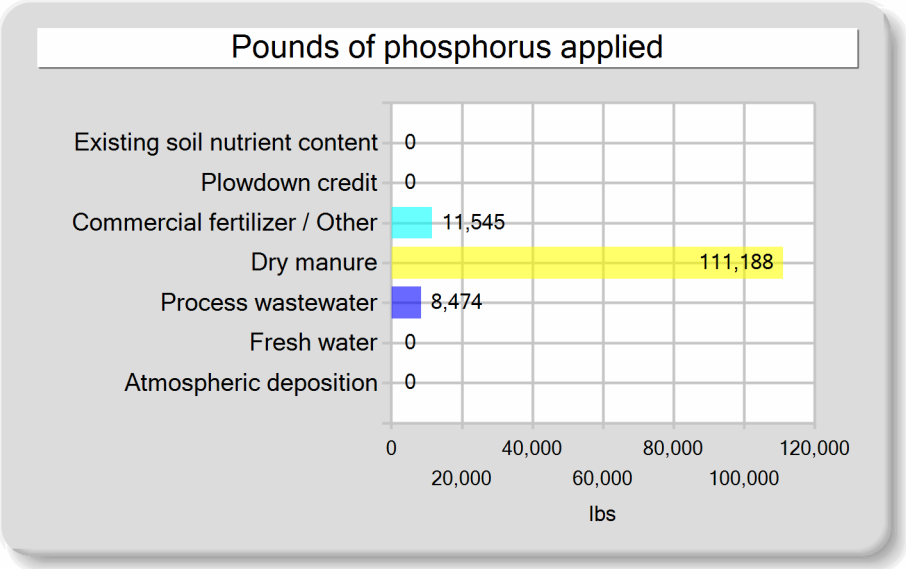
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	4,336.06	11,544.78	15,878.07	0.00
Dry manure	237,224.48	111,188.13	187,076.79	372,496.47
Process wastewater	92,311.37	8,474.28	103,492.86	649,005.15
Fresh water	6,797.23	0.00	0.00	398,117.13
Atmospheric deposition	13,944.00	0.00	0.00	0.00
Total nutrients applied	354,613.14	131,207.19	306,447.72	1,419,618.75
Anticipated crop nutrient removal	341,469.00	58,037.30	268,909.70	0.00
Actual crop nutrient removal	287,180.05	57,808.24	416,556.09	1,902,880.55
Nutrient balance	67,433.08	73,398.95	-110,108.37	-483,261.80
Applied to removed ratio	1.23	2.27	0.74	0.75

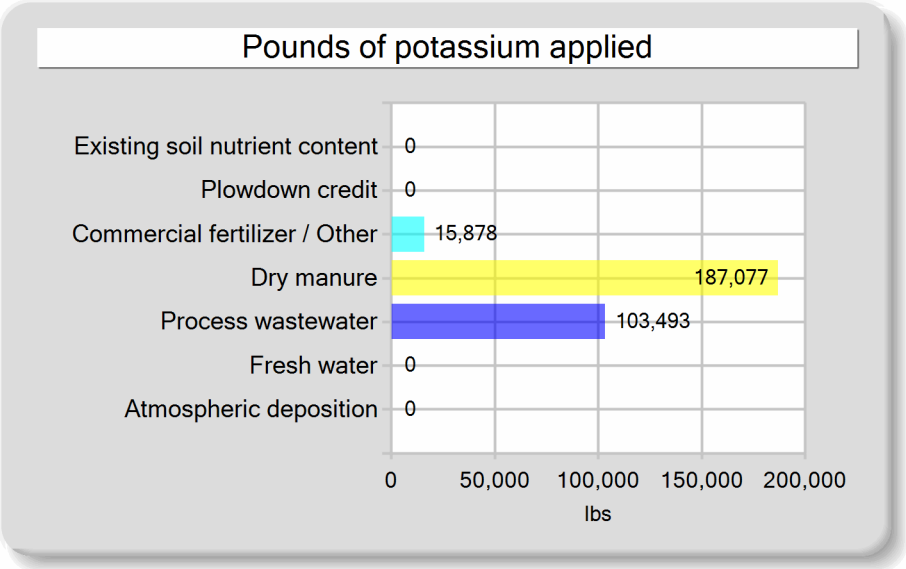
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

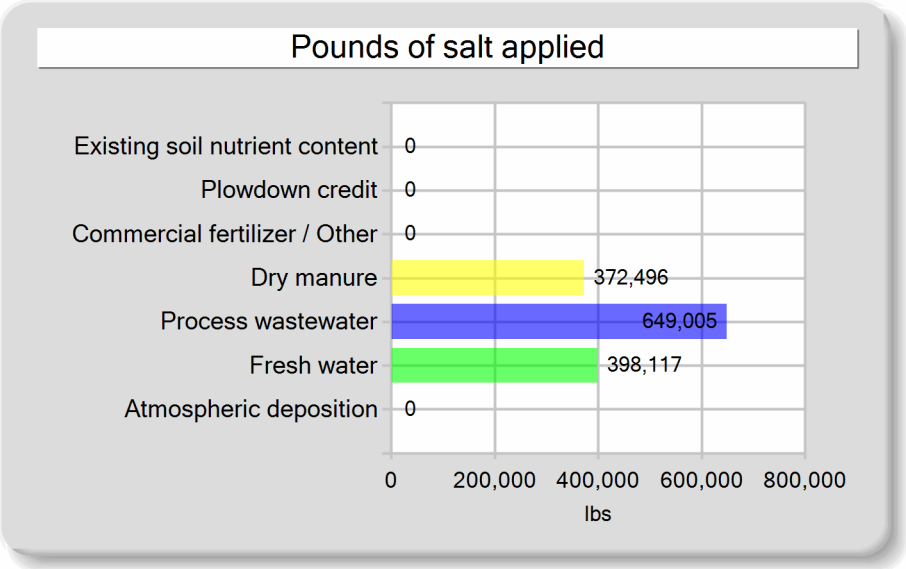


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

There were no manure or wastewater exports during the 2023 reporting year.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

John A Machado

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
John Machado	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6-26-24	
DATE	DATE

John A Machado Dairy Farms

24916 South Manteca Rd
Manteca, CA 95337

Source ID	Sample Date	Sample ID
Well #1	7/28/2023	1 Well #1
Well #2	7/28/2023	2 Well #2
Well #3	7/28/2023	3 Well #3
Well #6	7/28/2023	4 Well #6
Well #8	7/28/2023	5 Well #8
Well #9	7/28/2023	6 Well #9
Well #10	7/28/2023	7 Well #10
Well #11	7/28/2023	8 Well #11
Well #12	7/28/2023	9 Well #12
Well #13	7/28/2023	10 Well #13
Well #14	7/28/2023	11 Well #14
Well #15	7/28/2023	12 Well #15
Well #16	7/28/2023	Well #16
Well #17	7/28/2023	Well #17

BSK Associates Laboratory Fresno
687 N. Laverne Avenue
Fresno, CA 93727
559-497-2888 (Main)

AGG3796
8/17/2023
Invoice: AG19374

Michael Beerends
GHD Services Inc.
30 River Park Place West, Suite 220
Fresno, CA 93720

RE: Report for AGG3796 Ground Water

Dear Michael Beerends,

Thank you for using BSK Associates for your analytical testing needs. In the following pages, you will find the test results for the samples submitted to our laboratory on 7/28/2023. The results have been approved for release by our Laboratory Director as indicated by the authorizing signature below.

The samples were analyzed for the test(s) indicated on the Chain of Custody (see attached) and the results relate only to the samples analyzed. BSK certifies that the testing was performed in accordance with the quality system requirements specified in the 2016 TNI Standard. Any deviations from this standard or from the method requirements for each test procedure performed will be annotated alongside the analytical result or noted in the Case Narrative. Unless otherwise noted, the sample results are reported on an “as received” basis.

This certificate of analysis shall not be reproduced except in full, without written approval of the laboratory.

If additional clarification of any information is required, please contact your Project Manager, Misty Orton , at 559-497-2888.

Thank you again for using BSK Associates. We value your business and appreciate your loyalty.

Sincerely,

Misty Orton, Project Manager

Accredited in Accordance with NELAP
ORELAP #4021

The results in this report apply to the samples analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety.

Case Narrative

Project and Report Details

Client: GHD Services Inc.
Report To: Michael Beerends
Project #: 11218902 Wells
Received: 7/28/2023 - 13:05
Report Due: 8/11/2023

Invoice Details

Invoice To: GHD Services Inc.
Invoice Attn: Jeff Cloud
Project PO#: 340-013822

Sample Receipt Conditions

Cooler: Default Cooler
Temperature on Receipt ºC: 6.5

Containers Intact
COC/Labels Agree
Received On Wet Ice
Sample(s) arrived at lab on same day sampled.
Sample(s) were received in temperature range.
Initial receipt at BSK-FAL

Data Qualifiers

The following qualifiers have been applied to one or more analytical results:

None applied

Report Distribution

Recipient(s)	Report Format	CC:
Michael Beerends	FINAL.RPT	
Jeff Cloud	FINAL.RPT	karen.petryna@ghd.com
Matthew Brase	FINAL.RPT	Christina.majerowicz@ghd.com;Jeffrey.Cloud@ghd.com;Michael.beerends@ghd.com

The results in this report apply to the samples analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety.

AGG3796

Ground Water

11218902 Wells

Certificate of Analysis

Sample ID: AGG3796-01
Sampled By: Zackary Baskin
Sample Description: 12 Well 15

Sample Date - Time: 07/28/2023 - 09:30
Matrix: Ground Water
Sample Type: Grab

BSK Associates Laboratory Fresno

General Chemistry

Analyte	Method	Result	RL	Units	RL Mult	Batch	Prepared	Analyzed	Qual
Conductivity @ 25C	SM 2510B	**610**	1.0	umhos/cm	1	AGH0253	08/03/23	08/03/23	
Nitrate as N	EPA 300.0	**17**	0.23	mg/L	1	AGG1936	07/29/23 05:49	07/29/23	

The results in this report apply to the samples analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety.

AGG3796 FINAL 08172023 1748

Certificate of Analysis

Sample ID: AGG3796-02
Sampled By: Zackary Baskin
Sample Description: 11 Well 14

Sample Date - Time: 07/28/2023 - 09:30
Matrix: Ground Water
Sample Type: Grab

BSK Associates Laboratory Fresno

General Chemistry

Analyte	Method	Result	RL	Units	RL Mult	Batch	Prepared	Analyzed	Qual
Conductivity @ 25C	SM 2510B	**370**	1.0	umhos/cm	1	AGH0254	08/03/23	08/03/23	
Nitrate as N	EPA 300.0	**5.8**	0.23	mg/L	1	AGG1936	07/29/23 06:04	07/29/23	

The results in this report apply to the samples analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety.

Certificate of Analysis

Sample ID: AGG3796-03
Sampled By: Zackary Baskin
Sample Description: 10 Well 13

Sample Date - Time: 07/28/2023 - 09:30
Matrix: Ground Water
Sample Type: Grab

BSK Associates Laboratory Fresno

General Chemistry

Analyte	Method	Result	RL	Units	RL Mult	Batch	Prepared	Analyzed	Qual
Conductivity @ 25C	SM 2510B	**530**	1.0	umhos/cm	1	AGH0253	08/03/23	08/03/23	
Nitrate as N	EPA 300.0	ND	0.23	mg/L	1	AGG1936	07/29/23 05:17	07/29/23	

The results in this report apply to the samples analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety.

Certificate of Analysis

Sample ID: AGG3796-04
Sampled By: Zackary Baskin
Sample Description: 9 Well 12

Sample Date - Time: 07/28/2023 - 09:30
Matrix: Ground Water
Sample Type: Grab

BSK Associates Laboratory Fresno

General Chemistry

Analyte	Method	Result	RL	Units	RL Mult	Batch	Prepared	Analyzed	Qual
Conductivity @ 25C	SM 2510B	**770**	1.0	umhos/cm	1	AGH0253	08/03/23	08/03/23	
Nitrate as N	EPA 300.0	**2.4**	0.23	mg/L	1	AGG1936	07/29/23 05:33	07/29/23	

The results in this report apply to the samples analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety.

Certificate of Analysis

Sample ID: AGG3796-05
Sampled By: Zackary Baskin
Sample Description: 8 Well 11

Sample Date - Time: 07/28/2023 - 09:30
Matrix: Ground Water
Sample Type: Grab

BSK Associates Laboratory Fresno

General Chemistry

Analyte	Method	Result	RL	Units	RL Mult	Batch	Prepared	Analyzed	Qual
Conductivity @ 25C	SM 2510B	**620**	1.0	umhos/cm	1	AGH0253	08/03/23	08/03/23	
Nitrate as N	EPA 300.0	**12**	0.23	mg/L	1	AGG1936	07/29/23 04:15	07/29/23	

The results in this report apply to the samples analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety.

Certificate of Analysis

Sample ID: AGG3796-06
Sampled By: Zackary Baskin
Sample Description: 7 Well 10

Sample Date - Time: 07/28/2023 - 09:30
Matrix: Ground Water
Sample Type: Grab

BSK Associates Laboratory Fresno

General Chemistry

Analyte	Method	Result	RL	Units	RL Mult	Batch	Prepared	Analyzed	Qual
Conductivity @ 25C	SM 2510B	**490**	1.0	umhos/cm	1	AGH0253	08/03/23	08/03/23	
Nitrate as N	EPA 300.0	ND	0.23	mg/L	1	AGG1936	07/29/23 03:59	07/29/23	

The results in this report apply to the samples analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety.

Certificate of Analysis

Sample ID: AGG3796-07
Sampled By: Zackary Baskin
Sample Description: 6 Well 9

Sample Date - Time: 07/28/2023 - 09:30
Matrix: Ground Water
Sample Type: Grab

BSK Associates Laboratory Fresno

General Chemistry

Analyte	Method	Result	RL	Units	RL Mult	Batch	Prepared	Analyzed	Qual
Conductivity @ 25C	SM 2510B	**280**	1.0	umhos/cm	1	AGH0254	08/03/23	08/03/23	
Nitrate as N	EPA 300.0	ND	0.23	mg/L	1	AGG1936	07/29/23 06:36	07/29/23	

The results in this report apply to the samples analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety.

Certificate of Analysis

Sample ID: AGG3796-08
Sampled By: Zackary Baskin
Sample Description: 5 Well 8

Sample Date - Time: 07/28/2023 - 09:30
Matrix: Ground Water
Sample Type: Grab

BSK Associates Laboratory Fresno

General Chemistry

Analyte	Method	Result	RL	Units	RL Mult	Batch	Prepared	Analyzed	Qual
Conductivity @ 25C	SM 2510B	**1200**	1.0	umhos/cm	1	AGH0253	08/03/23	08/03/23	
Nitrate as N	EPA 300.0	**28**	0.23	mg/L	1	AGG1901	07/28/23 22:51	07/28/23	

The results in this report apply to the samples analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety.

Certificate of Analysis

Sample ID: AGG3796-09
Sampled By: Zackary Baskin
Sample Description: 4 Well 6

Sample Date - Time: 07/28/2023 - 09:30
Matrix: Ground Water
Sample Type: Grab

BSK Associates Laboratory Fresno

General Chemistry

Analyte	Method	Result	RL	Units	RL Mult	Batch	Prepared	Analyzed	Qual
Conductivity @ 25C	SM 2510B	**440**	1.0	umhos/cm	1	AGH0253	08/03/23	08/03/23	
Nitrate as N	EPA 300.0	**2.0**	0.23	mg/L	1	AGG1936	07/29/23 08:09	07/29/23	

The results in this report apply to the samples analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety.

Certificate of Analysis

Sample ID: AGG3796-10
Sampled By: Zackary Baskin
Sample Description: 3 Well 3

Sample Date - Time: 07/28/2023 - 09:30
Matrix: Ground Water
Sample Type: Grab

BSK Associates Laboratory Fresno

General Chemistry

Analyte	Method	Result	RL	Units	RL Mult	Batch	Prepared	Analyzed	Qual
Conductivity @ 25C	SM 2510B	**1000**	1.0	umhos/cm	1	AGH0253	08/03/23	08/03/23	
Nitrate as N	EPA 300.0	**9.7**	0.23	mg/L	1	AGG1936	07/29/23 06:20	07/29/23	

The results in this report apply to the samples analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety.

Certificate of Analysis

Sample ID: AGG3796-11
Sampled By: Zackary Baskin
Sample Description: 2 Well 2

Sample Date - Time: 07/28/2023 - 09:30
Matrix: Ground Water
Sample Type: Grab

BSK Associates Laboratory Fresno

General Chemistry

Analyte	Method	Result	RL	Units	RL Mult	Batch	Prepared	Analyzed	Qual
Conductivity @ 25C	SM 2510B	**1200**	1.0	umhos/cm	1	AGH0253	08/03/23	08/03/23	
Nitrate as N	EPA 300.0	**40**	0.23	mg/L	1	AGG1901	07/28/23 21:53	07/28/23	

The results in this report apply to the samples analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety.

Certificate of Analysis

Sample ID: AGG3796-12
Sampled By: Zackary Baskin
Sample Description: 1 Well 1

Sample Date - Time: 07/28/2023 - 09:30
Matrix: Ground Water
Sample Type: Grab

BSK Associates Laboratory Fresno

General Chemistry

Analyte	Method	Result	RL	Units	RL Mult	Batch	Prepared	Analyzed	Qual
Conductivity @ 25C	SM 2510B	**810**	1.0	umhos/cm	1	AGH0254	08/03/23	08/03/23	
Nitrate as N	EPA 300.0	**18**	0.23	mg/L	1	AGG1901	07/28/23 22:37	07/28/23	

The results in this report apply to the samples analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety.

Certificate of Analysis

Sample ID: AGG3796-13
Sampled By: Zackary Baskin
Sample Description: Well 16

Sample Date - Time: 07/28/2023 - 09:30
Matrix: Ground Water
Sample Type: Grab

BSK Associates Laboratory Fresno

General Chemistry

Analyte	Method	Result	RL	Units	RL Mult	Batch	Prepared	Analyzed	Qual
Chloride	EPA 300.0	**30**	1.0	mg/L	1	AGG1901	07/28/23	07/28/23	
Conductivity @ 25C	SM 2510B	**630**	1.0	umhos/cm	1	AGG1997	07/31/23	07/31/23	
Nitrate as N	EPA 300.0	**5.0**	0.23	mg/L	1	AGG1901	07/28/23 21:39	07/28/23	
Sulfate as SO4	EPA 300.0	**35**	1.0	mg/L	1	AGG1901	07/28/23	07/28/23	
Total Dissolved Solids	SM 2540C	**350**	5.0	mg/L	1	AGH0238	08/03/23	08/03/23	

Metals

Analyte	Method	Result	RL	Units	RL Mult	Batch	Prepared	Analyzed	Qual
Calcium	EPA 200.7	**52**	0.10	mg/L	1	AGH0235	08/03/23	08/04/23	
Magnesium	EPA 200.7	**14**	0.10	mg/L	1	AGH0235	08/03/23	08/04/23	
Sodium	EPA 200.7	**70**	1.0	mg/L	1	AGH0235	08/03/23	08/04/23	

The results in this report apply to the samples analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety.

Certificate of Analysis

Sample ID: AGG3796-13RE1
Sampled By: Zackary Baskin
Sample Description: Well 16

Sample Date - Time: 07/28/2023 - 09:30
Matrix: Ground Water
Sample Type: Grab

BSK Associates Laboratory Fresno

General Chemistry

Analyte	Method	Result	RL	Units	RL Mult	Batch	Prepared	Analyzed	Qual
Alkalinity as CaCO3	SM 2320B	**230**	3.0	mg/L	1	AGH0153	08/02/23	08/02/23	
Bicarbonate as CaCO3	SM 2320B	**230**	3.0	mg/L	1	AGH0153	08/02/23	08/02/23	
Carbonate as CaCO3	SM 2320B	ND	3.0	mg/L	1	AGH0153	08/02/23	08/02/23	
Hydroxide as CaCO3	SM 2320B	ND	3.0	mg/L	1	AGH0153	08/02/23	08/02/23	

The results in this report apply to the samples analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety.

Certificate of Analysis

Sample ID: AGG3796-14
Sampled By: Zackary Baskin
Sample Description: Well 17

Sample Date - Time: 07/28/2023 - 09:30
Matrix: Ground Water
Sample Type: Grab

BSK Associates Laboratory Fresno

General Chemistry

Analyte	Method	Result	RL	Units	RL Mult	Batch	Prepared	Analyzed	Qual
Chloride	EPA 300.0	**15**	1.0	mg/L	1	AGG1901	07/28/23	07/28/23	
Conductivity @ 25C	SM 2510B	**330**	1.0	umhos/cm	1	AGG1997	07/31/23	07/31/23	
Nitrate as N	EPA 300.0	**0.78**	0.23	mg/L	1	AGG1901	07/28/23 21:24	07/28/23	
Sulfate as SO4	EPA 300.0	**21**	1.0	mg/L	1	AGG1901	07/28/23	07/28/23	
Total Dissolved Solids	SM 2540C	**200**	5.0	mg/L	1	AGH0238	08/03/23	08/03/23	

Metals

Analyte	Method	Result	RL	Units	RL Mult	Batch	Prepared	Analyzed	Qual
Calcium	EPA 200.7	**31**	0.10	mg/L	1	AGH0235	08/03/23	08/04/23	
Magnesium	EPA 200.7	**7.7**	0.10	mg/L	1	AGH0235	08/03/23	08/04/23	
Sodium	EPA 200.7	**30**	1.0	mg/L	1	AGH0235	08/03/23	08/04/23	

The results in this report apply to the samples analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety.

Certificate of Analysis

Sample ID: AGG3796-14RE1
Sampled By: Zackary Baskin
Sample Description: Well 17

Sample Date - Time: 07/28/2023 - 09:30
Matrix: Ground Water
Sample Type: Grab

BSK Associates Laboratory Fresno

General Chemistry

Analyte	Method	Result	RL	Units	RL Mult	Batch	Prepared	Analyzed	Qual
Alkalinity as CaCO3	SM 2320B	**120**	3.0	mg/L	1	AGH0153	08/02/23	08/02/23	
Bicarbonate as CaCO3	SM 2320B	**120**	3.0	mg/L	1	AGH0153	08/02/23	08/02/23	
Carbonate as CaCO3	SM 2320B	ND	3.0	mg/L	1	AGH0153	08/02/23	08/02/23	
Hydroxide as CaCO3	SM 2320B	ND	3.0	mg/L	1	AGH0153	08/02/23	08/02/23	

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General Chemistry Quality Control Report

Analyte	Result	RL	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit	Date Analyzed	Qual
EPA 300.0 - Quality Control											
Batch: AGG1901										Prepared: 7/28/2023	
Prep Method: Method Specific Preparation										Analyst: GJA	
Blank (AGG1901-BLK1)											
Nitrate as N	ND	0.23	mg/L							07/28/23	
Chloride	ND	1.0	mg/L							07/28/23	
Sulfate as SO4	ND	1.0	mg/L							07/28/23	
Blank Spike (AGG1901-BS1)											
Nitrate as N	23	0.23	mg/L	23	ND	102	90-110			07/28/23	
Chloride	100	1.0	mg/L	100	ND	102	90-110			07/28/23	
Sulfate as SO4	100	1.0	mg/L	100	ND	101	90-110			07/28/23	
Matrix Spike (AGG1901-MS1), Source: AGG3651-03											
Nitrate as N	15	0.23	mg/L	11	3.9	102	80-120			07/28/23	
Chloride	51	1.0	mg/L	50	ND	100	80-120			07/28/23	
Sulfate as SO4	49	1.0	mg/L	50	ND	97	80-120			07/28/23	
Matrix Spike (AGG1901-MS2), Source: AGG3790-02											
Nitrate as N	18	0.23	mg/L	11	6.3	103	80-120			07/28/23	
Chloride	57	1.0	mg/L	50	6.3	102	80-120			07/28/23	
Sulfate as SO4	49	1.0	mg/L	50	ND	98	80-120			07/28/23	
Matrix Spike Dup (AGG1901-MSD1), Source: AGG3651-03											
Nitrate as N	15	0.23	mg/L	11	3.9	103	80-120	1	20	07/28/23	
Chloride	51	1.0	mg/L	50	ND	101	80-120	1	20	07/28/23	
Sulfate as SO4	49	1.0	mg/L	50	ND	98	80-120	1	20	07/28/23	
Matrix Spike Dup (AGG1901-MSD2), Source: AGG3790-02											
Nitrate as N	18	0.23	mg/L	11	6.3	103	80-120	0	20	07/28/23	
Chloride	57	1.0	mg/L	50	6.3	102	80-120	0	20	07/28/23	
Sulfate as SO4	49	1.0	mg/L	50	ND	98	80-120	0	20	07/28/23	
EPA 300.0 - Quality Control											
Batch: AGG1936										Prepared: 7/29/2023	
Prep Method: Method Specific Preparation										Analyst: DXR	
Blank (AGG1936-BLK1)											
Nitrate as N	ND	0.23	mg/L							07/29/23	
Blank Spike (AGG1936-BS1)											
Nitrate as N	23	0.23	mg/L	23	ND	102	90-110			07/29/23	
Matrix Spike (AGG1936-MS1), Source: AGG3793-01											
Nitrate as N	11	0.23	mg/L	11	ND	98	80-120			07/29/23	
Matrix Spike (AGG1936-MS2), Source: AGG3793-02											
Nitrate as N	11	0.23	mg/L	11	ND	99	80-120			07/29/23	

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General Chemistry Quality Control Report

Analyte	Result	RL	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit	Date Analyzed	Qual
EPA 300.0 - Quality Control											
Batch: AGG1936										Prepared: 7/29/2023	
Prep Method: Method Specific Preparation										Analyst: DXR	
Matrix Spike Dup (AGG1936-MSD1), Source: AGG3793-01											
Nitrate as N	11	0.23	mg/L	11	ND	100	80-120	2	20	07/29/23	
Matrix Spike Dup (AGG1936-MSD2), Source: AGG3793-02											
Nitrate as N	11	0.23	mg/L	11	ND	100	80-120	1	20	07/29/23	
SM 2320B - Quality Control											
Batch: AGG1997										Prepared: 7/31/2023	
Prep Method: Method Specific Preparation										Analyst: BAG	
Blank (AGG1997-BLK1)											
Alkalinity as CaCO3	ND	3.0	mg/L							07/31/23	
Bicarbonate as CaCO3	ND	3.0	mg/L							07/31/23	
Carbonate as CaCO3	ND	3.0	mg/L							07/31/23	
Hydroxide as CaCO3	ND	3.0	mg/L							07/31/23	
Blank Spike (AGG1997-BS1)											
Alkalinity as CaCO3	110	3.0	mg/L	100	ND	106	80-120			07/31/23	
Blank Spike Dup (AGG1997-BSD1)											
Alkalinity as CaCO3	100	3.0	mg/L	100	ND	102	80-120	3	20	07/31/23	
Duplicate (AGG1997-DUP1), Source: AGG3875-01											
Alkalinity as CaCO3	52	3.0	mg/L		53			2	10	07/31/23	
Bicarbonate as CaCO3	52	3.0	mg/L		53			2	10	07/31/23	
Carbonate as CaCO3	ND	3.0	mg/L		ND				10	07/31/23	
Hydroxide as CaCO3	ND	3.0	mg/L		ND				10	07/31/23	
SM 2320B - Quality Control											
Batch: AGH0153										Prepared: 8/2/2023	
Prep Method: Method Specific Preparation										Analyst: BAG	
Blank (AGH0153-BLK1)											
Alkalinity as CaCO3	ND	3.0	mg/L							08/02/23	
Bicarbonate as CaCO3	ND	3.0	mg/L							08/02/23	
Carbonate as CaCO3	ND	3.0	mg/L							08/02/23	
Hydroxide as CaCO3	ND	3.0	mg/L							08/02/23	
Blank Spike (AGH0153-BS1)											
Alkalinity as CaCO3	98	3.0	mg/L	100	ND	98	80-120			08/02/23	
Blank Spike Dup (AGH0153-BSD1)											
Alkalinity as CaCO3	100	3.0	mg/L	100	ND	102	80-120	4	20	08/02/23	
Duplicate (AGH0153-DUP1), Source: AGH0047-01											

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General Chemistry Quality Control Report

Analyte	Result	RL	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit	Date Analyzed	Qual
SM 2320B - Quality Control											
Batch: AGH0153										Prepared: 8/2/2023	
Prep Method: Method Specific Preparation										Analyst: BAG	
Duplicate (AGH0153-DUP1), Source: AGH0047-01											
Alkalinity as CaCO3	130	3.0	mg/L		130			1	10	08/02/23	
Bicarbonate as CaCO3	130	3.0	mg/L		130			1	10	08/02/23	
Carbonate as CaCO3	ND	3.0	mg/L		ND				10	08/02/23	
Hydroxide as CaCO3	ND	3.0	mg/L		ND				10	08/02/23	
SM 2510B - Quality Control											
Batch: AGG1997										Prepared: 7/31/2023	
Prep Method: Method Specific Preparation										Analyst: BAG	
Blank Spike (AGG1997-BS1)											
Conductivity @ 25C	1400	1.0	umhos/cm	1400	ND	100	90-110			07/31/23	
Blank Spike Dup (AGG1997-BSD1)											
Conductivity @ 25C	1400	1.0	umhos/cm	1400	ND	100	90-110	0	5	07/31/23	
Duplicate (AGG1997-DUP1), Source: AGG3875-01											
Conductivity @ 25C	710	1.0	umhos/cm		710			0	5	07/31/23	
SM 2510B - Quality Control											
Batch: AGH0253										Prepared: 8/3/2023	
Prep Method: Method Specific Preparation										Analyst: BAG	
Blank Spike (AGH0253-BS1)											
Conductivity @ 25C	1400	1.0	umhos/cm	1400	ND	96	90-110			08/03/23	
Blank Spike Dup (AGH0253-BSD1)											
Conductivity @ 25C	1400	1.0	umhos/cm	1400	ND	98	90-110	1	5	08/03/23	
Duplicate (AGH0253-DUP1), Source: AGG3592-01RE1											
Conductivity @ 25C	18	1.0	umhos/cm		19			3	5	08/03/23	
SM 2510B - Quality Control											
Batch: AGH0254										Prepared: 8/3/2023	
Prep Method: Method Specific Preparation										Analyst: BAG	
Blank Spike (AGH0254-BS1)											
Conductivity @ 25C	1300	1.0	umhos/cm	1400	ND	96	90-110			08/03/23	
Blank Spike Dup (AGH0254-BSD1)											
Conductivity @ 25C	1400	1.0	umhos/cm	1400	ND	96	90-110	1	5	08/03/23	
Duplicate (AGH0254-DUP1), Source: VGG0538-01											
Conductivity @ 25C	620	1.0	umhos/cm		620			0	5	08/03/23	

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BSK Associates Laboratory Fresno

General Chemistry Quality Control Report

Analyte	Result	RL	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit	Date Analyzed	Qual
SM 2540C - Quality Control											
Batch: AGH0238										Prepared: 8/3/2023	
Prep Method: Method Specific Preparation										Analyst: RRV	
Blank (AGH0238-BLK1)											
Total Dissolved Solids	ND	5.0	mg/L							08/03/23	
Blank Spike (AGH0238-BS1)											
Total Dissolved Solids	980		mg/L	1000		98	70-130			08/03/23	
Duplicate (AGH0238-DUP1), Source: AGH0055-01											
Total Dissolved Solids	1300	5.0	mg/L		1300			1	10	08/03/23	
Duplicate (AGH0238-DUP2), Source: AGH0024-01											
Total Dissolved Solids	990	5.0	mg/L		960			3	10	08/03/23	

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BSK Associates Laboratory Fresno

Metals Quality Control Report

EPA 200.7 - Quality Control

Batch: AGH0235 Prepared: 8/3/2023

Prep Method: EPA 200.2 Analyst: SAB

Analyte	Result	RL	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit	Date Analyzed	Qual
Blank (AGH0235-BLK2)											
Calcium	ND	0.10	mg/L							08/04/23	
Magnesium	ND	0.10	mg/L							08/04/23	
Sodium	ND	1.0	mg/L							08/04/23	
Blank Spike (AGH0235-BS2)											
Calcium	4.7	0.10	mg/L	4.8	ND	97	85-115			08/04/23	
Magnesium	4.6	0.10	mg/L	4.8	ND	95	85-115			08/04/23	
Sodium	5.3	1.0	mg/L	4.8	ND	110	85-115			08/04/23	
Blank Spike Dup (AGH0235-BSD2)											
Calcium	4.6	0.10	mg/L	4.8	ND	95	85-115	2	20	08/04/23	
Magnesium	4.4	0.10	mg/L	4.8	ND	92	85-115	3	20	08/04/23	
Sodium	5.1	1.0	mg/L	4.8	ND	107	85-115	3	20	08/04/23	
Matrix Spike (AGH0235-MS3), Source: AGG3721-01											
Calcium	17	0.10	mg/L	4.8	12	96	70-130			08/04/23	
Magnesium	10	0.10	mg/L	4.8	5.4	94	70-130			08/04/23	
Sodium	26	1.0	mg/L	4.8	22	88	70-130			08/04/23	
Matrix Spike (AGH0235-MS4), Source: AGG3796-14											
Calcium	35	0.10	mg/L	4.8	31	91	70-130			08/04/23	
Magnesium	12	0.10	mg/L	4.8	7.7	100	70-130			08/04/23	
Sodium	34	1.0	mg/L	4.8	30	91	70-130			08/04/23	
Matrix Spike Dup (AGH0235-MSD3), Source: AGG3721-01											
Calcium	17	0.10	mg/L	4.8	12	94	70-130	1	20	08/04/23	
Magnesium	10	0.10	mg/L	4.8	5.4	99	70-130	2	20	08/04/23	
Sodium	26	1.0	mg/L	4.8	22	87	70-130	0	20	08/04/23	
Matrix Spike Dup (AGH0235-MSD4), Source: AGG3796-14											
Calcium	34	0.10	mg/L	4.8	31	79	70-130	2	20	08/04/23	
Magnesium	12	0.10	mg/L	4.8	7.7	95	70-130	2	20	08/04/23	
Sodium	33	1.0	mg/L	4.8	30	81	70-130	1	20	08/04/23	

The results in this report apply to the samples analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety.

Certificate of Analysis

Notes:

- The Chain of Custody document and Sample Integrity Sheet are part of the analytical report.
- Any remaining sample(s) for testing will be disposed of according to BSK's sample retention policy unless other arrangements are made in advance.
- All positive results for EPA Methods 504.1 and 524.2 require the analysis of a Field Reagent Blank (FRB) to confirm that the results are not a contamination error from field sampling steps. If Field Reagent Blanks were not submitted with the samples, this method requirement has not been performed.
- Samples collected by BSK Analytical Laboratories were collected in accordance with the BSK Sampling and Collection Standard Operating Procedures.
- J-value is equivalent to DNQ (Detected, not quantified) which is a trace value. A trace value is an analyte detected between the MDL and the laboratory reporting limit. This result is of an unknown data quality and is only qualitative (estimated). Baseline noise, calibration curve extrapolation below the lowest calibrator, method blank detections, and integration artifacts can all produce apparent DNQ values, which contribute to the un-reliability of these values.
- (1) - Residual chlorine and pH analysis have a 15 minute holding time for both drinking and waste water samples as defined by the EPA and 40 CFR 136. Waste water and ground water (monitoring well) samples must be field filtered to meet the 15 minute holding time for dissolved metals.
- Field tests are outside the scope of laboratory accreditation and there is no certification available for field testing.
- Summations of analytes (i.e. Total Trihalomethanes) may appear to add individual amounts incorrectly, due to rounding of analyte values occurring before or after the total value is calculated, as well as rounding of the total value.
- RL Multiplier is the factor used to adjust the reporting limit (RL) due to variations in sample preparation procedures and dilutions required for matrix interferences.
- Due to the subjective nature of the Threshold Odor Method, all characterizations of the detected odor are the opinion of the panel of analysts. The characterizations can be found in Standard Methods 2170B Figure 2170:1.
- The MCLs provided in this report (if applicable) represent the primary MCLs for that analyte.
- (2) - Formerly known as Bis(2-Chloroisopropyl) ether.
  Unless otherwise noted, TOC results by SM 5310C method do not include purgeable organic carbon, which is removed along with the inorganic carbon interference. The POC contribution to TOC is considered to be negligible.

Certificate of Analysis

Definitions

mg/L:	Milligrams/Liter (ppm)	MDL:	Method Detection Limit	MDA95:	Min. Detected Activity
mg/Kg:	Milligrams/Kilogram (ppm)	RL:	Reporting Limit: DL x Dilution	MPN:	Most Probable Number
µg/L:	Micrograms/Liter (ppb)	ND:	None Detected below MRL/MDL	CFU:	Colony Forming Unit
µg/Kg:	Micrograms/Kilogram (ppb)	pCi/L:	PicoCuries per Liter	Absent:	Less than 1 CFU/100mLs
%:	Percent	RL Mult:	RL Multiplier	Present:	1 or more CFU/100mLs
NR:	Non-Reportable	MCL:	Maximum Contaminant Limit	U:	The analyte was not detected at or above the reported sample quantitation limit.

Please see the individual Subcontract Lab's report for applicable certifications.

The following parameters are not available for certification through CA ELAP:

Odor Diisopropyl ether (DIPE) by EPA 524.2

The following parameters are calculated values and are outside the scope of our NELAP accreditation:

Total Nitrogen Aggressive Index Trivalent Chromium

BSK is not accredited under the NELAP program for the following additional parameters: **NA**

Certifications: **Please refer to our website for a copy of our Accredited Fields of Testing under each certification.**

Fresno

State of California - ELAP	1180	State of Hawaii	4021
Los Angeles CSD	9254479	NELAP certified	4021-021
State of Nevada	CA000792022-1	State of Oregon - NELAP	4021-021
EPA UCMR5	CA00079	State of Washington	C997-23

Sacramento

State of California - ELAP	1180-S1

San Bernardino

State of California - ELAP	1180-S2	Los Angeles CSD	9254478
NELAP certified	4119-007	State of Oregon - NELAP	4119-007

Vancouver

NELAP certified	WA100008-016	State of Oregon - NELAP	WA100008-016
State of Washington	C824-22		

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BSK Associates SR-FL-0002-24

Sample Integrity

BSK Bottles: Yes No Page 1 of 1

COC Info				
Was temperature within range? Chemistry ≤ 6°C Micro < 8°C	Yes No NA	Were correct containers and preservatives received for the tests requested?	Yes No NA
If samples were taken today, is there evidence that chilling has begun?	Yes No NA	Bubbles Present VOAs (524.2/TTHM/TCP)? TB Received? (Check Method Below)	Yes No NA Yes No NA
Did all bottles arrive unbroken and intact?	Yes No	Was a sufficient amount of sample received?	Yes No
Did all bottle labels agree with COC?	Yes No	Do samples have a hold time <72 hours?	Yes No
Was sodium thiosulfate added to CN sample(s) until chlorine was no longer present?	Yes NA	Was PM notified of discrepancies? PM: By/Time:	Yes No NA

Bottles Received

"—" means preservation/chlorine checks are either N/A or are performed in the lab

250ml(A) 500ml(B) 1Liter(C) 40mlVOA(V) 125ml(D)	Checks*	Passed?	1-12	13-14			
Bacti $Na_2S_2O_3$	—	—					
None (P) White Cap	—	—	1A	1B,1C			
Cr6 (P) Lt. Green Label/Blue Cap NH4OH(NH4)2SO4 **DW**	Cl, pH > 8	P F					
Cr6 (P) Pink Label/Blue Cap NH4OH(NH4)2SO4 **WW**	pH 9.3-9.7	P F					
Cr6 (P) Black Label/Blue Cap NH4OH(NH4)2SO4 **7199** ***24 HOUR HOLD TIME***	pH 9.0-9.5	P F					
HNO_3 (P) Red Cap or HCl (P) Purple Cap/Lt. Blue Label	—	—		1B			
H_2SO_4 (P) or (AG) Yellow Cap/Label	pH < 2	P F					
NaOH (P) Green Cap	Cl, pH >10	P F					
NaOH + ZnAc (P)	pH > 9	P F					
Dissolved Oxygen 300ml (g)	—	—					
None (AG) 608/8081/8082, 625, 632/8321, 8151, 8270	—	—					
HCl (AG) Lt. Blue Label O&G, Diesel, **TCP**	—	—					
Ascorbic, EDTA, KH_2Ct (AG) Pink Label 525	—	—					
Na_2SO_3 250mL (AG) Neon Green Label 515	—	—					
$Na_2S_2O_3$ 1 Liter (Brown P) 549	—	—					
$Na_2S_2O_3$ (AG) Blue Label 548, THM, 524	—	—					
$Na_2S_2O_3$ (CG) Blue Label 504, 505, 547	—	—					
$Na_2S_2O_3$ + MCAA (CG) Orange Label 531	pH < 3	P F					
NH_4Cl (AG) Purple Label 552	—	—					
EDA (P) or (AG) Brown Label DBPs	—	—					
HCL (CG) 524.2,BTEX,Gas, MTBE, 8260/624	—	—					
Buffer pH 4 (CG)	—	—					
H_3PO_4 (CG) Salmon Label	—	—					
Trizma – EPA 537.1 Light Blue Label **FB**	---	---					
Ammonia Acetate - EPA 533 Purple Label **FB**	---	---					
Bottled Water	—	—					
Asbestos 1L (P) w/ Foil / LL Metals Bottle	—	—					
Clear Glass	—	—					
OTHER:	—	—					

N+17-28-23

Split

	Container	Preservative	Lot #	Initials	Date/Time
S P					
S P					

Preservation Check
pH Lot #
Cl Lot #

Comments

***Preservation check completed by lab performing analysis.**

✓ **Indicates Blanks Received**

504 ___ 524.2 ___ TTHM ___ 537/533 ___ TCP___

✓ **MS/MSD Received Method:** _____

Labeled by: **Labels Checked by:**

Scanned: Cen Rush/Short HT Page: _____ Time: _____

BSK ASSOCIATES
1414 Stanislaus St., Fresno, CA 93706
(559) 497-2888 · Fax (559) 497-2893
www.bskassociates.com

Turnaround Time Request
- [x] Standard - 10 business days
- [ ] Rush (Surcharge may apply)
- Date needed:

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*Required Fields

Temp: 6.5° 4.5

Company/Client Name*: GHD

Report Attention*: Mike Beerends

Additional cc's: Michael.Beerends@GHD.com

Invoice To*:

PO#:

Phone*: 559-326-5900 Fax: 559-412-2820

E-mail*: Matthew.Brase@ghd.com

Address*: 30 River Park Place West Suite 220 City*: Fresno State*: CA Zip*: 93720

Project: 11218902

Project #: 11218902

How would you like to receive your completed results?* [x] E-Mail [ ] Fax [ ] Mail

Reporting Options: [ ] Trace (J-Flag) [ ] Swamp [ ] EDD Type: _____

Sampler Name (Printed/Signature)*: Zackary Baskin

Regulatory Carbon Copies: [ ] SWRCB (Drinking Water) [ ] Merced Co [ ] Fresno Co [ ] Madera Co [ ] Tulare Co [ ] Other: _____

Regulatory Compliance: [ ] EDT to California SWRCB (Drinking Water) System Number*: _____ [ ] Geotracker #: _____

Matrix Types: SW=Surface Water BW=Bottled Water GW=Ground Water WW=Waste Water STW=Storm Water DW=Drinking Water SO=Solid

#	Sample Description*	Sampled* Date	Sampled* Time	Matrix*	Comments / Station Code / WTRAX	NO3-N (Nitrate - Nitrogen)	EC	TDS	Total Nitrogen	Calcium	Magnesium	Sodium	Chloride	Sulfate	Bicarbonate and Carbonate
1	12 Well 15	7/28/23	0930	GW		X	X								
2	11 Well 14				18 bottles	X	X								
3	10 Well 13				total	X	X								
4	9 Well 12					X	X								
5	8 Well 11					X	X								
6	7 Well 10					X	X								
7	6 Well 9					X	X								
8	5 Well 8					X	X								
9	4 Well 6					X	X								
10	3 Well 3					X	X								
11	2 Well 2					X	X								
12	1 Well 1					X	X								
13	Well 16					X	X	X		X	X	X	X	X	X
14	Well 17					X	X	X		X	X	X	X	X	X

Relinquished by: (Signature and Printed Name)	Company	Date	Time	Received by: (Signature and Printed Name)	Company
Zackary Baskin	GHD	7/28/23	13:05		
Relinquished by: (Signature and Printed Name)	Company	Date	Time	Received by: (Signature and Printed Name)	Company

Received for Lab by: (Signature and Printed Name) N Hammond Date 7-28-23 Time 1305

Payment Received at Delivery: Check / Cash
Date: Amount: PIA#: Init.

Shipping Method: ONTRAC UPS GSO WALK-IN FED EX Courier: _____ Custody Seal: Y/N

Cooling Method: Wet Blue None Chilling Process Begun: Y/N

Payment for services rendered as noted herein are due in full within 30 days from the date invoiced. If not so paid, account balances are deemed delinquent. Delinquent balances are subject to monthly service charges and interest specified in BSK's current Standard Terms and Conditions for Laboratory Services. The person signing for the Client/Company acknowledges that they are either the Client or an authorized agent to the Client, that the Client agrees to be responsible for payment for the services on this Chain of Custody, and agrees to BSK's terms and conditions for laboratory services unless contractually bound otherwise. BSK's current terms and conditions can be found at www.bskassociates.com/BSKLabTermsConditions.pdf